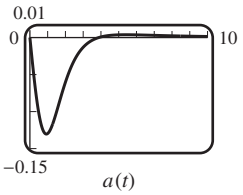
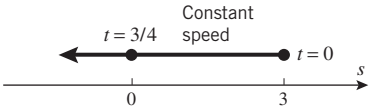
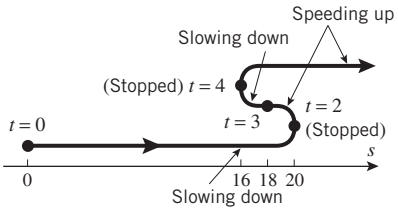
(c) speeding up for $\sqrt{5} < t < \sqrt{15}$; slowing down for $0 < t < \sqrt{5}$ and $\sqrt{15} < t$



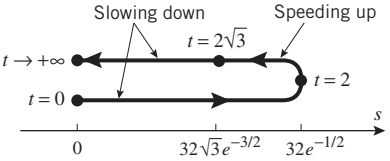
25.



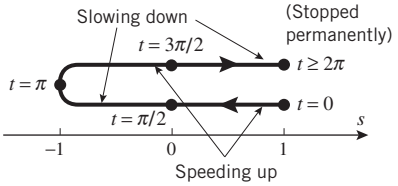
27.



29.



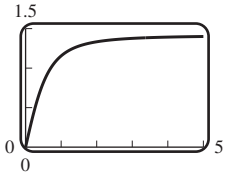
31.



33. **(a)** 12 ft/s **(b)** $t = 2.2$ s, $s = -24.2$ ft

35. **(a)** $t = 2 \pm 1/\sqrt{3}$, $s = \ln 2$, $v = \pm\sqrt{3}$ **(b)** $t = 2$, $s = 0$, $a = 6$

37. **(a)**



(b) $\sqrt{2}$

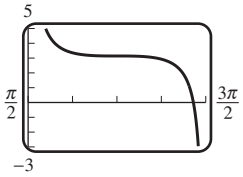
39. **(b)** $\frac{2}{3}$ unit **(c)** $0 \le t < 1$ and $t > 2$

Exercise Set 4.7 (Page 300)

1. 1.414213562 **3.** 1.817120593 **5.** $x \approx 1.76929$

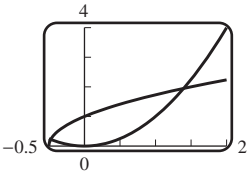
7. $x \approx 1.224439550$ **9.** $x \approx -1.24962$ **11.** $x \approx 1.02987$

13. $x \approx 4.493409458$



15. $x \approx 0.68233$

17. -0.474626618, 1.395336994 **19.** $x \approx 0.58853$ or 3.09636



Responses to True–False questions may be abridged to save space.

21. True; $x = x_{n+1}$ is the x-intercept of the tangent line to $y = f(x)$ at $x = x_n$.

23. False; for example, if $f(x) = x(x-3)^2$, Newton's Method fails (analogous to Figure 4.7.4) with $x_1 = 1$ and approximates the root $x = 3$ for $x_1 > 1$.

25. **(b)** 3.162277660 **27.** -4.098859132

29. $x = -1$ or $x \approx 0.17951$ **31.** (0.589754512, 0.347810385)

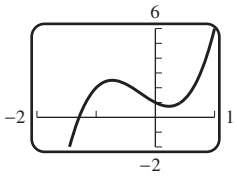
33. **(b)** $\theta \approx 2.99156$ rad or $171°$ **35.** -1.220744085, 0.724491959

37. $i = 0.053362$ or 5.33% **39.** **(a)** The values do not converge.

Exercise Set 4.8 (Page 308)

1. $c = 4$ **3.** $c = \pi$ **5.** $c = 1$ **7.** $c = -\sqrt{5}$

9. **(a)** $[-2, 1]$
(b) $c \approx -1.29$
(c) -1.2885843



Responses to True–False questions may be abridged to save space.

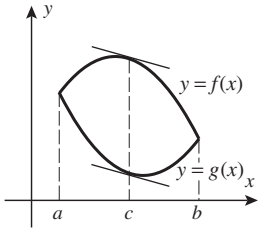
11. False; Rolle's Theorem requires the additional hypothesis that f is differentiable on (a, b) and $f(a) = f(b) = 0$; see Example 2.

13. False; the Constant Difference Theorem applies to two functions with equal derivatives on an interval to conclude that the functions differ by a constant on the interval.

15. **(b)** $\tan x$ is not continuous on $[0, \pi]$. **25.** $f(x) = xe^x - e^x + 2$

35. **(b)** $f(x) = \sin x$, $g(x) = \cos x$

37.



41. $a = 6$, $b = -3$

Chapter 4 Review Exercises (Page 310)

1. **(a)** $f(x_1) < f(x_2)$; $f(x_1) > f(x_2)$; $f(x_1) = f(x_2)$
(b) $f' > 0$; $f' < 0$; $f' = 0$

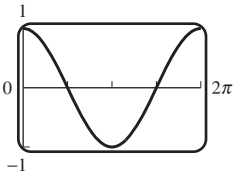
3. **(a)** $\left[\frac{5}{2}, +\infty\right)$ **(b)** $\left(-\infty, \frac{5}{2}\right]$ **(c)** $(-\infty, +\infty)$ **(d)** none **(e)** none

5. **(a)** $[0, +\infty)$ **(b)** $(-\infty, 0]$ **(c)** $(-\sqrt{2/3}, \sqrt{2/3})$
(d) $(-\infty, -\sqrt{2/3})$, $(\sqrt{2/3}, +\infty)$ **(e)** $-\sqrt{2/3}$, $\sqrt{2/3}$

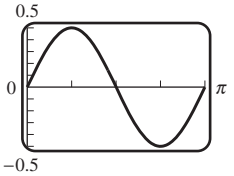
7. **(a)** $[-1, +\infty)$ **(b)** $(-\infty, -1]$ **(c)** $(-\infty, 0)$, $(2, +\infty)$
(d) $(0, 2)$ **(e)** 0, 2

9. **(a)** $(-\infty, 0]$ **(b)** $[0, +\infty)$ **(c)** $(-\infty, -1/\sqrt{2})$, $(1/\sqrt{2}, +\infty)$
(d) $(-1/\sqrt{2}, 1/\sqrt{2})$ **(e)** $\pm 1/\sqrt{2}$

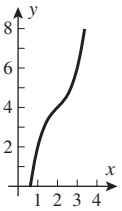
11. increasing on $[\pi, 2\pi]$;
decreasing on $[0, \pi]$;
concave up on $(\pi/2, 3\pi/2)$;
concave down on $(0, \pi/2)$, $(3\pi/2, 2\pi)$;
inflection points: $(\pi/2, 0)$, $(3\pi/2, 0)$



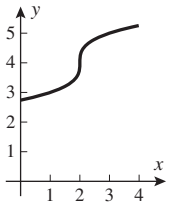
13. increasing on $[0, \pi/4]$, $[3\pi/4, \pi]$;
decreasing on $[\pi/4, 3\pi/4]$;
concave up on $(\pi/2, \pi)$;
concave down on $(0, \pi/2)$;
inflection point: $(\pi/2, 0)$



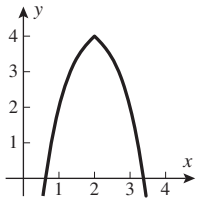
15. **(a)**



(b)



(c)

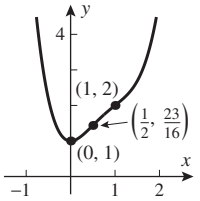


17. $-\dfrac{b}{2a} \le 0$ **19.** $x = -1$ **21.** **(a)** at an inflection point

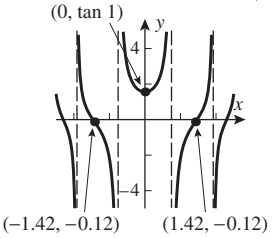
25. **(a)** $x = \pm\sqrt{2}$ (stationary points) **(b)** $x = 0$ (stationary point)

27. **(a)** relative max at $x = 1$, relative min at $x = 7$, neither at $x = 0$
(b) relative max at $x = \pi/2, 3\pi/2$; relative min at $x = 7\pi/6, 11\pi/6$
(c) relative max at $x = 5$

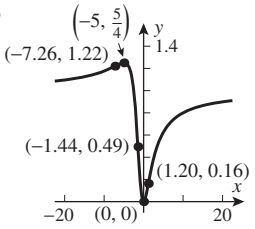
29. $\lim\limits_{x\to-\infty} f(x) = +\infty$, $\lim\limits_{x\to+\infty} f(x) = +\infty$;
relative min at $x = 0$;
points of inflection at $x = \frac{1}{2}, 1$;
no asymptotes



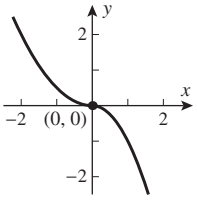
31. $\lim\limits_{x\to\pm\infty} f(x)$ does not exist; critical point at $x = 0$; relative min at $x = 0$; point of inflection when $1 + 4x^2\tan(x^2 + 1) = 0$; vertical asymptotes at $x = \pm\sqrt{\pi\left(n + \frac{1}{2}\right) - 1}$, $n = 0, 1, 2, \ldots$



33. critical points at $x = -5, 0$; relative max at $x = -5$, relative min at $x = 0$; points of inflection at $x \approx -7.26, -1.44, 1.20$; horizontal asymptote $y = 1$ for $x \to \pm\infty$



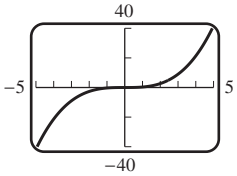
35. $\lim\limits_{x\to-\infty} f(x) = +\infty$, $\lim\limits_{x\to+\infty} f(x) = -\infty$;
critical point at $x = 0$;
no extrema;
inflection point at $x = 0$
(f changes concavity);
no asymptotes



37. no relative extrema **39.** relative min of 0 at $x = 0$

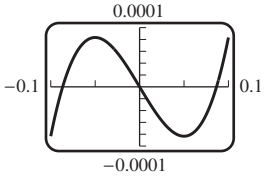
41. relative min of 0 at $x = 0$ **43.** relative min of 0 at $x = 0$

45. **(a)**

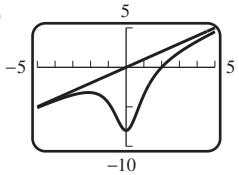


(b) relative max at $x = -\frac{1}{20}$;
relative min at $x = \frac{1}{20}$

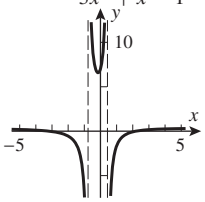
(c) The finer details can be seen when graphing over a much smaller x-window.



47. **(a)**



49. $f(x) = \dfrac{x^2 + x - 7}{3x^2 + x - 1}$, $x \ne \dfrac{1}{2}$

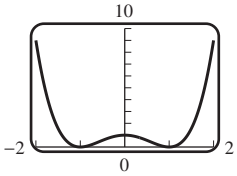


horizontal asymptote $y = 1/3$
vertical asymptotes at
$x = (-1 \pm \sqrt{13})/6$

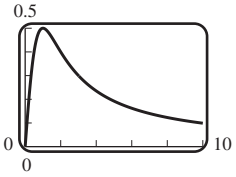
53. **(a)** true **(b)** false

55. **(a)** no max; min $= -13/4$ at $x = 3/2$ **(b)** no max or min
(c) no max; min $m = e^2/4$ at $x = 2$
(d) no max; min $m = e^{-1/e}$ at $x = 1/e$

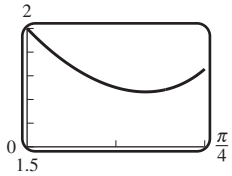
57. **(a)** minimum value 0 for $x = \pm 1$; no maximum



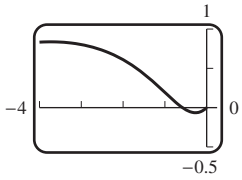
(b) max $= 1/2$ at $x = 1$; min $= 0$ at $x = 0$



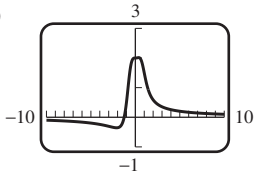
(c) maximum value 2 at $x = 0$; minimum value $\sqrt{3}$ at $x = \pi/6$



(d) maximum value $f(-2 - \sqrt{3}) \approx 0.84$; minimum value $f(-2 + \sqrt{3}) \approx -0.06$



59. **(a)**



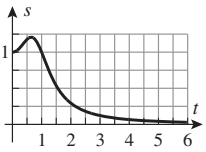
(b) minimum:
$(-2.111985, -0.355116)$;
maximum:
$(0.372591, 2.012931)$

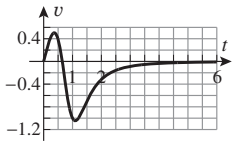
61. width $= 4\sqrt{2}$, height $= 3\sqrt{2}$ **63.** 2 in square

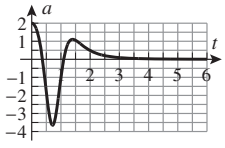
65. **(a)** yes **(b)** yes

67. **(a)** $v = -2\dfrac{t(t^4 + 2t^2 - 1)}{(t^4 + 1)^2}$, $a = 2\dfrac{3t^8 + 10t^6 - 12t^4 - 6t^2 + 1}{(t^4 + 1)^3}$

(b)





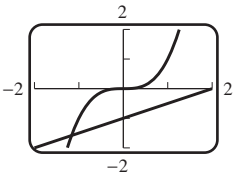


(c) $t \approx 0.64$, $s \approx 1.2$ **(d)** $0 \le t \le 0.64$

(e) speeding up when $0 \le t < 0.36$ and $0.64 < t < 1.1$, otherwise slowing down **(f)** maximum speed ≈ 1.05 when $t \approx 1.10$

69. $x \approx -2.11491, 0.25410, 1.86081$

71. $x \approx -1.165373043$



73. 249×10^6 km

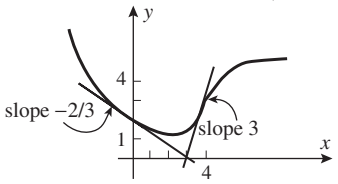
75. **(a)** yes, $c = 0$ **(b)** no **(c)** yes, $c = \sqrt{\pi/2}$

77. use Rolle's Theorem

Chapter 4 Making Connections (Page 314)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

1. **(a)** no zeros **(b)** one **(c)** $\lim_{x \to +\infty} g'(x) = 0$



2. **(a)** $(-2.2, 4), (2, 1.2), (4.2, 3)$
(b) critical numbers at $x = -5.1, -2, 0.2, 2$;
local min at $x = -5.1, 2$; local max at $x = -2$;
no extrema at $x = 0.2$; $f''(1) \approx -1.2$

3. $x = -4, 5$ **4.** **(d)** $f(c) = 0$

6. **(a)** route (i): 10 s; route (iv): 10 s
(b) $2 \le x \le 5$; $\dfrac{4\sqrt{10}}{2.1} + \dfrac{5}{0.7} \approx 13.166$ s
(c) $0 \le x \le 2$; 10 s
(d) route (i) or (iv); 10 s

Exercise Set 5.1 (Page 321)

1.

n	2	5	10	50	100
A_n	0.853553	0.749739	0.710509	0.676095	0.671463

3.

n	2	5	10	50	100
A_n	1.57080	1.93376	1.98352	1.99935	1.99984

5.

n	2	5	10	50	100
A_n	0.583333	0.645635	0.668771	0.688172	0.690653

7.

n	2	5	10	50	100
A_n	0.433013	0.659262	0.726130	0.774567	0.780106

9.

n	2	5	10	50	100
A_n	3.71828	2.85174	2.59327	2.39772	2.37398

11.

n	2	5	10	50	100
A_n	1.04720	0.75089	0.65781	0.58730	0.57894

13. $3(x-1)$ **15.** $x(x+2)$ **17.** $(x+3)(x-1)$

Responses to True–False questions may be abridged to save space.

19. False; the limit would be the area of the circle 4π.

21. True; this is the basis of the antiderivative method.

23. area $= A(6) - A(3)$ **27.** $f(x) = 2x$; $a = 2$

Exercise Set 5.2 (Page 330)

1. **(a)** $\displaystyle\int \frac{x}{\sqrt{1+x^2}}\,dx = \sqrt{1+x^2} + C$

(b) $\displaystyle\int (x+1)e^x\,dx = xe^x + C$

5. $\dfrac{d}{dx}\left[\sqrt{x^3+5}\right] = \dfrac{3x^2}{2\sqrt{x^3+5}}$, so $\displaystyle\int \frac{3x^2}{2\sqrt{x^3+5}}\,dx = \sqrt{x^3+5} + C$.

7. $\dfrac{d}{dx}[\sin(2\sqrt{x})] = \dfrac{\cos(2\sqrt{x})}{\sqrt{x}}$, so $\displaystyle\int \frac{\cos(2\sqrt{x})}{\sqrt{x}}\,dx = \sin(2\sqrt{x}) + C$.

9. **(a)** $(x^9/9) + C$ **(b)** $\frac{7}{12}x^{12/7} + C$ **(c)** $\frac{2}{9}x^{9/2} + C$

11. $\dfrac{5}{2}x^2 - \dfrac{1}{6x^4} + C$ **13.** $-\dfrac{1}{2}x^{-2} - \dfrac{12}{5}x^{5/4} + \dfrac{8}{3}x^3 + C$

15. $(x^2/2) + (x^5/5) + C$ **17.** $3x^{4/3} - \frac{12}{7}x^{7/3} + \frac{3}{10}x^{10/3} + C$

19. $\dfrac{x^2}{2} - \dfrac{2}{x} + \dfrac{1}{3x^3} + C$ **21.** $2\ln|x| + 3e^x + C$

23. $-3\cos x - 2\tan x + C$ **25.** $\tan x + \sec x + C$

27. $\tan\theta + C$ **29.** $\sec x + C$ **31.** $\theta - \cos\theta + C$

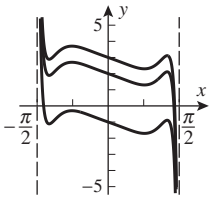
33. $\frac{1}{2}\sin^{-1} x - 3\tan^{-1} x + C$ **35.** $\tan x - \sec x + C$

Responses to True–False questions may be abridged to save space.

37. True; this is Equations (1) and (2).

39. False; the initial condition is not satisfied since $y(0) = 2$.

41.



43. **(a)** $y(x) = \frac{3}{4}x^{4/3} + \frac{5}{4}$
(b) $y = -\cos t + t + 1 - \pi/3$
(c) $y(x) = \frac{2}{3}x^{3/2} + 2x^{1/2} - \frac{8}{3}$

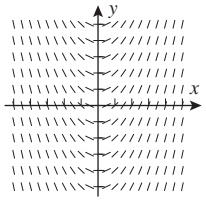
45. **(a)** $y = 4e^x - 3$ **(b)** $y = \ln|t| + 5$

47. $s(t) = 16t^2 + 20$ **49.** $s(t) = 2t^{3/2} - 15$

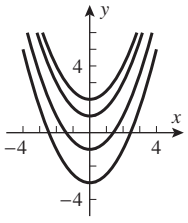
51. $f(x) = \frac{4}{15}x^{5/2} + C_1x + C_2$ **53.** $y = x^2 + x - 6$

55. $f(x) = \cos x + 1$ **57.** $y = x^3 - 6x + 7$

59. **(a)**

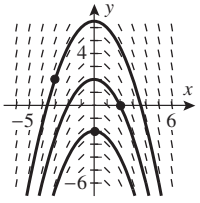


(b)

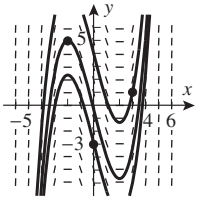


(c) $f(x) = \dfrac{x^2}{2} - 1$

61. (b)



63. (c)



67. **(b)** $\pi/2$ **69.** $\tan x - x + C$

71. **(a)** $\frac{1}{2}(x - \sin x) + C$ **(b)** $\frac{1}{2}(x + \sin x) + C$

73. $v = \dfrac{1087}{\sqrt{273}}T^{1/2}$ ft/s

Exercise Set 5.3 (Page 338)

1. **(a)** $\dfrac{(x^2+1)^{24}}{24} + C$ **(b)** $-\dfrac{\cos^4 x}{4} + C$

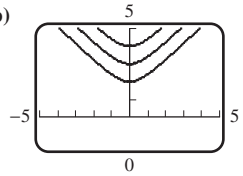
3. **(a)** $\frac{1}{4}\tan(4x+1) + C$ **(b)** $\frac{1}{6}(1+2y^2)^{3/2} + C$

5. **(a)** $-\frac{1}{2}\cot^2 x + C$ **(b)** $\frac{1}{10}(1+\sin t)^{10} + C$

7. **(a)** $\frac{2}{7}(1+x)^{7/2} - \frac{4}{5}(1+x)^{5/2} + \frac{2}{3}(1+x)^{3/2} + C$
(b) $-\cot(\sin x) + C$

9. **(a)** $\ln|\ln x| + C$ **(b)** $-\frac{1}{5}e^{-5x} + C$

11. **(a)** $\frac{1}{3}\tan^{-1}(x^3)+C$ **(b)** $\sin^{-1}(\ln x)+C$
15. $\frac{1}{40}(4x-3)^{10}+C$ **17.** $-\frac{1}{7}\cos 7x+C$ **19.** $\frac{1}{4}\sec 4x+C$
21. $\frac{1}{2}e^{2x}+C$ **23.** $\frac{1}{2}\sin^{-1}(2x)+C$ **25.** $\frac{1}{21}(7t^2+12)^{3/2}+C$
27. $\dfrac{3}{2(1-2x)^2}+C$ **29.** $-\dfrac{1}{40(5x^4+2)^2}+C$ **31.** $e^{\sin x}+C$
33. $-\frac{1}{6}e^{-2x^3}+C$ **35.** $\tan^{-1}e^x+C$ **37.** $\frac{1}{5}\cos(5/x)+C$
39. $-\frac{1}{15}\cos^5 3t+C$ **41.** $\frac{1}{2}\tan(x^2)+C$ **43.** $-\frac{1}{6}(2-\sin 4\theta)^{3/2}+C$
45. $\sin^{-1}(\tan x)+C$ **47.** $\frac{1}{6}\sec^3 2x+C$ **49.** $-e^{-x}+C$
51. $-e^{-2\sqrt{x}}+C$ **53.** $\frac{1}{6}(2y+1)^{3/2}-\frac{1}{2}(2y+1)^{1/2}+C$
55. $-\frac{1}{2}\cos 2\theta+\frac{1}{6}\cos^3 2\theta+C$ **57.** $t+\ln|t|+C$
59. $\int[\ln(e^x)+\ln(e^{-x})]\,dx=C$
61. **(a)** $\sin^{-1}\left(\frac{1}{3}x\right)+C$ **(b)** $\frac{1}{\sqrt{5}}\tan^{-1}\left(\frac{x}{\sqrt{5}}\right)+C$
(c) $\dfrac{1}{\sqrt{\pi}}\sec^{-1}\left(\dfrac{x}{\sqrt{\pi}}\right)+C$
63. $\dfrac{1}{b}\dfrac{(a+bx)^{n+1}}{n+1}+C$ **65.** $\dfrac{1}{b(n+1)}\sin^{n+1}(a+bx)+C$
67. **(a)** $\frac{1}{2}\sin^2 x+C_1;\ -\frac{1}{2}\cos^2 x+C_2$ **(b)** They differ by a constant.
69. $\frac{2}{15}(5x+1)^{3/2}-\frac{158}{15}$ **71.** $y=-\frac{1}{2}e^{2t}+\frac{13}{2}$
73. **(a)** $\sqrt{x^2+1}+C$ **75.** $f(x)=\frac{2}{9}(3x+1)^{3/2}+\frac{7}{9}$
77. 1280
(b)



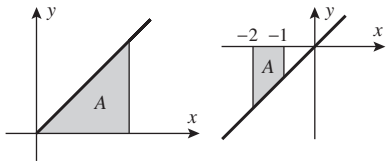
Exercise Set 5.4 (Page 350)

1. **(a)** 36 **(b)** 55 **(c)** 40 **(d)** 6 **(e)** 11 **(f)** 0 **3.** $\sum_{k=1}^{10} k$ **5.** $\sum_{k=1}^{10} 2k$
7. $\sum_{k=1}^{6}(-1)^{k+1}(2k-1)$ **9.** **(a)** $\sum_{k=1}^{50} 2k$ **(b)** $\sum_{k=1}^{50}(2k-1)$ **11.** 5050
13. 2870 **15.** 214,365 **17.** $\frac{3}{2}(n+1)$ **19.** $\frac{1}{4}(n-1)^2$
Responses to True–False questions may be abridged to save space.
21. True; by parts (a) and (c) of Theorem 5.4.2.
23. False; consider $[a,b]=[-1,0]$.
25. **(a)** $\left(2+\frac{3}{n}\right)^4\cdot\frac{3}{n},\ \left(2+\frac{6}{n}\right)^4\cdot\frac{3}{n},\ \left(2+\frac{9}{n}\right)^4\cdot\frac{3}{n},$
$\left(2+\frac{3(n-1)}{n}\right)^4\cdot\frac{3}{n},\ (2+3)^4\cdot\frac{3}{n}$ **(b)** $\sum_{k=0}^{n-1}\left(2+k\cdot\frac{3}{n}\right)^4\frac{3}{n}$
27. **(a)** 46 **(b)** 52 **(c)** 58 **29.** **(a)** $\frac{\pi}{4}$ **(b)** 0 **(c)** $-\frac{\pi}{4}$
31. **(a)** 0.7188, 0.7058, 0.6982 **(b)** 0.6928, 0.6931, 0.6931
(c) 0.6688, 0.6808, 0.6882
33. **(a)** 4.8841, 5.1156, 5.2488 **(b)** 5.3471, 5.3384, 5.3346
(c) 5.6841, 5.5156, 5.4088
35. $\frac{15}{4}$ **37.** 18 **39.** 320 **41.** $\frac{15}{4}$ **43.** 18 **45.** 16 **47.** $\frac{1}{3}$ **49.** 0
51. $\frac{2}{3}$ **53.** **(b)** $\frac{1}{4}(b^4-a^4)$
55. $\dfrac{n^2+2n}{4}$ if n is even; $\dfrac{(n+1)^2}{4}$ if n is odd **57.** $3^{17}-3^4$ **59.** $-\frac{399}{400}$
61. **(b)** $\frac{1}{2}$ **65.** **(a)** yes **(b)** yes

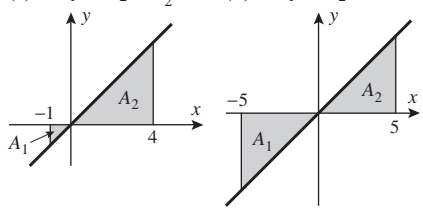
Exercise Set 5.5 (Page 360)

1. **(a)** $\frac{71}{6}$ **(b)** 2 **3.** **(a)** $-\frac{117}{16}$ **(b)** 3 **5.** $\int_{-1}^{2} x^2\,dx$
7. $\int_{-3}^{3} 4x(1-3x)\,dx$
9. **(a)** $\lim_{\max\Delta x_k\to 0}\sum_{k=1}^{n} 2x_k^*\,\Delta x_k;\ a=1,\ b=2$
(b) $\lim_{\max\Delta x_k\to 0}\sum_{k=1}^{n}\dfrac{x_k^*}{x_k^*+1}\Delta x_k;\ a=0,\ b=1$

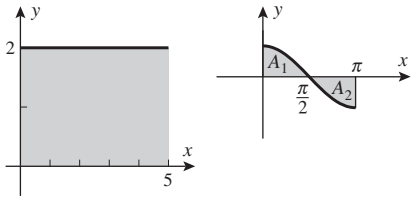
13. **(a)** $A=\frac{9}{2}$ **(b)** $-A=-\frac{3}{2}$



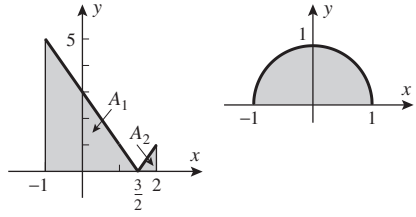
(c) $-A_1+A_2=\frac{15}{2}$ **(d)** $-A_1+A_2=0$



15. **(a)** $A=10$ **(b)** $A_1-A_2=0$ by symmetry



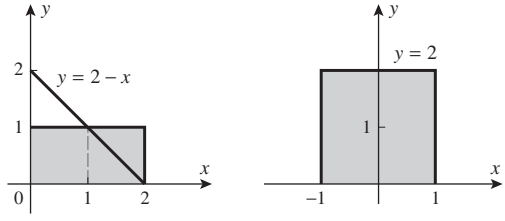
(c) $A_1+A_2=\frac{13}{2}$ **(d)** $\pi/2$



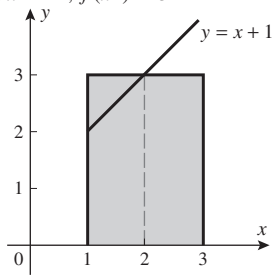
17. **(a)** 2
(b) 4
(c) 10
(d) 10
19. **(a)** 0.8
(b) −2.6
(c) −1.8
(d) −0.3
21. −1 **23.** 3 **25.** −4 **27.** $(1+\pi)/2$
Responses to True–False questions may be abridged to save space.
29. False; see Theorem 5.5.8(a).
31. False; consider $f(x)=x-2$ on $[0,3]$.
33. **(a)** negative **(b)** positive **37.** $\frac{25}{2}\pi$ **39.** $\frac{5}{2}$
45. **(a)** integrable **(b)** integrable **(c)** not integrable **(d)** integrable

Exercise Set 5.6 (Page 373)

1. **(a)** $\int_0^2(2-x)\,dx=2$ **(b)** $\int_{-1}^{1}2\,dx=4$ **(c)** $\int_1^3(x+1)\,dx=6$
3. **(a)** $x^*=f(x^*)=1$ **(b)** x^* is any point in $[-1,1]$, $f(x^*)=2$



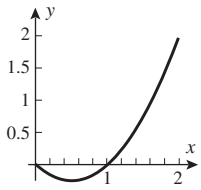
(c) $x^* = 2$, $f(x^*) = 3$

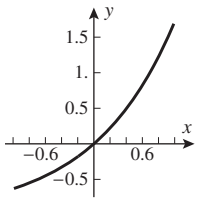


5. $\frac{65}{4}$ **7.** 14 **9.** $\frac{3}{2}$ **11.** **(a)** $\frac{4}{3}$ **(b)** -7 **13.** 48 **15.** 3 **17.** $\frac{844}{5}$
19. 0 **21.** $\sqrt{2}$ **23.** $5e^3 - 10$ **25.** $\pi/4$ **27.** $\pi/12$ **29.** -12
31. **(a)** $5/2$ **(b)** $2 - \frac{\sqrt{2}}{2}$ **33.** **(a)** $e + (1/e) - 2$ **(b)** 1
35. **(a)** $\frac{17}{6}$ **(b)** $F(x) = \begin{cases} \dfrac{x^2}{2}, & x \le 1 \\ \dfrac{x^3}{3} + \dfrac{1}{6}, & x > 1 \end{cases}$

Responses to True–False questions may be abridged to save space.

37. False; since $|x|$ is continuous, it has an antiderivative.
39. True; by the Fundamental Theorem of Calculus.
41. 0.6659; $\frac{2}{3}$ **43.** 3.1060; $2 \tan 1$ **45.** 12 **47.** $\frac{9}{2}$
49. area $= 1$ **51.** area $= e + e^{-1} - 2$



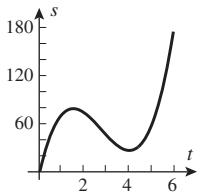


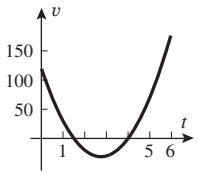
53. **(b)** degree mode, 0.93
55. **(a)** change in height from age 0 to age 10 years; inches
(b) change in radius from time $t = 1$ s to time $t = 2$ s; centimeters
(c) difference between speed of sound at 100°F and at 32°F; feet per second **(d)** change in position from time t_1 to time t_2; centimeters
57. **(a)** $3x^2 - 3$ **59.** **(a)** $\sin(x^2)$ **(b)** $e^{\sqrt{x}}$ **61.** $-x \sec x$
63. **(a)** 0 **(b)** 5 **(c)** $\frac{4}{5}$
65. **(a)** $x = 3$ **(b)** increasing on $[3, +\infty)$, decreasing on $(-\infty, 3]$
(c) concave up on $(-1, 7)$, concave down on $(-\infty, -1)$ and $(7, +\infty)$
67. **(a)** $(0, +\infty)$ **(b)** $x = 1$
69. **(a)** 120 gal **(b)** 420 gal **(c)** 2076.36 gal **71.** 1

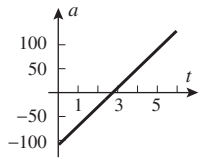
Exercise Set 5.7 (Page 382)

1. **(a)** displacement $= 3$; distance $= 3$
(b) displacement $= -3$; distance $= 3$
(c) displacement $= -\frac{1}{2}$; distance $= \frac{3}{2}$
(d) displacement $= \frac{3}{2}$; distance $= 2$
3. **(a)** 35.3 m/s **(b)** 51.4 m/s **5.** **(a)** $t^3 - t^2 + 1$ **(b)** $4t + 3 - \frac{1}{3}\sin 3t$
7. **(a)** $\frac{3}{2}t^2 + t - 4$ **(b)** $t + 1 - \ln t$
9. **(a)** displacement $= 1$ m; distance $= 1$ m
(b) displacement $= -1$ m; distance $= 3$ m
11. **(a)** displacement $= \frac{9}{4}$ m; distance $= \frac{11}{4}$ m
(b) displacement $= 2\sqrt{3} - 6$ m; distance $= 6 - 2\sqrt{3}$ m
13. 4, 13/3 **15.** 296/27, 296/27
17. **(a)** $s = 2/\pi$, $v = 1$, $|v| = 1$, $a = 0$
(b) $s = \frac{1}{2}$, $v = -\frac{3}{2}$, $|v| = \frac{3}{2}$, $a = -3$ **19.** $t \approx 1.27$ s

21.

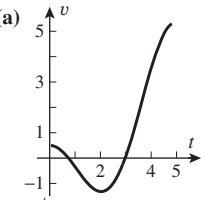






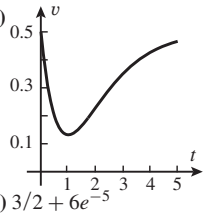
Responses to True–False questions may be abridged to save space.

23. True; if $a(t) = a_0$, then $v(t) = a_0 t + v_0$.
25. False; consider $v(t) = \sin t$ on $[0, 2\pi]$.
27. **(a)**



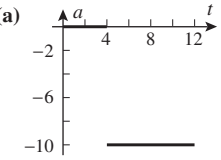
(b) $5/2 - \sin(5) + 5\cos(5)$

29. **(a)**

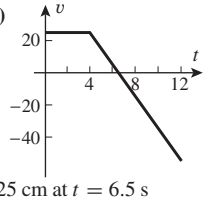


(b) $3/2 + 6e^{-5}$

31. **(a)**



(b)

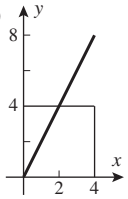


(c) 120 cm, -20 cm **(d)** 131.25 cm at $t = 6.5$ s

33. **(a)** $-\frac{121}{5}$ ft/s^2 **(b)** $\frac{70}{33}$ s **(c)** $\frac{60}{11}$ s **35.** 280 m **37.** 50 s, 5000 ft
39. **(a)** 16 ft/s, -48 ft/s **(b)** 196 ft **(c)** 112 ft/s
41. **(a)** 1 s **(b)** $\frac{1}{2}$ s **43.** **(a)** 6.122 s **(b)** 183.7 m **(c)** 6.122 s **(d)** 60 m/s
45. **(a)** 5 s **(b)** 272.5 m **(c)** 10 s **(d)** -49 m/s
(e) 12.46 s **(f)** 73.1 m/s

Exercise Set 5.8 (Page 388)

1. **(a)** 4 **(b)** 2 **(c)**



3. 6
5. $2/\pi$
7. $\dfrac{1}{e - 1}$
9. $\dfrac{\pi}{12(\sqrt{3} - 1)}$
11. $\dfrac{1 - e^{-8}}{8}$ **13.** **(a)** 5.28 **(b)** 4.305 **(c)** 4 **15.** **(a)** $-\frac{1}{6}$ **(b)** $\frac{1}{2}$

Responses to True–False questions may be abridged to save space.

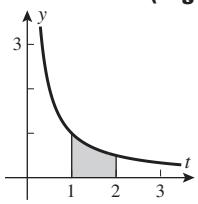
19. False; let $g(x) = \cos x$; $f(x) = 0$ on $[0, 3\pi/2]$.
21. True; see Theorem 5.5.4(*b*).
23. **(a)** $\frac{263}{4}$ **(b)** 31 **25.** 1404π lb **27.** 97 cars/min **31.** 27

Exercise Set 5.9 (Page 393)

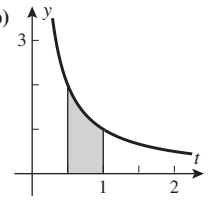
1. **(a)** $\frac{1}{2}\int_1^5 u^3\,du$ **(b)** $\frac{3}{2}\int_9^{25} \sqrt{u}\,du$ **(c)** $\dfrac{1}{\pi}\displaystyle\int_{-\pi/2}^{\pi/2} \cos u\,du$
(d) $\int_1^2 (u+1)u^5\,du$ **3.** **(a)** $\frac{1}{2}\int_{-1}^1 e^u\,du$ **(b)** $\int_1^2 u\,du$
5. 10 **7.** 0 **9.** $\frac{1192}{15}$ **11.** $8 - (4\sqrt{2})$ **13.** $-\frac{1}{48}$ **15.** $\ln\frac{21}{13}$
17. $\pi/6$ **19.** $25\pi/6$ **21.** $\pi/8$ **23.** $2/\pi$ m **25.** 6 **27.** $\pi/18$
29. 1/21 **31.** 2 **33.** $\frac{2}{3}(\sqrt{10} - 2\sqrt{2})$ **35.** $2(\sqrt{7} - \sqrt{3})$ **37.** 1
39. 0 **41.** $(\sqrt{3} - 1)/3$ **43.** $\frac{106}{405}$ **45.** $(\ln 3)/2$ **47.** $\pi/(6\sqrt{3})$
49. $\pi/9$ **51.** **(a)** $\frac{23}{4480}$ **53.** **(a)** $\frac{5}{3}$ **(b)** $\frac{5}{3}$ **(c)** $-\frac{1}{2}$ **57.** Method 2
59. Method 1 **61.** $\approx 48{,}233{,}500{,}000$ **63.** **(a)** 0.45 **(b)** 0.461
65. $(\ln 7)/2$ **67.** **(a)** $2/\pi$ **71.** **(b)** $\frac{3}{2}$ **(c)** $\pi/4$

Exercise Set 5.10 (Page 406)

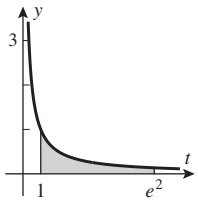
1. (a)



(b)



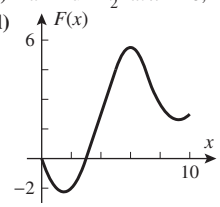
(c)



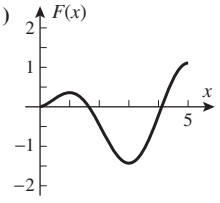
3. (a) 7 (b) -5 (c) -3 (d) 6
5. 1.603210678; magnitude of error is < 0.0063
7. (a) $x^{-1}, x > 0$ (b) $x^2, x \neq 0$ (c) $-x^2, -\infty < x < +\infty$ (d) $-x, -\infty < x < +\infty$ (e) $x^3, x > 0$ (f) $\ln x + x, x > 0$ (g) $x - \sqrt[3]{x}, -\infty < x < +\infty$ (h) $e^x/x, x > 0$
9. (a) $e^{\pi \ln 3}$ (b) $e^{\sqrt{2}\ln 2}$ 11. (a) $\sqrt{e}$ (b) e^2 13. $x^2 - x$
15. (a) $3/x$ (b) 1 17. (a) 0 (b) 0 (c) 1

Responses to True–False questions may be abridged to save space.

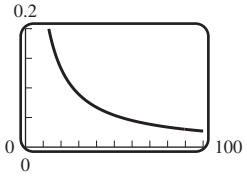
19. True; both equal $-\ln a$.
21. False; the integrand is unbounded on $[-1, e]$ and thus the integrand is undefined.
23. (a) $2x^3\sqrt{1+x^2}$ (b) $-\frac{2}{3}(x^2+1)^{3/2} + \frac{2}{5}(x^2+1)^{5/2} - \frac{4\sqrt{2}}{15}$
25. (a) $-\cos(x^3)$ (b) $-\tan^2 x$ 27. $-3\dfrac{3x-1}{9x^2+1} + 2x\dfrac{x^2-1}{x^4+1}$
29. (a) $3x^2\sin^2(x^3) - 2x\sin^2(x^2)$ (b) $\dfrac{2}{1-x^2}$
31. (a) $F(0) = 0, F(3) = 0, F(5) = 6, F(7) = 6, F(10) = 3$
(b) increasing on $\left[\frac{3}{2}, 6\right]$ and $\left[\frac{37}{4}, 10\right]$, decreasing on $\left[0, \frac{3}{2}\right]$ and $\left[6, \frac{37}{4}\right]$
(c) maximum $\frac{15}{2}$ at $x = 6$, minimum $-\frac{9}{4}$ at $x = \frac{3}{2}$
(d)



33. $F(x) = \begin{cases} (1-x^2)/2, & x < 0 \\ (1+x^2)/2, & x \geq 0 \end{cases}$ 35. $y(x) = x^2 + \ln x + 1$
37. $y(x) = \tan x + \cos x - (\sqrt{2}/2)$
39. $P(x) = P_0 + \int_0^x r(t)\,dt$ individuals 41. I is the derivative of II.
43. (a) $t = 3$ (b) $t = 1, 5$
(c) $t = 5$ (d) $t = 3$
(e) F is concave up on $\left(0, \frac{1}{2}\right)$ and $(2, 4)$, concave down on $\left(\frac{1}{2}, 2\right)$ and $(4, 5)$.
(f)

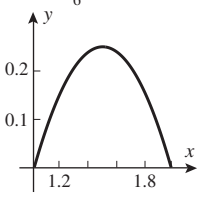


45. (a) relative maxima at $x = \pm\sqrt{4k+1}, k = 0, 1, \ldots$; relative minima at $x = \pm\sqrt{4k-1}, k = 1, 2, \ldots$
(b) $x = \pm\sqrt{2k}, k = 1, 2, \ldots$, and at $x = 0$
47. $f(x) = 2e^{2x}, a = \ln 2$ 49. 0.06



Chapter 5 Review Exercises (Page 408)

1. $-\dfrac{1}{4x^2} + \dfrac{8}{3}x^{3/2} + C$ 3. $-4\cos x + 2\sin x + C$
5. $3x^{1/3} - 5e^x + C$ 7. $\tan^{-1} x + 2\sin^{-1} x + C$
9. (a) $y(x) = 2\sqrt{x} - \frac{2}{3}x^{3/2} - \frac{4}{3}$ (b) $y(x) = \sin x - 5e^x + 5$
(c) $y(x) = \frac{5}{4} + \frac{3}{4}x^{4/3}$ (d) $y(x) = \frac{1}{2}e^{x^2} - \frac{1}{2}$
13. $\frac{1}{2}\sec^{-1}(x^2 - 1) + C$ 15. $\frac{1}{3}\sqrt{5 + 2\sin 3x} + C$
17. $-\dfrac{1}{3a}\dfrac{1}{ax^3+b} + C$
19. (a) $\displaystyle\sum_{k=0}^{14}(k+4)(k+1)$ (b) $\displaystyle\sum_{k=5}^{19}(k-1)(k-4)$
21. $\frac{32}{3}$ 23. 0.35122, 0.42054, 0.38650
27. (a) $\frac{3}{4}$ (b) $-\frac{3}{2}$ (c) $-\frac{35}{4}$ (d) -2 (e) not enough information
(f) not enough information
29. (a) $2 + (\pi/2)$ (b) $\dfrac{1}{3}(10^{3/2} - 1) - \dfrac{9\pi}{4}$ (c) $\pi/8$
31. 48 33. $\frac{2}{3}$ 35. $\frac{3}{2} - \sec 1$ 37. $\frac{5}{2}$ 39. $\frac{52}{3}$ 41. $e^3 - e$
43. area $= \frac{1}{6}$



45. $\frac{22}{3}$
47. (a) $x^3 + 1$
49. e^{x^2}
51. $|x - 1|$
53. $\dfrac{\cos x}{1 + \sin^3 x}$
57. (b) $\dfrac{\pi}{2}$; $\tan^{-1} x + \tan^{-1}\left(\dfrac{1}{x}\right) = \dfrac{\pi}{2}$
59. (a) $F(x)$ is 0 if $x = 1$, positive if $x > 1$, and negative if $x < 1$.
(b) $F(x)$ is 0 if $x = -1$, positive if $-1 < x \leq 2$, and negative if $-2 \leq x < -1$.
61. (a) $\frac{4}{3}$ (b) $e - 1$ 63. $\frac{3}{10}$ 67. $\frac{1}{4}t^4 - \frac{2}{3}t^3 + t + 1$
69. $t^2 - 3t + 7$ 71. 12 m, 20 m 73. $\frac{1}{3}$ m, $\frac{10}{3} - 2\sqrt{2}$ m
75. displacement $= -6$ m; distance $= \frac{13}{2}$ m
77. (a) 2.2 s (b) 387.2 ft 79. $v_0/2$ ft/s 81. $\frac{121}{5}$ 83. $\frac{2}{3}$ 85. 0
87. $2 - 2/\sqrt{e}$ 89. (a) e^2 (b) $e^{1/3}$

Chapter 5 Making Connections (Page 412)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

1. (b) $b^2 - a^2$ 2. 16/3 3. 12
4. (a) the sum for f is m times that for g
(b) $m\displaystyle\int_0^1 g(x)\,dx = \int_0^m f(x)\,du$
5. (a) they are equal (b) $\displaystyle\int_2^3 g(x)\,dx = \int_4^9 f(u)\,du$

Exercise Set 6.1 (Page 419)

1. 9/2 3. 1 5. (a) 4/3 (b) 4/3 7. 49/192 9. 1/2 11. $\sqrt{2}$
13. $\frac{1}{2}$ 15. $\pi - 1$ 17. 24 19. 37/12 21. $4\sqrt{2}$ 23. $\frac{1}{2}$
25. $\ln 2 - \frac{1}{2}$

Responses to True–False questions may be abridged to save space.

27. True; use area Formula (1) with $f(x) = g(x) + c$.
29. True; the integrand must assume both positive and negative values. By the Intermediate-Value Theorem, the integrand must be equal to 0 somewhere in $[a, b]$.
31. $k \approx 0.9973$ 33. 9152/105 35. $9/\sqrt[3]{4}$
37. (a) 4/3 (b) $m = 2 - \sqrt[3]{4}$ 39. 1.180898334
41. 0.4814, 2.3639, 1.1897 43. 2.54270

45. racer 1's lead over racer 2 at time $t = 0$
47. the increase in population from 1960 to 2010 **49.** $a^2/6$

Exercise Set 6.2 (Page 428)

1. 8π **3.** $13\pi/6$ **5.** $(1-\sqrt{2}/2)\pi$ **7.** 8π **9.** $32/5$ **11.** $256\pi/3$
13. $2048\pi/15$ **15.** 4π **17.** $\pi^2/4$ **19.** $3/5$ **21.** 2π
23. $72\pi/5$ **25.** $\frac{\pi}{2}(e^2-1)$

Responses to True–False questions may be abridged to save space.

27. False; see the solids associated with Exercises 9 and 10.
29. False; see Example 2 where the cross-sectional area is a linear function of x.
31. $4\pi ab^2/3$ **33.** π **35.** $\int_a^b \pi[f(x)-k]^2\,dx$ **37.** **(b)** $40\pi/3$
39. $648\pi/5$ **41.** $\pi/2$ **43.** $\pi/15$ **45.** $40{,}000\pi\ \text{ft}^3$ **47.** $1/30$
49. **(a)** $2\pi/3$ **(b)** $16/3$ **(c)** $4\sqrt{3}/3$ **51.** 0.710172176 **53.** π
57. **(b)** left ≈ 11.157; right ≈ 11.771; $V \approx$ average $= 11.464\ \text{cm}^3$
59. $V = \begin{cases} 3\pi h^2, & 0 \le h < 2 \\ \frac{1}{3}\pi(12h^2 - h^3 - 4), & 2 \le h \le 4 \end{cases}$ **61.** $\frac{2}{3}r^3\tan\theta$ **63.** $16r^3/3$

Exercise Set 6.3 (Page 436)

1. $15\pi/2$ **3.** $\pi/3$ **5.** $2\pi/5$ **7.** 4π **9.** $20\pi/3$ **11.** $\pi\ln 2$ **13.** $\pi/2$
15. $\pi/5$

Responses to True–False questions may be abridged to save space.

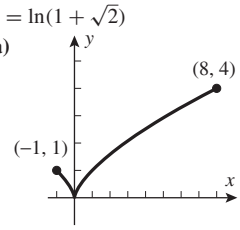
17. True; this is a restatement of Formula (1).
19. True; see Formula (2).
21. $2\pi e^2$ **23.** 1.73680 **25.** **(a)** $7\pi/30$ **(b)** easier
27. **(a)** $\int_0^1 2\pi(1-x)x\,dx$ **(b)** $\int_0^1 2\pi(1+y)(1-y)\,dy$
29. $7\pi/4$ **31.** $\pi r^2 h/3$ **33.** $V = \frac{4}{3}\pi(L/2)^3$ **35.** $b = 1$

Exercise Set 6.4 (Page 441)

1. $L = \sqrt{5}$ **3.** $(85\sqrt{85} - 8)/243$ **5.** $\frac{1}{27}(80\sqrt{10} - 13\sqrt{13})$ **7.** $\frac{17}{6}$

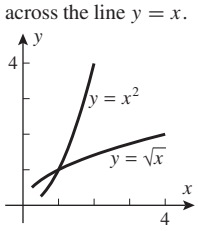
Responses to True–False questions may be abridged to save space.

9. False; f' is undefined at the endpoints ± 1.
11. True; if $f(x) = mx + c$ over $[a, b]$, then $L = \sqrt{1+m^2}(b-a)$, which is equal to the given sum.
13. $L = \ln(1+\sqrt{2})$
15. **(a)**



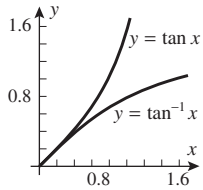
(b) dy/dx does not exist at $x = 0$.
(c) $L = (13\sqrt{13} + 80\sqrt{10} - 16)/27$

17. **(a)** They are mirror images across the line $y = x$.



(b) $\int_{1/2}^{2} \sqrt{1+4x^2}\,dx$, $\int_{1/4}^{4} \sqrt{1+\frac{1}{4x}}\,dx$, $x = \sqrt{u}$ transforms the first integral into the second.
(c) $\int_{1/4}^{4} \sqrt{1+\frac{1}{4y}}\,dy$, $\int_{1/2}^{2} \sqrt{1+4y^2}\,dy$
(d) 4.0724, 4.0716
(e) The first: Both are underestimates of the arc length, so the larger one is more accurate.
(f) 4.0724, 4.0662 **(g)** 4.0729

19. **(a)** They are mirror images across the line $y = x$.



(b) $\int_0^{\pi/3} \sqrt{1+\sec^4 x}\,dx$, $\int_0^{\sqrt{3}} \sqrt{1+\frac{1}{(1+x^2)^2}}\,dx$, $x = \tan^{-1} u$ transforms the first integral into the second.
(c) $\int_0^{\sqrt{3}} \sqrt{1+\frac{1}{(1+y^2)^2}}\,dy$, $\int_0^{\pi/3} \sqrt{1+\sec^4 y}\,dy$
(d) 2.0566, 2.0567
(e) The second: Both are underestimates of the arc length, so the larger one is more accurate. **(f)** 2.0509, 2.0571 **(g)** 2.0570
23. 4354 ft **25.** 196.31 yards **27.** $(2\sqrt{2}-1)/3$
29. π **31.** $L = \sqrt{2}(e^{\pi/2} - 1)$ **33.** **(b)** 9.69 **(c)** 5.16 cm

Exercise Set 6.5 (Page 447)

1. $35\pi\sqrt{2}$ **3.** 8π **5.** $40\pi\sqrt{82}$ **7.** 24π **9.** $16\pi/9$
11. $16,911\pi/1024$ **13.** $S \approx 14.42$ **15.** $S \approx 22.94$

Responses to True–False questions may be abridged to save space.

17. True; use Formula (1) with $r_1 = 0$, $r_2 = r$, $l = \sqrt{r^2+h^2}$.
19. True; the sum telescopes to the surface area of a cylinder.
21. 14.39 **23.** $S = \int_a^b 2\pi[f(x)+k]\sqrt{1+[f'(x)]^2}\,dx$
33. $\frac{8}{3}\pi(17\sqrt{17} - 1)$ **35.** $\frac{\pi}{24}(17\sqrt{17} - 1)$

Exercise Set 6.6 (Page 456)

1. 7.5 ft·lb **3.** $d = 7/4$ **5.** 100 ft·lb **7.** 160 J **9.** 20 lb/ft

Responses to True–False questions may be abridged to save space.

11. False; the work done is the same.
13. True; joules **15.** $47{,}385\pi$ ft·lb **17.** 261,600 J
19. **(a)** 926,640 ft·lb **(b)** hp of motor $= 0.468$ **21.** 75,000 ft·lb
23. 120,000 ft·tons **25.** **(a)** $2{,}400{,}000{,}000/x^2$ lb
(b) $(9.6 \times 10^{10})/(x+4000)^2$ lb **(c)** 2.5344×10^{10} ft·lb
27. $v_f = 100$ m/s
29. **(a)** decrease of 4.5×10^{14} J **(b)** ≈ 0.107 **(c)** ≈ 8.24 bombs

Exercise Set 6.7 (Page 465)

1. **(a)** positive: m_2 is at the fulcrum, so it can be ignored; masses m_1 and m_3 are equidistant from position 5, but $m_1 < m_3$, so the beam will rotate clockwise. **(b)** The fulcrum should be placed $\frac{50}{7}$ units to the right of m_1.
3. $(\frac{1}{2}, \frac{1}{2})$ **5.** $(1, \frac{1}{2})$ **7.** $(\frac{2}{3}, \frac{1}{3})$ **9.** $(\frac{5}{14}, \frac{38}{35})$ **11.** $(\frac{2}{3}, \frac{1}{3})$
13. $(-\frac{1}{2}, 4)$ **15.** $(\frac{1}{2}, \frac{8}{5})$ **17.** $(\frac{9}{20}, \frac{9}{20})$ **19.** $(\frac{49}{48}, \frac{7}{3} - \ln 2)$
23. $\frac{4}{3}$; $(\frac{3}{5}, \frac{3}{8})$ **25.** 3; $(0, \frac{2}{3})$ **27.** 8; $(\frac{\pi}{2}, \frac{\pi}{8})$
29. $\ln 4 - 1$; $\left(\frac{4\ln 4 - 3}{4\ln 4 - 4}, \frac{(\ln 2)^2 + 1 - \ln 4}{\ln 4 - 1}\right)$

Responses to True–False questions may be abridged to save space.

31. True; use symmetry. **33.** True; use symmetry.
35. $\left(\frac{2a}{3}, 0\right)$ **37.** $(\bar{x}, \bar{y}) = \left(0, \frac{(a+2b)c}{3(a+b)}\right)$
41. $2\pi^2 abk$ **43.** $(a/3, b/3)$

Exercise Set 6.8 (Page 472)

1. **(a)** $F = 31{,}200$ lb; $P = 312\ \text{lb/ft}^2$
(b) $F = 2{,}452{,}500$ N; $P = 98.1$ kPa
3. 499.2 lb **5.** 8.175×10^5 N **7.** 1,098,720 N **9.** yes
11. $\rho a^3/\sqrt{2}$ lb

Responses to True–False questions may be abridged to save space.

13. True; this is a consequence of inequalities (4).

15. False; by Equation (7) the force can be arbitrarily large for a fixed volume of water.
17. 61,748 lb **19.** 9.81×10^9 N **21.** **(b)** $80\rho_0$ lb/min

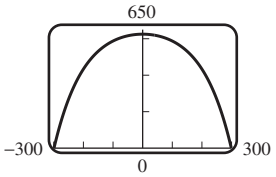
Exercise Set 6.9 (Page 482)

1. **(a)** ≈ 10.0179 **(b)** ≈ 3.7622 **(c)** $15/17 \approx 0.8824$ **(d)** ≈ -1.4436 **(e)** ≈ 1.7627 **(f)** ≈ 0.9730
3. **(a)** $\frac{4}{3}$ **(b)** $\frac{5}{4}$ **(c)** $\frac{312}{313}$ **(d)** $-\frac{63}{16}$
5.

	$\sinh x_0$	$\cosh x_0$	$\tanh x_0$	$\coth x_0$	$\operatorname{sech} x_0$	$\operatorname{csch} x_0$
(a)	2	$\sqrt{5}$	$2/\sqrt{5}$	$\sqrt{5}/2$	$1/\sqrt{5}$	1/2
(b)	3/4	5/4	3/5	5/3	4/5	4/3
(c)	4/3	5/3	4/5	5/4	3/5	3/4

9. $4\cosh(4x-8)$ **11.** $-\frac{1}{x}\operatorname{csch}^2(\ln x)$
13. $\frac{1}{x^2}\operatorname{csch}\left(\frac{1}{x}\right)\coth\left(\frac{1}{x}\right)$ **15.** $\frac{2+5\cosh(5x)\sinh(5x)}{\sqrt{4x+\cosh^2(5x)}}$
17. $x^{5/2}\tanh(\sqrt{x})\operatorname{sech}^2(\sqrt{x})+3x^2\tanh^2(\sqrt{x})$
19. $\frac{1}{\sqrt{9+x^2}}$ **21.** $\frac{1}{(\cosh^{-1}x)\sqrt{x^2-1}}$ **23.** $-\frac{(\tanh^{-1}x)^{-2}}{1-x^2}$
25. $\frac{\sinh x}{|\sinh x|}=\begin{cases}1, & x>0\\ -1, & x<0\end{cases}$ **27.** $-\frac{e^x}{2x\sqrt{1-x}}+e^x\operatorname{sech}^{-1}x$
29. $\frac{1}{7}\sinh^7 x+C$ **31.** $\frac{2}{3}(\tanh x)^{3/2}+C$ **33.** $\ln(\cosh x)+C$
35. $37/375$ **37.** $\frac{1}{3}\sinh^{-1}3x+C$ **39.** $-\operatorname{sech}^{-1}(e^x)+C$
41. $-\operatorname{csch}^{-1}|2x|+C$ **43.** $\frac{1}{2}\ln 3$

Responses to True–False questions may be abridged to save space.
45. True; see Figure 6.9.1 **47.** True; $f(x)=\sinh x$
49. $16/9$ **51.** 5π **53.** $\frac{3}{4}$
55. **(a)** $+\infty$ **(b)** $-\infty$ **(c)** 1 **(d)** -1 **(e)** $+\infty$ **(f)** $+\infty$
63. $|u|<1$: $\tanh^{-1}u+C$; $|u|>1$: $\tanh^{-1}(1/u)+C$
65. **(a)** $\ln 2$ **(b)** $1/2$ **71.** 405.9 ft
73. **(a)**



(b) 1480.2798 ft
(c) ± 283.6249 ft
(d) $82°$

75. **(b)** 14.44 m **(c)** $15\ln 3 \approx 16.48$ m

Chapter 6 Review Exercises (Page 485)

7. **(a)** $\int_a^b (f(x)-g(x))\,dx+\int_b^c (g(x)-f(x))\,dx+\int_c^d (f(x)-g(x))\,dx$ **(b)** $11/4$
9. $4352\pi/105$ **11.** $3/2+\ln 4$ **13.** 9 **15.** $\frac{\pi}{6}\left(65^{3/2}-37^{3/2}\right)$
17. $3\sqrt{3}$ **19.** **(a)** $W=\frac{1}{16}$ J **(b)** 5 m **21.** $\left(\frac{8}{5},0\right)$
23. **(a)** $F=\int_0^1 \rho x 3\,dx$ N **(b)** $F=\int_1^4 \rho(1+x)2x\,dx$ lb/ft^2
(c) $F=\int_{-10}^0 9810|y|2\sqrt{\frac{125}{8}(y+10)}\,dy$ N

Chapter 6 Making Connections (Page 487)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

1. **(a)** πA_1 **(b)** $a=\frac{A_1}{2A_2}$ **2.** 1,010,807 ft·lb **3.** $\int_0^a 2\pi r f(r)\,dr$

Exercise Set 7.1 (Page 490)

1. $-2(x-3)^4+C$ **3.** $\frac{1}{2}\tan(x^2)+C$ **5.** $-\frac{1}{3}\ln(2+\cos 3x)+C$
7. $\cosh(e^x)+C$ **9.** $e^{\tan x}+C$ **11.** $-\frac{1}{42}\cos^6 7x+C$
13. $\ln(e^x+\sqrt{e^{2x}+4})+C$ **15.** $2e^{\sqrt{x-1}}+C$ **17.** $2\sinh\sqrt{x}+C$
19. $-\frac{2}{\ln 3}3^{-\sqrt{x}}+C$ **21.** $\frac{1}{2}\coth\frac{2}{x}+C$ **23.** $-\frac{1}{4}\ln\left|\frac{2+e^{-x}}{2-e^{-x}}\right|+C$
25. $\sin^{-1}(e^x)+C$ **27.** $-\frac{1}{2}\cos(x^2)+C$ **29.** $-\frac{1}{\ln 25}5^{-x^2}+C$
31. **(b)** $\ln\left|\tan\frac{x}{2}\right|+C$ **(c)** $\ln\left|\cot\left(\frac{\pi}{4}-\frac{x}{2}\right)\right|+C$
33. **(a)** $\frac{1}{2}\sin^2 x+C$ **(b)** $-\frac{1}{4}\cos 2x+C$

Exercise Set 7.2 (Page 498)

1. $-e^{-2x}\left(\frac{x}{2}+\frac{1}{4}\right)+C$ **3.** $x^2e^x-2xe^x+2e^x+C$
5. $x^2\sin x+2x\cos x-2\sin x+C$ **7.** $-\frac{1}{3}x\cos 3x+\frac{1}{9}\sin 3x+C$
9. $\frac{x^2}{2}\ln x-\frac{x^2}{4}+C$ **11.** $x(\ln x)^2-2x\ln x+2x+C$
13. $x\ln(3x-2)-x-\frac{2}{3}\ln(3x-2)+C$
15. $x\sin^{-1}2x+\frac{1}{2}\sqrt{1-4x^2}+C$
17. $x\tan^{-1}(3x)-\frac{1}{6}\ln(1+9x^2)+C$
19. $(x/2)[\sin(\ln x)-\cos(\ln x)]+C$ **21.** $\frac{1}{2}e^x(\sin x-\cos x)+C$
23. $x\tan x+\ln|\cos x|+C$ **25.** $\frac{1}{2}x^2e^{x^2}-\frac{1}{2}e^{x^2}+C$
27. $\frac{1}{9}(5e^6+1)$ **29.** $(2e^3+1)/9$ **31.** $3\ln 3-2$ **33.** $\frac{5\pi}{6}-\sqrt{3}+1$
35. $-\pi/2$ **37.** $\frac{1}{3}\left(2\sqrt{3}\pi-\frac{\pi}{2}-2+\ln 2\right)$

Responses to True–False questions may be abridged to save space.
39. True; see the subsection "Guidelines for Integration by Parts."
41. False; $\ln x$ isn't a factor of the integrand.
43. $-2\sqrt{x}\cos\sqrt{x}+2\sin\sqrt{x}+C$
47. $(4x^3-6x)\sin 2x-(2x^4-6x^2+3)\cos 2x+C$
49. $-(3x^2+5x+7)e^{-x}+C$
51. $\frac{e^{ax}}{a^2+b^2}(a\sin bx-b\cos bx)+C$ **53.** **(a)** $\frac{1}{2}\sin^2 x+C$
55. **(a)** $A=1$ **(b)** $V=\pi(e-2)$ **57.** $V=2\pi^2$ **59.** π^2-4
61. **(a)** $-\frac{1}{4}\sin^3 x\cos x-\frac{3}{8}\sin x\cos x+\frac{3}{8}x+C$ **(b)** $8/15$
65. **(a)** $\frac{1}{2}\tan^2 x+\ln|\cos(x)|+C$ **(b)** $\frac{1}{3}\sec^2 x\tan x+\frac{2}{3}\tan x+C$
(c) $x^3e^x-3x^2e^x+6xe^x-6e^x+C$
69. $(x+1)\ln(x+1)-x+C$ **71.** $\frac{1}{2}(x^2+1)\tan^{-1}x-\frac{1}{2}x+C$

Exercise Set 7.3 (Page 506)

1. $-\frac{1}{3}\cos^3 x+C$ **3.** $\frac{\theta}{2}-\frac{1}{16}\sin 8\theta+C$
5. $\frac{1}{3a}\cos^3 a\theta-\cos a\theta+C$ **7.** $\frac{1}{2b}\sin^2 bx+C$
9. $\frac{1}{3}\sin^3 t-\frac{1}{5}\sin^5 t+C$ **11.** $\frac{1}{8}x-\frac{1}{32}\sin 4x+C$
13. $-\frac{1}{14}\cos 7x+\frac{1}{2}\cos x+C$ **15.** $-\frac{1}{3}\cos(3x/2)-\cos(x/2)+C$
17. $2/3$ **19.** 0 **21.** $3/8$ **23.** $\frac{1}{2}\tan(2x-1)+C$
25. $\ln|\cos(e^{-x})|+C$ **27.** $\frac{1}{7}\ln|\sec 7x+\tan 7x|+C$
29. $\frac{1}{3}\tan^3 x+C$ **31.** $\frac{1}{16}\sec^4 4x+C$ **33.** $\frac{1}{5}\sec^5 x-\frac{1}{3}\sec^3 x+C$
35. $\frac{1}{4}\sec^3 x\tan x-\frac{5}{8}\sec x\tan x+\frac{3}{8}\ln|\sec x+\tan x|+C$
37. $\frac{1}{3}\sec^3 t+C$ **39.** $\tan x+\frac{1}{3}\tan^3 x+C$
41. $\frac{1}{8}\tan^2 4x+\frac{1}{4}\ln|\cos 4x|+C$ **43.** $\frac{2}{3}\tan^{3/2}x+\frac{2}{7}\tan^{7/2}x+C$
45. $\frac{1}{2}-\frac{\pi}{8}$ **47.** $-\frac{1}{2}+\ln 2$ **49.** $-\frac{1}{5}\csc^5 x+\frac{1}{3}\csc^3 x+C$
51. $-\frac{1}{2}\csc^2 x-\ln|\sin x|+C$

Responses to True–False questions may be abridged to save space.
53. False; use this identity to help evaluate integrals of the form $\int \sin mx\cos nx\,dx$.
55. True; $\int \sin^5 x\cos^8 x\,dx=\int \sin x(1-\cos^2 x)^2\cos^8 x\,dx=-\int(1-u^2)^2u^8\,du=-\int(u^8-2u^{10}+u^{12})\,du$
59. $L=\ln(\sqrt{2}+1)$ **61.** $V=\pi/2$

67. $-\dfrac{1}{\sqrt{a^2+b^2}}\ln\left[\dfrac{\sqrt{a^2+b^2}+a\cos x-b\sin x}{a\sin x+b\cos x}\right]+C$

69. **(a)** $\frac{2}{3}$ **(b)** $3\pi/16$ **(c)** $\frac{8}{15}$ **(d)** $5\pi/32$

▶ Exercise Set 7.4 (Page 513)

1. $\frac{1}{2}\sin^{-1}(x/3)+\frac{1}{2}x\sqrt{9-x^2}+C$ 3. $2\sin^{-1}\left(\dfrac{x}{2}\right)-\dfrac{1}{2}x\sqrt{4-x^2}+C$

5. $\frac{1}{16}\tan^{-1}(x/2)+\dfrac{x}{8(4+x^2)}+C$ 7. $\sqrt{x^2-9}-3\sec^{-1}(x/3)+C$

9. $-(x^2+2)\sqrt{1-x^2}+C$ 11. $\dfrac{\sqrt{9x^2-4}}{4x}+C$ 13. $\dfrac{x}{\sqrt{1-x^2}}+C$

15. $\ln|\sqrt{x^2-16}+x|+C$ 17. $\dfrac{-x}{9\sqrt{4x^2-9}}+C$

19. $\frac{1}{2}\sin^{-1}(e^x)+\frac{1}{2}e^x\sqrt{1-e^{2x}}+C$ 21. $2/3$ 23. $(\sqrt{3}-\sqrt{2})/2$

25. $\dfrac{10\sqrt{3}+18}{243}$

Responses to True–False questions may be abridged to save space.

27. False; use the substitution $x=a\sec\theta$ with $0\le\theta<\pi/2$ $(x\ge a)$ or $\pi/2\le\theta<\pi$ $(x\le -a)$.

29. True; with the restriction $-\pi/2\le\theta\le\pi/2$, this substitution gives $\sqrt{a^2-x^2}=a\cos\theta$ and $dx=a\cos\theta\,d\theta$.

31. $\frac{1}{2}\ln(x^2+9)+C$

33. $L=\sqrt{10}-\sqrt{2}+\ln(\frac{1}{3}(\sqrt{10}-1))-\ln(\sqrt{2}-1)$

35. $S=\dfrac{\pi}{32}\left[18\sqrt{5}-\ln(2+\sqrt{5})\right]$

37. $\dfrac{1}{2}\tan^{-1}\left(\dfrac{x-2}{2}\right)+C$ 39. $\sin^{-1}\left(\dfrac{x-1}{3}\right)+C$

41. $\ln(x-3+\sqrt{(x-3)^2+1})+C$

43. $8\sin^{-1}\left(\dfrac{x+1}{4}\right)+\dfrac{1}{2}(x+1)\sqrt{15-2x-x^2}+C$

45. $\dfrac{1}{\sqrt{14}}\tan^{-1}\left(\sqrt{2/7}(x+1)\right)+C$ 47. $\sin^{-1}\left(\dfrac{2}{3}\right)-\sin^{-1}\left(\dfrac{1}{3}\right)$

49. $u=\sin^2 x$, $\frac{1}{2}\displaystyle\int\sqrt{1-u^2}du$

$=\frac{1}{4}[\sin^2 x\sqrt{1-\sin^4 x}+\sin^{-1}(\sin^2 x)]+C$

51. **(a)** $\sinh^{-1}(x/3)+C$ **(b)** $\ln\left(\dfrac{\sqrt{x^2+9}}{3}+\dfrac{x}{3}\right)+C$

▶ Exercise Set 7.5 (Page 521)

1. $\dfrac{A}{x-3}+\dfrac{B}{x+6}$ 3. $\dfrac{A}{x}+\dfrac{B}{x^2}+\dfrac{C}{x-1}$

5. $\dfrac{A}{x}+\dfrac{B}{x^2}+\dfrac{C}{x^3}+\dfrac{Dx+E}{x^2+2}+\dfrac{Fx+G}{(x^2+2)^2}$ 7. $\dfrac{Ax+B}{x^2-2}+\dfrac{Cx+D}{(x^2-2)^2}$

9. $\dfrac{1}{7}\ln\left|\dfrac{x-5}{x+2}\right|+C$ 11. $9\ln|2x-1|+2\ln|x+5|+C$

13. $\ln\left|\dfrac{x(x+3)^2}{x-3}\right|+C$ 15. $\dfrac{x^2}{2}-3x+2\ln|x+3|+C$

17. $3x+12\ln|x-2|-\dfrac{2}{x-2}+C$

19. $\ln|x^2-3x-10|+C$

21. $x+\dfrac{x^3}{3}+\ln\left|\dfrac{(x-1)^2(x+1)}{x^2}\right|+C$

23. $3\ln|x|-\ln|x-1|-\dfrac{5}{x-1}+C$

25. $\dfrac{2}{x-3}+\ln|x-3|+\ln|x+1|+C$

27. $-\dfrac{1}{x+1}+\dfrac{1}{2(x+1)^2}+C$

29. $-\frac{7}{34}\ln|4x-1|+\frac{6}{17}\ln(x^2+1)+\frac{3}{17}\tan^{-1}x+C$

31. $3\tan^{-1}x+\frac{1}{2}\ln(x^2+3)+C$

33. $\dfrac{x^2}{2}-2x+\dfrac{1}{2}\ln(x^2+1)+C$

Responses to True–False questions may be abridged to save space.

35. True; $\dfrac{2x+3}{x^2}=\dfrac{2x}{x^2}+\dfrac{3}{x^2}=\dfrac{2}{x}+\dfrac{3}{x^2}$.

37. True; partial fractions rewrites proper rational functions $P(x)/Q(x)$ as a sum of terms of the form $\dfrac{A}{(Bx+C)^k}$ and/or $\dfrac{Dx+E}{(Fx^2+Gx+H)^k}$.

39. $\ln\left(\dfrac{1-\sin\theta}{6+\sin\theta}\right)+C$ 41. $e^x-2\tan^{-1}\left(\frac{1}{2}e^x\right)+C$

43. $V=\pi\left(\frac{19}{5}-\frac{9}{4}\ln 5\right)$ 45. $\dfrac{1}{\sqrt{2}}\tan^{-1}\left(\dfrac{x+1}{\sqrt{2}}\right)+\dfrac{1}{x^2+2x+3}+C$

47. $\frac{1}{8}\ln|x-1|-\frac{1}{5}\ln|x-2|+\frac{1}{12}\ln|x-3|-\frac{1}{120}\ln|x+3|+C$

▶ Exercise Set 7.6 (Page 531)

1. Formula (60): $\frac{4}{5}x+\frac{4}{25}\ln|5x-1|+C$

3. Formula (65): $\dfrac{1}{5}\ln\left|\dfrac{x}{5+7x}\right|+C$

5. Formula (102): $\frac{1}{20}(2x-1)(4x+3)^{3/2}+C$

7. Formula (108): $\dfrac{1}{2}\ln\left|\dfrac{\sqrt{4-5x}-2}{\sqrt{4-5x}+2}\right|+C$

9. Formula (69): $\dfrac{1}{6}\ln\left|\dfrac{x+3}{x-3}\right|+C$

11. Formula (73): $\dfrac{x}{2}\sqrt{x^2-2}-\ln|x+\sqrt{x^2-2}|+C$

13. Formula (95): $\dfrac{x}{2}\sqrt{x^2+4}-2\ln(x+\sqrt{x^2+4})+C$

15. Formula (74): $\dfrac{x}{2}\sqrt{4-x^2}+2\sin^{-1}\dfrac{x}{2}+C$

17. Formula (79): $\sqrt{1-x^2}-\ln\left|\dfrac{1+\sqrt{1-x^2}}{x}\right|+C$

19. Formula (38): $-\dfrac{\sin 8x}{16}+\dfrac{1}{4}\sin 2x+C$

21. Formula (50): $\dfrac{x^5}{25}(5\ln x-1)+C$

23. Formula (42): $\dfrac{e^{-2x}}{29}(-2\sin(5x)-5\cos(5x))+C$

25. Formula (62): $\dfrac{1}{2}\displaystyle\int\dfrac{u\,du}{(4-3u)^2}=\dfrac{1}{18}\left[\dfrac{4}{4-3e^{2x}}+\ln|4-3e^{2x}|\right]+C$

27. Formula (68): $\dfrac{2}{3}\displaystyle\int\dfrac{du}{u^2+4}=\dfrac{1}{3}\tan^{-1}\dfrac{3\sqrt{x}}{2}+C$

29. Formula (76): $\dfrac{1}{2}\displaystyle\int\dfrac{du}{\sqrt{u^2-25}}=\dfrac{1}{2}\ln|2x+\sqrt{4x^2-25}|+C$

31. Formula (81): $\dfrac{1}{4}\displaystyle\int\dfrac{u^2}{\sqrt{2-u^2}}du=-\dfrac{1}{4}x^2\sqrt{2-4x^4}$

$+\dfrac{1}{4}\sin^{-1}\left(\sqrt{2}x^2\right)+C$

33. Formula (27): $\displaystyle\int\cos^2 u\,du=\frac{1}{2}\ln x+\frac{1}{4}\sin(2\ln x)+C$

35. Formula (51): $\dfrac{1}{4}\displaystyle\int ue^u\,du=\dfrac{1}{4}(-2x-1)e^{-2x}+C$

37. $u=\sin 3x$, Formula (67): $\dfrac{1}{3}\displaystyle\int\dfrac{du}{u(u+1)^2}$

$=\dfrac{1}{3}\left(\dfrac{1}{\sin 3x+1}+\left|\dfrac{\sin 3x}{\sin 3x+1}\right|\right)+C$

39. $u=4x^2$, Formula (70): $\dfrac{1}{8}\displaystyle\int\dfrac{du}{u^2-1}=\dfrac{1}{16}\ln\left|\dfrac{4x^2-1}{4x^2+1}\right|+C$

41. $u=3x$, Formula (112): $\dfrac{1}{3}\displaystyle\int\sqrt{\tfrac{5}{3}u-u^2}\,du=\dfrac{18x-5}{36}\sqrt{5x-9x^2}$

$+\dfrac{25}{216}\sin^{-1}\left(\dfrac{18x-5}{5}\right)+C$

43. $u=2e^x$, Formula (74): $\dfrac{1}{2}\displaystyle\int\sqrt{3-u^2}\,du=\dfrac{1}{2}e^x\sqrt{3-4e^{2x}}$

$+\dfrac{3}{4}\sin^{-1}\left(\dfrac{2e^x}{\sqrt{3}}\right)+C$

45. $u = 3x$, Formula (44): $\int u \sin u \, du = \sin 3x - 3x \cos 3x + C$

47. $u = -\sqrt{x}$, Formula (51): $2\int ue^u \, du = -2(\sqrt{x}+1)e^{-\sqrt{x}} + C$

49. $x^2 + 6x - 7 = (x+3)^2 - 16$, $u = x + 3$, Formula (70):
$$\int \frac{du}{u^2 - 16} = \frac{1}{8}\ln\left|\frac{x-1}{x+7}\right| + C$$

51. $x^2 - 4x - 5 = (x-2)^2 - 9$, $u = x - 2$, Formula (77):
$$\int \frac{u+2}{\sqrt{9-u^2}}\,du = -\sqrt{5+4x-x^2} + 2\sin^{-1}\left(\frac{x-2}{3}\right) + C$$

53. $u = \sqrt{x-3}$, $\frac{2}{5}(x-3)^{5/2} + 2(x-3)^{3/2} + C$

55. $u = \sqrt{x^3+1}$,
$$\frac{2}{3}\int u^2(u^2-1)\,du = \frac{2}{15}(x^3+1)^{5/2} - \frac{2}{9}(x^3+1)^{3/2} + C$$

57. $u = x^{1/3}$, $\int \frac{3u^2}{u^3 - u}\,du = \frac{3}{2}\ln|x^{2/3} - 1| + C$

59. $u = x^{1/4}$, $4\int \frac{1}{u(1-u)}\,du = 4\ln\frac{x^{1/4}}{|1 - x^{1/4}|} + C$

61. $u = x^{1/6}$,
$$6\int \frac{u^3}{u-1}\,du = 2x^{1/2} + 3x^{1/3} + 6x^{1/6} + 6\ln|x^{1/6} - 1| + C$$

63. $u = \sqrt{1+x^2}$, $\int (u^2 - 1)\,du = \frac{1}{3}(1+x^2)^{3/2} - (1+x^2)^{1/2} + C$

65.
$$\int \frac{1}{1 + \frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2}} \frac{2}{1+u^2}\,du = \int \frac{1}{u+1}\,du$$
$$= \ln|\tan(x/2) + 1| + C$$

67. $\int \frac{d\theta}{1 - \cos\theta} = \int \frac{1}{u^2}\,du = -\cot(\theta/2) + C$

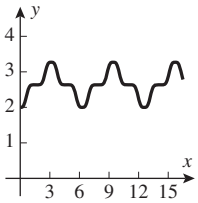
69.
$$\int \frac{1}{\frac{2u}{1+u^2} + \frac{2u}{1+u^2}\cdot\frac{1+u^2}{1-u^2}} \cdot \frac{2}{1+u^2}\,du = \int \frac{1-u^2}{2u}\,du$$
$$= \frac{1}{2}\ln|\tan(x/2)| - \frac{1}{4}\tan^2(x/2) + C$$

71. $x = \frac{6e^3}{1+e^3}$ **73.** $A = 4\sqrt{5} + 18\sin^{-1}\frac{2}{3}$ **75.** $A = \frac{1}{40}\ln 9$

77. $V = \pi(\pi - 2)$ **79.** $V = 2\pi(1 - 4e^{-3})$

81. $L = \sqrt{65} + \frac{1}{8}\ln(8 + \sqrt{65})$ **83.** $S = 2\pi\left[\sqrt{2} + \ln(1+\sqrt{2})\right]$

85.



91. $\frac{1}{31}\cos^{31} x \sin^{31} x + C$

93. $-\frac{1}{9}\ln|1 + x^{-9}| + C$

Exercise Set 7.7 (Page 544)

1. $\int_0^8 \sqrt{x+1}\,dx = \frac{52}{3} \approx 17.3333333$
(a) $M_{10} = 17.34206$; $|E_M| \approx .0087237$
(b) $T_{10} = 17.31575$; $|E_T| \approx .0175868$
(c) $S_{20} = 17.33329$; $|E_S| \approx .00004648$

3. $\int_0^{\pi/2} \cos x\,dx = 1$
(a) $M_{10} = 1.00103$; $|E_M| \approx 0.00102882$
(b) $T_{10} = 0.997943$; $|E_T| \approx 0.00205701$
(c) $S_{20} = 1.00000$; $|E_S| \approx 2.11547 \times 10^{-7}$

5. $\int_1^3 e^{-2x}\,dx = \frac{-1+e^4}{2e^6} \approx 0.0664283$
(a) $M_{10} = 0.0659875$; $|E_M| \approx 0.000440797$
(b) $T_{10} = 0.0673116$; $|E_T| \approx 0.000883357$
(c) $S_{20} = 0.0664289$; $|E_S| \approx 5.87673 \times 10^{-7}$

7. **(a)** $|E_M| \le \frac{9}{3200} = 0.0028125$
(b) $|E_T| \le \frac{9}{1600} = 0.005625$
(c) $|E_S| \le \frac{81}{10{,}240{,}000} \approx 7.91016 \times 10^{-6}$

9. **(a)** $|E_M| \le \frac{\pi^3}{19{,}200} \approx 0.00161491$
(b) $|E_T| \le \frac{\pi^3}{9600} \approx 0.00322982$
(c) $|E_S| \le \frac{\pi^5}{921{,}600{,}000} \approx 3.32053 \times 10^{-7}$

11. **(a)** $|E_M| \le \frac{1}{75e^2} \approx 0.00180447$
(b) $|E_T| \le \frac{2}{75e^2} \approx 0.00360894$
(c) $|E_S| \le \frac{1}{56{,}250e^2} \approx 2.40596 \times 10^{-6}$

13. **(a)** $n = 24$ **(b)** $n = 34$ **(c)** $n = 8$

15. **(a)** $n = 13$ **(b)** $n = 18$ **(c)** $n = 4$

17. **(a)** $n = 43$ **(b)** $n = 61$ **(c)** $n = 8$

Responses to True–False questions may be abridged to save space.

19. False; $S_{50} = \frac{2}{3}M_{25} + \frac{1}{3}T_{25}$

21. False; T_n is the average of L_n and R_n.

23. $g(x) = \frac{1}{24}x^2 - \frac{3}{8}x + \frac{13}{12}$

25. $S_{10} = 1.76414975$; $\int_{-2}^{2} e^{-x^2} \approx 1.76416278$

27. $S_{10} = 14.5496435$; $\int_{-1}^{3} x\sqrt{2+x^3} \approx 14.59935195$

29. $S_{10} = 0.3102678$; $\int_0^1 \sin(x^2)\,dx \approx 0.31026830$

31. **(a)** $M_{10} = 3.14243$; error $E_M \approx -0.000833331$
(b) $T_{10} = 3.13993$; error $E_T \approx 0.00166666$
(c) $S_{20} = 3.14159$; error $E_S \approx 6.20008 \times 10^{-10}$

33. $S_{14} = 0.693147984$, $|E_S| \approx 0.000000803 = 8.03 \times 10^{-7}$

35. $n = 238$ **39.** 1.91009889 **41.** 1071 ft **43.** 37.9 mi **45.** 9.3 L

47. **(a)** $\max|f''(x)| \approx 3.844880$ **(b)** $n = 18$ **(c)** 0.904741

49. **(a)** The maximum value of $|f^{(4)}(x)|$ is approximately 141.09327.
(b) $n = 10$ **(c)** $S_{10} = 0.962257248$

Exercise Set 7.8 (Page 554)

1. **(a)** improper; infinite discontinuity at $x = 4$ **(b)** not improper
(c) improper; infinite discontinuity at $x = 0$
(d) improper; infinite interval of integration
(e) improper; infinite interval of integration and infinite discontinuity at $x = 1$ **(f)** not improper

3. $\frac{1}{3}$ **5.** $\frac{3}{2}\ln 2$ **7.** $\frac{1}{2}$ **9.** $-\frac{1}{4}$ **11.** $\frac{1}{4}$ **13.** divergent **15.** 0

17. divergent **19.** divergent **21.** $\pi/2$ **23.** 1 **25.** divergent

27. $\frac{9}{2}$ **29.** divergent **31.** $\pi/2$

Responses to True–False questions may be abridged to save space.

33. False; the integrand $\frac{1}{x(x-3)}$ is continuous on $[1, 2]$.

35. True; see Theorem 7.8.2 with $p = \frac{4}{3} > 1$.

37. 2 **39.** 2 **41.** $-\frac{1}{2}$

43. **(a)** 2.664069 **(b)** 2.804364 **(c)** 0.219384 **(d)** 0.504067 **45.** 27

47. -1 **49.** $\frac{1}{3}$ **51.** **(a)** $V = \pi/2$ **(b)** $S = \pi[\sqrt{2} + \ln(1+\sqrt{2})]$

53. **(b)** $1/e$ **(c)** It is convergent. **55.** $V = \pi$

59. $\frac{2\pi NI}{kr}\left(1 - \frac{a}{\sqrt{r^2+a^2}}\right)$

61. **(b)** 2.4×10^7 mi·lb **63.** **(a)** $\frac{1}{s^2}$ **(b)** $\frac{2}{s^3}$ **(c)** $\frac{e^{-3s}}{s}$

67. **(a)** 1.047 **71.** 0.621

Chapter 7 Review Exercises (Page 557)

1. $\frac{2}{15}(4+5x)^{3/2}+C$ 3. $-\frac{2}{3}\cos^{3/2}\theta + C$ 5. $\frac{1}{6}\tan^3(x^2)+C$
7. **(a)** $2\sin^{-1}(\sqrt{x/2})+C$; $-2\sin^{-1}(\sqrt{2-x}/\sqrt{2})+C$; $\sin^{-1}(x-1)+C$
9. $-xe^{-x}-e^{-x}+C$ 11. $x\ln(2x+7)-x+\frac{7}{2}\ln(2x+7)+C$
13. $(4x^4-12x^2+6)\sin(2x)+(8x^3-12x)\cos(2x)+C$
15. $\frac{1}{2}\theta-\frac{1}{24}\sin 12\theta + C$ 17. $-\frac{1}{8}\cos 4x+\frac{1}{4}\cos 2x+C$
19. $-\frac{1}{8}\sin^3(2x)\cos 2x-\frac{3}{16}\cos 2x\sin 2x+\frac{3}{8}x+C$
21. $2\sin^{-1}\frac{x}{2}-\frac{1}{2}x\sqrt{4-x^2}+C$ 23. $\ln|x+\sqrt{x^2-1}|+C$
25. $\dfrac{x\sqrt{x^2+9}}{2}-\dfrac{9\ln(|\sqrt{x^2+9}+x|)}{2}+C$ 27. $\dfrac{1}{6}\ln\left|\dfrac{x-1}{x+5}\right|+C$
29. $\frac{1}{2}x^2-2x+6\ln|x+2|+C$
31. $\ln|x+2|+\dfrac{4}{x+2}-\dfrac{2}{(x+2)^2}+C$
35. Formula (40): $-\frac{1}{34}\cos 17x+\frac{1}{6}\cos 3x+C$
37. Formula (113): $\frac{1}{24}(8x^2-2x-3)\sqrt{x-x^2}+\frac{1}{16}\sin^{-1}(2x-1)+C$
39. Formula (28): $\frac{1}{2}\tan 2x - x + C$
41. $\displaystyle\int_1^3 \frac{1}{\sqrt{x+1}} = 4-2\sqrt{2}\approx 1.17157$
(a) $M_{10}=1.17138$; $|E_M|\approx 0.000190169$
(b) $T_{10}=1.17195$; $|E_T|\approx 0.000380588$
(c) $S_{20}=1.17157$; $|E_S|\approx 8.35151\times 10^{-8}$
43. **(a)** $|E_M|\le \dfrac{1}{1600\sqrt{2}}\approx 0.000441942$
(b) $|E_T|\le \dfrac{1}{800\sqrt{2}}\approx 0.000883883$
(c) $|E_S|\le \dfrac{7}{15{,}360{,}000\sqrt{2}}\approx 3.22249\times 10^{-7}$
45. **(a)** $n=22$ **(b)** $n=30$ **(c)** $n=6$ 47. 1 49. 6
51. e^{-1} 53. $a=\pi/2$ 55. $\dfrac{x}{3\sqrt{3+x^2}}+C$ 57. $\frac{5}{12}-\frac{1}{2}\ln 2$
59. $\frac{1}{6}\sin^3 2x-\frac{1}{10}\sin^5 2x+C$
61. $\frac{2}{13}e^{2x}\cos 3x+\frac{3}{13}e^{2x}\sin 3x+C$
63. $-\frac{1}{6}\ln|x-1|+\frac{1}{15}\ln|x+2|+\frac{1}{10}\ln|x-3|+C$
65. $4-\pi$ 67. $\ln\dfrac{\sqrt{e^x+1}-1}{\sqrt{e^x+1}+1}+C$ 69. $\dfrac{\pi}{12}+\dfrac{\sqrt{3}}{2}-1$
71. $\sqrt{x^2+2x+2}+2\ln(\sqrt{x^2+2x+2}+x+1)+C$ 73. $\dfrac{1}{2(a^2+1)}$

Chapter 7 Making Connections (Page 559)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

3. **(a)** $\Gamma(1)=1$ **(c)** $\Gamma(2)=1, \Gamma(3)=2, \Gamma(4)=6$
5. **(b)** 1.37078 seconds

Exercise Set 8.1 (Page 566)

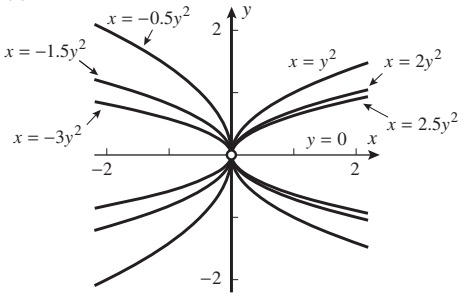
3. **(a)** first order **(b)** second order

Responses to True–False questions may be abridged to save space.

5. False; only first-order derivatives appear.
7. True; it is third order.
15. $y(x)=e^{-2x}+3e^x$ 17. $y(x)=-e^{-2x}+2e^{3x}$
19. $y(x)=-e^{-2x}(\sin 3x+\cos 3x)$ 21. $y(x)=-2x^2+2x+5$
23. $y(x)=2/(3-2x)$ 25. $y(x)=3/x^2$
27. **(a)** $\dfrac{dy}{dt}=ky^2$, $y(0)=y_0$ $(k>0)$
(b) $\dfrac{dy}{dt}=-ky^2$, $y(0)=y_0$ $(k>0)$
29. **(a)** $\dfrac{ds}{dt}=\dfrac{1}{3}s$ **(b)** $\dfrac{d^2s}{dt^2}=2\dfrac{ds}{dt}$ 33. **(b)** $L/2$

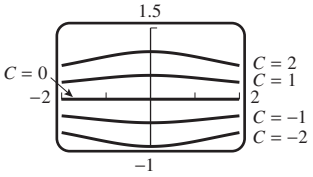
Exercise Set 8.2 (Page 575)

1. $y=Cx$ 3. $y=Ce^{-\sqrt{1+x^2}}-1, C\ne 0$
5. $2\ln|y|+y^2=e^x+C$ 7. $y=\ln(\sec x+C)$
9. $y=\dfrac{1}{1-C(\csc x-\cot x)}, y=0$
11. $y^2+\sin y=x^3+\pi^2$ 13. $y^2-2y=t^2+t-1$
15. **(a)**

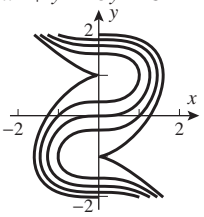


(b) $y^2=x/3$

17. $y=\dfrac{C}{\sqrt{x^2+4}}$

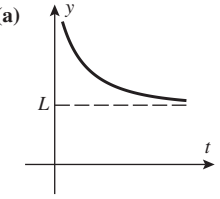


19. $x^3+y^3-3y=C$

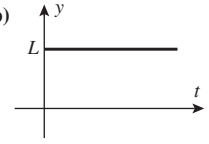


Responses to True–False questions may be abridged to save space.

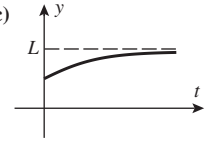
21. True; since $\dfrac{1}{f(y)}\,dy=dx$. 23. True; since $\left(\frac{1}{2}\right)^5 32=1$.
27. $y=\ln\left(\dfrac{x^2}{2}-7\right)$
29. **(a)** $y'(t)=y(t)/50$, $y(0)=20{,}000$ **(b)** $y(t)=20{,}000e^{t/50}$
(c) $50\ln 2\approx 34.66$ hr **(d)** $50\ln(2.25)\approx 40.55$ hr
31. **(a)** $\dfrac{dy}{dt}=-ky$, $k\approx 0.1810$ **(b)** $y=5.0\times 10^7 e^{-0.181t}$
(c) $\approx 219{,}000$ atoms **(d)** 12.72 days
33. $50\ln(50)\approx 196$ days 35. 3.30 days
39. **(b)** 35 years **(c)** 20 years **(d)** 7%
43. **(a)** no **(b)** same, $r\%$ 45. **(b)** $\ln(2)/\ln(1.3)\approx 2.64$ hr
47. **(a)** \$1419.07 **(b)** \$4965.85 **(c)** 9.9 years
51. **(a)**



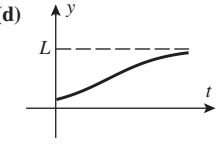
(b)



(c)



(d)

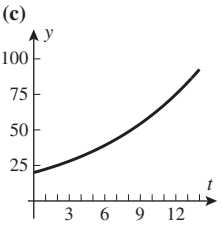


53. $y_0\approx 2$, $L\approx 8$, $k\approx 0.5493$
55. **(a)** $y_0=5$ **(b)** $L=12$ **(c)** $k=1$ **(d)** $t=0.3365$
(e) $\dfrac{dy}{dt}=\dfrac{1}{2}y(12-y)$, $y(0)=5$
57. **(a)** $y=\dfrac{1000}{1+49e^{-0.115t}}$

(b)

t	0	1	2	3	4	5	6	7
$y(t)$	20	22	25	28	31	35	39	44
t	8	9	10	11	12	13	14	
$y(t)$	49	54	61	67	75	83	93	

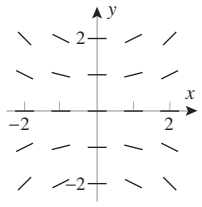
(c)



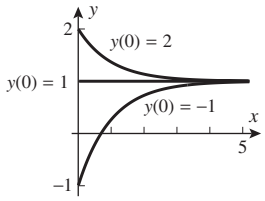
59. **(a)** $T = 21 + 74e^{-kt}$ **(b)** 6.22 min
61. **(a)** $v = c\ln\dfrac{m_0}{m_0 - kt} - gt$ **(b)** 2030 m/s
63. **(a)** $h \approx (2 - 0.003979t)^2$ **(b)** 8.4 min
65. **(a)** $v = 64/(2t+1)$, $x = 32\ln(2t+1)$

Exercise Set 8.3 (Page 584)

1.



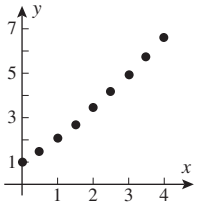
3.



5. $y \to 1$ as $x \to +\infty$

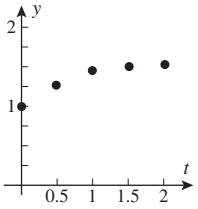
7.

n	0	1	2	3	4	5	6	7	8
x_n	0	0.5	1	1.5	2	2.5	3	3.5	4
y_n	1.00	1.50	2.07	2.71	3.41	4.16	4.96	5.82	6.72



9.

n	0	1	2	3	4
t_n	0.00	0.50	1.00	1.50	2.00
y_n	1.00	1.27	1.42	1.49	1.53



11. 0.63

Responses to True–False questions may be abridged to save space.

13. True; the derivative is positive.
15. True; $y = y_0$ is a solution if y_0 is a root of p.
17. **(b)** $y(1/2) = \sqrt{15}/2$ 19. **(b)** The x-intercept is $\ln 3$.
23. **(a)** $y' = \dfrac{2xy - y^3}{3xy^2 - x^2}$ **(c)** $xy^3 - x^2y = 6$
25. **(b)** $\lim\limits_{n\to+\infty} y_n = \lim\limits_{n\to+\infty}\left(\dfrac{n+1}{n}\right)^n = e$

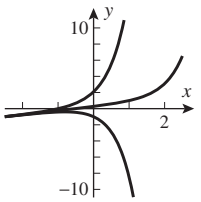
Exercise Set 8.4 (Page 592)

1. $y = \frac{1}{2}e^{-3x} + Ce^{-5x}$ 3. $y = e^{-x}\cos(e^x) + Ce^{-x}$
5. $y = \dfrac{C}{\sqrt{x^2+1}}$ 7. $y = \dfrac{x}{2} + \dfrac{5}{2x}$ 9. $y = 5e^{x^2} - 2$

Responses to True–False questions may be abridged to save space.

11. True; it will approach the concentration of the entering fluid.
13. False; $y = x^2$ is a solution to $dy/dx = 2x$, but $y + y = 2x^2$ is not.

15.



17. $\lim\limits_{x\to+\infty} y = \begin{cases} +\infty & \text{if } y_0 \ge 1/4 \\ -\infty, & \text{if } y_0 < 1/4 \end{cases}$

19. **(a)**

n	0	1	2	3	4	5
x_n	0	0.2	0.4	0.6	0.8	1.0
y_n	1	1.20	1.48	1.86	2.35	2.98

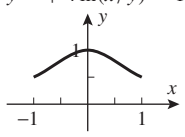
(b) $y = -(x+1) + 2e^x$

x_n	0	0.2	0.4	0.6	0.8	1.0
$y(x_n)$	1	1.24	1.58	2.04	2.65	3.44
Absolute error	0	0.04	0.10	0.19	0.30	0.46
Percentage error	0	3	6	9	11	13

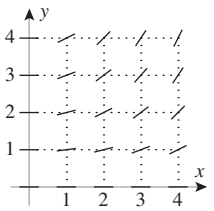
21. **(a)** $240 - 210e^{-t/30}$ oz **(b)** 163 oz 23. 25 lb
27. **(a)** $I(t) = 2 - 2e^{-2t}$ A **(b)** $I(t) \to 2$ A

Chapter 8 Review Exercises (Page 594)

3. $y = \tan(x^3/3 + C)$ 5. $\ln|y| + y^2/2 = e^x + C$ and $y = 0$
7. $y^{-4} + 4\ln(x/y) = 1$
9.

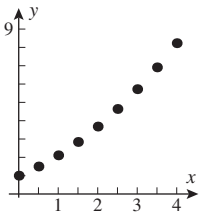


11.



13.

n	0	1	2	3	4	5	6	7	8
x_n	0	0.5	1	1.5	2	2.5	3	3.5	4
y_n	1	1.50	2.11	2.84	3.68	4.64	5.72	6.91	8.23



15. $y(1) \approx 1.00$

n	0	1	2	3	4	5
t_n	0	0.2	0.4	0.6	0.8	1.0
y_n	1.00	1.20	1.26	1.10	0.94	1.00

17. about 2106.5 years 19. $y = \frac{1}{2}e^{-2x} + Ce^{-4x}$
21. $y = -1 + 6e^{x^2/2}$ 23. $y = 2\operatorname{sech} x + \frac{1}{2}(x\operatorname{sech} x + \sinh x)$
25. **(a)** linear **(b)** both **(c)** separable **(d)** neither 27. about 646 oz

Chapter 8 Making Connections (Page 595)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

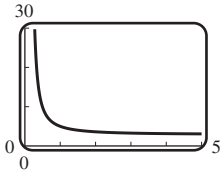
1. **(b)** $y = 2 - e^{-2x}$
3. **(a)** $du/dx = (f(u) - u)/x$ **(b)** $x^2 - 2xy - y^2 = C$

Exercise Set 9.1 (Page 605)

1. (a) $\frac{1}{5^{n-1}}$ (b) $\frac{(-1)^{n-1}}{3^{n-1}}$ (c) $\frac{2n-1}{2n}$ (d) $\frac{n^2}{\pi^{1/(n+1)}}$
3. (a) 2, 0, 2, 0 (b) 1, −1, 1, −1 (c) $2(1+(-1)^n)$; $2+2\cos n\pi$
5. (a) The limit doesn't exist due to repeated oscillation between −1 and 1. (b) 0; 0; 0; 0; 0 (c) yes
7. $\frac{1}{4}, \frac{2}{5}, \frac{3}{6}, \frac{4}{7}, \frac{5}{8}$; converges, $\lim_{n\to+\infty} \frac{n}{n+3} = 1$
9. 3, 3, 3, 3, 3; converges, $\lim_{n\to+\infty} 3 = 3$
11. $\frac{\ln 1}{1}, \frac{\ln 2}{2}, \frac{\ln 3}{3}, \frac{\ln 4}{4}, \frac{\ln 5}{5}$; converges, $\lim_{n\to+\infty} \frac{\ln n}{n} = 0$
13. 0, 2, 0, 2, 0; diverges 15. $-1, \frac{16}{9}, -\frac{54}{28}, \frac{128}{65}, -\frac{250}{126}$; diverges
17. $0, \frac{1}{2}, \frac{10}{18}, \frac{18}{32}, \frac{28}{50}$, converges; $\lim_{n\to+\infty} \frac{1}{2}\left(1-\frac{1}{n}\right)\left(1+\frac{2}{n}\right) = \frac{1}{2}$
19. $e^{-1}, 4e^{-2}, 9e^{-3}, 16e^{-4}, 25e^{-5}$; converges, $\lim_{n\to+\infty} n^2e^{-n} = 0$
21. $2, \left(\frac{5}{3}\right)^2, \left(\frac{6}{4}\right)^3, \left(\frac{7}{5}\right)^4, \left(\frac{8}{6}\right)^5$; converges, $\lim_{n\to+\infty} \left[\frac{n+3}{n+1}\right]^n = e^2$
23. $\left\{\frac{2n-1}{2n}\right\}_{n=1}^{+\infty}$; converges, $\lim_{n\to+\infty} \frac{2n-1}{2n} = 1$
25. $\left\{(-1)^{n+1}\frac{1}{3^n}\right\}_{n=1}^{+\infty}$; converges, $\lim_{n\to+\infty} (-1)^{n+1}\frac{1}{3^n} = 0$
27. $\left\{(-1)^{n+1}\left(\frac{1}{n}-\frac{1}{n+1}\right)\right\}_{n=1}^{+\infty}$;
converges, $\lim_{n\to+\infty} (-1)^{n+1}\left(\frac{1}{n}-\frac{1}{n+1}\right) = 0$
29. $\{\sqrt{n+1}-\sqrt{n+2}\}_{n=1}^{+\infty}$; converges, $\lim_{n\to+\infty} (\sqrt{n+1}-\sqrt{n+2}) = 0$

Responses to True–False questions may be abridged to save space.

31. False; for example, $\{(-1)^{n+1}\}$ diverges with terms that oscillate between 1 and −1.
33. True; a sequence is a function whose domain is a set of integers.
35. The limit is 0.
37. for example, $\{(-1)^n\}_{n=1}^{+\infty}$ and $\{\sin(\pi n/2)+1/n\}_{n=1}^{+\infty}$
39. (a) 1, 2, 1, 4, 1, 6 (b) $a_n = \begin{cases} n, & n \text{ odd} \\ 1/2^n, & n \text{ even} \end{cases}$
(c) $a_n = \begin{cases} 1/n, & n \text{ odd} \\ 1/(n+1); & n \text{ even} \end{cases}$
(d) (a) diverges; (b) diverges; (c) $\lim_{n\to+\infty} a_n = 0$
43. (a) $(0.5)^{2n}$ (c) $\lim_{n\to+\infty} a_n = 0$ (d) $-1 \le a_0 \le 1$
45. (a)



(b) $\lim_{n\to+\infty} (2^n+3^n)^{1/n} = 3$
49. (a) $N = 4$ (b) $N = 10$ (c) $N = 1000$

Exercise Set 9.2 (Page 613)

1. strictly decreasing 3. strictly increasing 5. strictly decreasing
7. strictly increasing 9. strictly decreasing 11. strictly increasing

Responses to True–False questions may be abridged to save space.

13. True; $a_{n+1} - a_n > 0$ for all n is equivalent to $a_1 < a_2 < a_3 < \cdots < a_n < \cdots$.
15. False; for example, $\{(-1)^{n+1}\} = \{1, -1, 1, -1, \ldots\}$ is bounded but diverges.
17. strictly increasing 19. strictly increasing
21. eventually strictly increasing 23. eventually strictly increasing
25. Yes; the limit lies in the interval [1, 2].
27. (a) $\sqrt{2}, \sqrt{2+\sqrt{2}}, \sqrt{2+\sqrt{2+\sqrt{2}}}$ (e) $L = 2$

Exercise Set 9.3 (Page 621)

1. (a) $2, \frac{12}{5}, \frac{62}{25}, \frac{312}{125}; \frac{5}{2}\left(1-\left(\frac{1}{5}\right)^n\right)$; $\lim_{n\to+\infty} s_n = \frac{5}{2}$ (converges)
(b) $\frac{1}{4}, \frac{3}{4}, \frac{7}{4}, \frac{15}{4}; -\frac{1}{4}(1-2^n)$; $\lim_{n\to+\infty} s_n = +\infty$ (diverges)
(c) $\frac{1}{6}, \frac{1}{4}, \frac{3}{10}, \frac{1}{3}; \frac{1}{2} - \frac{1}{n+2}$; $\lim_{n\to+\infty} s_n = \frac{1}{2}$ (converges)
3. $\frac{4}{7}$ 5. $\frac{63}{10}$ 7. $\frac{1}{4}$ 9. $\frac{1}{6}$ 11. diverges 13. $\frac{448}{3}$
15. (a) Exercise 5 (b) Exercise 3 (c) Exercise 7 (d) Exercise 9

Responses to True–False questions may be abridged to save space.

17. False; an infinite series converges if its sequence of *partial sums* converges.
19. True; the sequence of partial sums $\{s_n\}$ for the harmonic series satisfies $s_{2^n} > \frac{n+1}{2}$, so this series diverges.
21. 1 23. $\frac{559}{99}$ 27. 50 m
29. (a) $S_n = -\ln(n+1)$; $\lim_{n\to+\infty} S_n = -\infty$ (diverges)
(b) $S_n = \sum_{k=2}^{n+1}\left[\ln\frac{k-1}{k} - \ln\frac{k}{k+1}\right]$, $\lim_{n\to+\infty} S_n = -\ln 2$
31. (a) converges for $|x| < 1$; $S = \frac{x}{1+x^2}$
(b) converges for $|x| > 2$; $S = \frac{1}{x^2-2x}$
(c) converges for $x > 0$; $S = \frac{1}{e^x-1}$
33. $a_n = \frac{1}{2^{n-1}}a_1 + \frac{1}{2^{n-1}} + \frac{1}{2^{n-2}} + \cdots + \frac{1}{2}$, $\lim_{n\to+\infty} a_n = 1$

Exercise Set 9.4 (Page 629)

1. (a) $\frac{4}{3}$ (b) $-\frac{3}{4}$
3. (a) $p = 4$, converges (b) $p = \frac{1}{2}$, diverges
(c) $p = 1$, diverges (d) $p = \frac{2}{3}$, diverges
5. (a) diverges (b) diverges (c) diverges (d) no information
7. (a) diverges (b) converges
9. diverges 11. diverges 13. diverges 15. diverges 17. diverges
19. converges 21. diverges 23. converges 25. converges for $p > 1$
29. (a) diverges (b) diverges

Responses to True–False questions may be abridged to save space.

31. False; for example, $\sum_{k=0}^{\infty} 2^{-k}$ converges to 2, but $\sum \frac{1}{2^{-k}} = \sum 2^k$ diverges.
33. True; see Theorem 9.4.4.
35. (a) $(\pi^2/2) - (\pi^4/45)$ (b) $(\pi^2/6) - (5/4)$ (c) $\pi^4/90$
37. (d) $\frac{1}{11} < \frac{1}{6}\pi^2 - s_{10} < \frac{1}{10}$
39. (a) $\int_n^{+\infty} \frac{1}{x^4}\,dx = \frac{1}{3n^3}$; apply Exercise 36(b) (b) $n = 6$
(c) $\frac{\pi^4}{90} \approx 1.08238$
41. converges

Exercise Set 9.5 (Page 636)

1. (a) converges (b) diverges 5. converges 7. converges
9. diverges 11. converges 13. inconclusive 15. diverges
17. diverges 19. converges

Responses to True–False questions may be abridged to save space.

21. False; the limit comparison test uses a limit of the quotient of corresponding terms taken from two different sequences.
23. True; use the limit comparison test with the convergent series $\sum(1/k^2)$.
25. converges 27. converges 29. converges 31. diverges
33. converges 35. converges 37. diverges 39. converges
41. diverges 43. converges 45. converges 47. converges

49. $u_k = \dfrac{k!}{1\cdot 3\cdot 5\cdots(2k-1)}$; $\rho = \lim\limits_{k\to+\infty}\dfrac{k+1}{2k+1} = \dfrac{1}{2}$; converges
51. **(a)** converges **(b)** diverges **53.** **(a)** converges

Exercise Set 9.6 (Page 646)

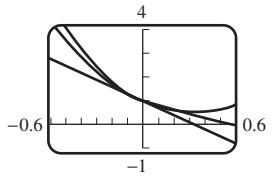
3. diverges **5.** converges **7.** converges absolutely **9.** diverges
11. converges absolutely **13.** conditionally convergent **15.** divergent
17. conditionally convergent **19.** conditionally convergent
21. divergent **23.** conditionally convergent **25.** converges absolutely
27. converges absolutely

Responses to True–False questions may be abridged to save space.

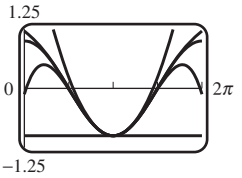
29. True; if a series converges but diverges absolutely, then it converges conditionally.
31. False; an alternating series has terms that alternate between positive and negative.
33. $|\text{error}| < 0.143$ **35.** $|\text{error}| < 0.1$ **37.** $n = 9999$
39. $n = 39{,}999$ **41.** $|\text{error}| < 0.00074$; $s_{10} \approx 0.4995$; $S = 0.5$
43. 0.84 **45.** 0.41 **47.** **(c)** $n = 50$
49. **(a)** If $a_k = \dfrac{(-1)^k}{\sqrt{k}}$, then $\sum a_k$ converges and $\sum a_k^2$ diverges.
If $a_k = \dfrac{(-1)^k}{k}$, then $\sum a_k$ converges and $\sum a_k^2$ also converges.
(b) If $a_k = \dfrac{1}{k}$, then $\sum a_k^2$ converges and $\sum a_k$ diverges. If $a_k = \dfrac{1}{k^2}$, then $\sum a_k^2$ converges and $\sum a_k$ also converges.

Exercise Set 9.7 (Page 657)

1. **(a)** $1 - x + \frac{1}{2}x^2,\ 1 - x$ **(b)** $1 - \frac{1}{2}x^2,\ 1$
3. **(a)** $1 + \frac{1}{2}(x-1) - \frac{1}{8}(x-1)^2$ **(b)** 1.04875 **5.** 1.66434809
7. $p_0(x) = 1,\ p_1(x) = 1 - x,\ p_2(x) = 1 - x + \frac{1}{2}x^2,$
$p_3(x) = 1 - x + \dfrac{1}{2}x^2 - \dfrac{1}{3!}x^3,$
$p_4(x) = 1 - x + \dfrac{1}{2}x^2 - \dfrac{1}{3!}x^3 + \dfrac{1}{4!}x^4;\ \sum\limits_{k=0}^{n}\dfrac{(-1)^k}{k!}x^k$
9. $p_0(x) = 1,\ p_1(x) = 1,\ p_2(x) = 1 - \dfrac{\pi^2}{2!}x^2,\ p_3(x) = 1 - \dfrac{\pi^2}{2!}x^2,$
$p_4(x) = 1 - \dfrac{\pi^2}{2!}x^2 + \dfrac{\pi^4}{4!}x^4;\ \sum\limits_{k=0}^{\lfloor n/2\rfloor}\dfrac{(-1)^k\pi^{2k}}{(2k)!}x^{2k}$ (See Exercise 70 of Section 0.2.)
11. $p_0(x) = 0,\ p_1(x) = x,\ p_2(x) = x - \frac{1}{2}x^2,\ p_3(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3,$
$p_4(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4;\ \sum\limits_{k=1}^{n}\dfrac{(-1)^{k+1}}{k}x^k$
13. $p_0(x) = 1,\ p_1(x) = 1,\ p_2(x) = 1 + \dfrac{x^2}{2},$
$p_3(x) = 1 + \dfrac{x^2}{2},\ p_4(x) = 1 + \dfrac{x^2}{2} + \dfrac{x^4}{4!};\ \sum\limits_{k=0}^{\lfloor n/2\rfloor}\dfrac{1}{(2k)!}x^{2k}$ (See Exercise 70 of Section 0.2.)
15. $p_0(x) = 0,\ p_1(x) = 0,\ p_2(x) = x^2,\ p_3(x) = x^2,$
$p_4(x) = x^2 - \frac{1}{6}x^4;\ \sum\limits_{k=0}^{\lfloor n/2\rfloor-1}\dfrac{(-1)^k}{(2k+1)!}x^{2k+2}$ (See Exercise 70 of Section 0.2.)
17. $p_0(x) = e,\ p_1(x) = e + e(x-1),$
$p_2(x) = e + e(x-1) + \dfrac{e}{2}(x-1)^2,$
$p_3(x) = e + e(x-1) + \dfrac{e}{2}(x-1)^2 + \dfrac{e}{3!}(x-1)^3,$
$p_4(x) = e + e(x-1) + \dfrac{e}{2}(x-1)^2 + \dfrac{e}{3!}(x-1)^3 + \dfrac{e}{4!}(x-1)^4;$
$\sum\limits_{k=0}^{n}\dfrac{e}{k!}(x-1)^k$
19. $p_0(x) = -1,\ p_1(x) = -1 - (x+1),$
$p_2(x) = -1 - (x+1) - (x+1)^2,$
$p_3(x) = -1 - (x+1) - (x+1)^2 - (x+1)^3,$
$p_4(x) = -1 - (x+1) - (x+1)^2 - (x+1)^3 - (x+1)^4;$
$\sum\limits_{k=0}^{n}(-1)(x+1)^k$
21. $p_0(x) = p_1(x) = 1,\ p_2(x) = p_3(x) = 1 - \dfrac{\pi^2}{2}\left(x - \dfrac{1}{2}\right)^2,$
$p_4(x) = 1 - \dfrac{\pi^2}{2}\left(x - \dfrac{1}{2}\right)^2 + \dfrac{\pi^4}{4!}\left(x - \dfrac{1}{2}\right)^4;$
$\sum\limits_{k=0}^{\lfloor n/2\rfloor}\dfrac{(-1)^k\pi^{2k}}{(2k)!}\left(x - \dfrac{1}{2}\right)^{2k}$ (See Exercise 70 of Section 0.2.)
23. $p_0(x) = 0,\ p_1(x) = (x-1),\ p_2(x) = (x-1) - \frac{1}{2}(x-1)^2,$
$p_3(x) = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3,$
$p_4(x) = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4;$
$\sum\limits_{k=1}^{n}\dfrac{(-1)^{k-1}}{k}(x-1)^k$
25. **(a)** $p_3(x) = 1 + 2x - x^2 + x^3$
(b) $p_3(x) = 1 + 2(x-1) - (x-1)^2 + (x-1)^3$
27. $p_0(x) = 1,\ p_1(x) = 1 - 3x,$
$p_2(x) = 1 - 3x + \frac{9}{2}x^2,$
$p_3(x) = 1 - 3x + \frac{9}{2}x^2 - \frac{9}{2}x^3$

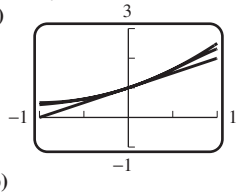


29. $p_0(x) = -1,\ p_2(x) = -1 + \frac{1}{2}(x-\pi)^2,$
$p_4(x) = -1 + \frac{1}{2}(x-\pi)^2 - \frac{1}{24}(x-\pi)^4,$
$p_6(x) = -1 + \frac{1}{2}(x-\pi)^2 - \frac{1}{24}(x-\pi)^4 + \frac{1}{720}(x-\pi)^6$



Responses to True–False questions may be abridged to save space.

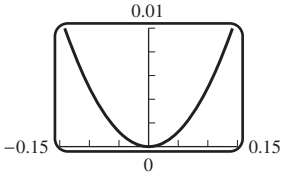
31. True; $y = f(x_0) + f'(x_0)(x - x_0)$ is the first-degree Taylor polynomial for f about $x = x_0$.
33. False; $p_6^{(4)}(x_0) = f^{(4)}(x_0)$ **35.** 1.64870 **37.** IV
39. **(a)**



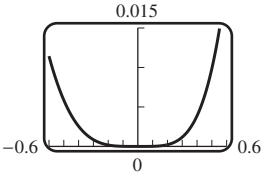
(b)

x	−1.000	−0.750	−0.500	−0.250	0.000	0.250	0.500	0.750	1.000
$f(x)$	0.431	0.506	0.619	0.781	1.000	1.281	1.615	1.977	2.320
$p_1(x)$	0.000	0.250	0.500	0.750	1.000	1.250	1.500	1.750	2.000
$p_2(x)$	0.500	0.531	0.625	0.781	1.000	1.281	1.625	2.031	2.500

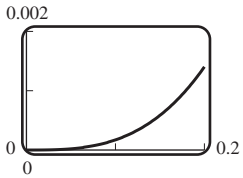
(c) $|e^{\sin x} - (1+x)| < 0.01$ for $-0.14 < x < 0.14$



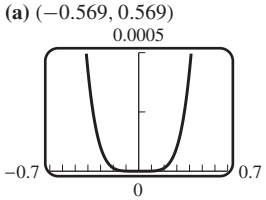
(d) $\left|e^{\sin x} - \left(1 + x + \dfrac{x^2}{2}\right)\right| < 0.01$ for $-0.50 < x < 0.50$



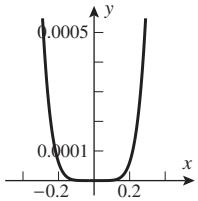
41. **(a)** $[0, 0.137]$ **(b)**



43. **(a)** $(-0.569, 0.569)$



45. $(-0.311, 0.311)$



Exercise Set 9.8 (Page 667)

1. $\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} x^k$ **3.** $\sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k}}{(2k)!} x^{2k}$ **5.** $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k$

7. $\sum_{k=0}^{\infty} \frac{1}{(2k)!} x^{2k}$ **9.** $\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+2}$ **11.** $\sum_{k=0}^{\infty} \frac{1}{ek!} (x+1)^k$

13. $\sum_{k=0}^{\infty} (-1)(x+1)^k$ **15.** $\sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k}}{(2k)!} \left(x - \frac{1}{2}\right)^{2k}$

17. $\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} (x-1)^k$ **19.** $-1 < x < 1$; $\frac{1}{1+x}$

21. $1 < x < 3$; $\frac{1}{3-x}$ **23.** **(a)** $-2 < x < 2$ **(b)** $f(0) = 1$; $f(1) = \frac{2}{3}$

Responses to True–False questions may be abridged to save space.

25. True; see Theorem 9.8.2(c).

27. True; the polynomial *is* the Maclaurin series and converges for all x.

29. $R = 1$; $[-1, 1)$ **31.** $R = +\infty$; $(-\infty, +\infty)$ **33.** $R = \frac{1}{6}$; $\left[-\frac{1}{6}, \frac{1}{6}\right]$

35. $R = 1$; $[-1, 1]$ **37.** $R = 1$; $(-1, 1]$ **39.** $R = +\infty$; $(-\infty, +\infty)$

41. $R = 1$; $[-1, 1]$ **43.** $R = 1$; $(-2, 0]$ **45.** $R = \frac{4}{3}$; $\left(-\frac{19}{3}, -\frac{11}{3}\right)$

47. $R = +\infty$; $(-\infty, +\infty)$ **49.** $(-\infty, +\infty)$ **55.** radius $= R$

61. **(a)** $n = 5$; $s_5 \approx 1.1026$ **(b)** $\zeta(3.7) \approx 1.10629$

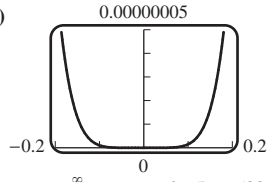
Exercise Set 9.9 (Page 676)

3. 0.069756 **5.** 0.99500 **7.** 0.99619 **9.** 0.52109

11. **(a)** $\sum_{k=1}^{\infty} 2\frac{(1/11)^{2k-1}}{2k-1}$ **(b)** 0.1823

13. **(a)** 0.4635; 0.3218 **(b)** 3.1412 **(c)** no

15. **(a)** error $\le 9 \times 10^{-8}$ **(b)**



17. **(a)** $\sum_{k=0}^{\infty} (-1)^k x^k$ **(b)** $1 + \frac{x}{3} + \sum_{k=2}^{\infty} (-1)^{k-1} \frac{2 \cdot 5 \cdots (3k-4)}{3^k k!} x^k$

(c) $\sum_{k=0}^{\infty} (-1)^k \frac{(k+2)(k+1)}{2} x^k$ **23.** 23.406%

Exercise Set 9.10 (Page 686)

1. **(a)** $1 - x + x^2 - \cdots + (-1)^k x^k + \cdots$; $R = 1$

(b) $1 + x^3 + x^6 + \cdots + x^{3k} + \cdots$; $R = 1$

(c) $1 + 2x + 4x^2 + \cdots + 2^k x^k + \cdots$; $R = \frac{1}{2}$

(d) $\frac{1}{3} + \frac{1}{3^2}x + \frac{1}{3^3}x^2 + \cdots + \frac{1}{3^{k+1}}x^k + \cdots$; $R = 3$

3. **(a)** $(2+x)^{-1/2} = \frac{1}{2^{1/2}} - \frac{1}{2^{5/2}}x + \frac{1 \cdot 3}{2^{9/2} \cdot 2!}x^2 - \frac{1 \cdot 3 \cdot 5}{2^{13/2} \cdot 3!}x^3 + \cdots$

(b) $(1-x^2)^{-2} = 1 + 2x^2 + 3x^4 + 4x^6 + \cdots$

5. **(a)** $3x - \frac{3^3}{3!}x^3 + \frac{3^5}{5!}x^5 - \frac{3^7}{7!}x^7 + \cdots$; $R = +\infty$

(b) $1 - 2x + 2x^2 - \frac{4}{3}x^3 + \cdots$; $R = +\infty$

(c) $1 + x^2 + \frac{1}{2!}x^4 + \frac{1}{3!}x^6 + \cdots$; $R = +\infty$

(d) $x^2 - \frac{\pi^2}{2}x^4 + \frac{\pi^4}{4!}x^6 - \frac{\pi^6}{6!}x^8 + \cdots$; $R = +\infty$

7. **(a)** $x^2 - 4x^3 + 16x^4 - 64x^5 + \cdots$; $R = \frac{1}{4}$

(b) $2x^2 + \frac{2^3}{3!}x^4 + \frac{2^5}{5!}x^6 + \frac{2^7}{7!}x^8 + \cdots$; $R = +\infty$

(c) $x - \frac{3}{2}x^3 + \frac{3}{8}x^5 + \frac{1}{16}x^7 + \cdots$; $R = 1$

9. **(a)** $x^2 - \frac{2^3}{4!}x^4 + \frac{2^5}{6!}x^6 - \frac{2^7}{8!}x^8 + \cdots$

(b) $12x^3 - 6x^6 + 4x^9 - 3x^{12} + \cdots$

11. **(a)** $1 - (x-1) + (x-1)^2 - \cdots + (-1)^k(x-1)^k + \cdots$ **(b)** $(0,2)$

13. **(a)** $x + x^2 + \frac{x^3}{3} - \frac{x^5}{30} + \cdots$ **(b)** $x - \frac{x^3}{24} + \frac{x^4}{24} - \frac{71}{1920}x^5 + \cdots$

15. **(a)** $1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + \frac{61}{720}x^6 + \cdots$ **(b)** $x - 2x^2 + \frac{11}{6}x^3 - x^4 + \cdots$

19. $2 - 4x + 2x^2 - 4x^3 + 2x^4 + \cdots$

25. $[-1, 1]$; $[-1, 1)$ **27.** **(a)** $\sum_{k=0}^{\infty} x^{2k+1}$ **(b)** $f^{(5)}(0) = 5!$, $f^{(6)}(0) = 0$

(c) $f^{(n)}(0) = n!c_n = \begin{cases} n! & \text{if } n \text{ odd} \\ 0 & \text{if } n \text{ even} \end{cases}$

29. **(a)** 1 **(b)** $-\frac{1}{3}$ **31.** 0.9046 **33.** 0.200

35. **(a)** $\sum_{k=0}^{\infty} \frac{x^{5k}}{k!}$; $R = +\infty$ **37.** **(a)** $5/16$ **(b)** $\ln(4/3)$

39. **(a)** $x - \frac{1}{6}x^3 + \frac{3}{40}x^5 - \frac{5}{112}x^7 + \cdots$

(b) $x + \sum_{k=1}^{\infty} (-1)^k \frac{1 \cdot 3 \cdot 5 \cdots (2k-1)}{2^k k!(2k+1)} x^{2k+1}$ **(c)** $R = 1$

41. **(a)** $y(t) = y_0 \sum_{k=0}^{\infty} \frac{(-1)^k (0.000121)^k t^k}{k!}$ **(c)** $0.9998790073y_0$

43. $2\pi\sqrt{\frac{L}{g}}\left(1 + \frac{k^2}{4} + \frac{9k^4}{64}\right)$

Chapter 9 Review Exercises (Page 689)

9. **(a)** true **(b)** sometimes false **(c)** sometimes false
(d) true **(e)** sometimes false **(f)** sometimes false
(g) false **(h)** sometimes false **(i)** true
(j) true **(k)** sometimes false **(l)** sometimes false

11. **(a)** $\left\{\frac{n+2}{(n+1)^2 - n^2}\right\}_{n=1}^{+\infty}$; converges, $\lim_{n\to+\infty} \frac{n+2}{(n+1)^2 - n^2} = \frac{1}{2}$

(b) $\left\{(-1)^{n+1}\frac{n}{2n+1}\right\}_{n=1}^{+\infty}$; diverges

15. **(a)** converges **(b)** converges **17.** **(a)** converges **(b)** diverges

19. **(a)** diverges **(b)** converges **21.** $\frac{1}{4 \cdot 5^{99}}$

23. **(a)** 2 **(b)** diverges **(c)** $11/18$ **(d)** $\pi/4$ **25.** $p > 1$

29. **(a)** $p_0(x) = 1$, $p_1(x) = 1 - 6x$, $p_2(x) = 1 - 6x + 5x^2$,
$p_3(x) = 1 - 6x + 5x^2 + 3x^3$, $p_4(x) = 1 - 6x + 5x^2 + 4x^3$

33. **(a)** $e^3 - 1$ **(b)** 0 **(c)** $\cos e$ **(d)** $\frac{1}{3}$

37. **(a)** $x - \frac{2}{3}x^3 + \frac{2}{15}x^5 - \frac{4}{315}x^7$ **(b)** $x - \frac{2}{3}x^3 + \frac{2}{15}x^5 - \frac{4}{315}x^7$

Chapter 9 Making Connections (Page 691)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

1. **(a)** $\dfrac{a\sin\theta}{1-\cos\theta}$ **(b)** $a\csc\theta$ **(c)** $a\cot\theta$

2. **(a)** $A = 1,\ B = -2$ **(b)** $s_n = 2 - \dfrac{2^{n+1}}{3^{n+1} - 2^{n+1}}$; 2

4. **(a)** $124.58 < d < 124.77$ **(b)** $1243 < s < 1424$

6. **(b)** $v(t) \approx v_0 - \left(\dfrac{cv_0}{m} + g\right)t + \dfrac{c^2}{2m^2}\left(v_0 + \dfrac{mg}{c}\right)t^2$

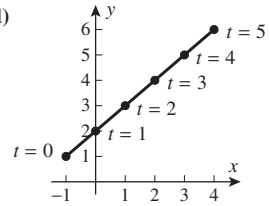
Exercise Set 10.1 (Page 700)

1. **(a)** $y = x + 2\ (-1 \le x \le 4)$

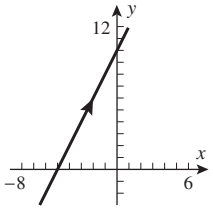
(c)

t	0	1	2	3	4	5
x	−1	0	1	2	3	4
y	1	2	3	4	5	6

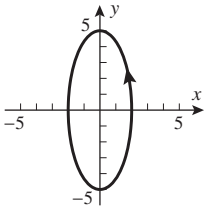
(d)



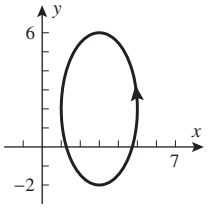
3.



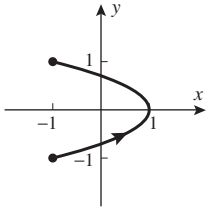
5.



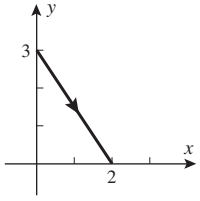
7.



9.



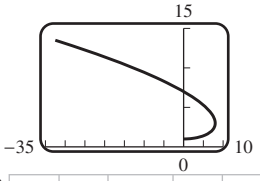
11.



13. $x = 5\cos t,\ y = -5\sin t\ (0 \le t \le 2\pi)$ **15.** $x = 2,\ y = t$

17. $x = t^2,\ y = t\ (-1 \le t \le 1)$

19. **(a)**

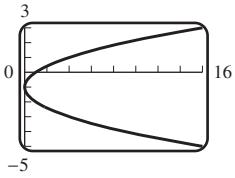


(b)

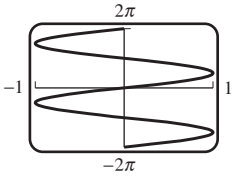
t	0	1	2	3	4	5
x	0	5.5	8	4.5	−8	−32.5
y	1	1.5	3	5.5	9	13.5

(c) $t = 0, 2\sqrt{3}$ **(d)** $0 < t < 2\sqrt{2}$ **(e)** $t = 2$

21. **(a)**



(b)

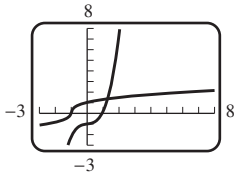


23. **(a)** II **(b)** I **(c)** IV **(d)** III **(e)** VI **(f)** V **25.** **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{3}$

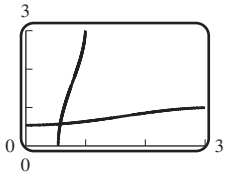
27. **(b)** from (x_0, y_0) to (x_1, y_1)

(c) $x = 3 - 2(t-1),\ y = -1 + 5(t-1)$

29.



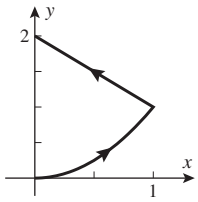
31.



Responses to True–False questions may be abridged to save space.

33. True; $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} = \dfrac{12t^3 - 6t^2}{x'(t)}$

35. False; $x = \sin t,\ y = \cos^2 t$ describe only the portion of the parabola $y = 1 - x^2$ with $-1 \le x \le 1$. **37.**



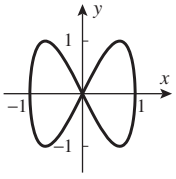
39. **(a)** $x = 4\cos t,\ y = 3\sin t$ **(b)** $x = -1 + 4\cos t,\ y = 2 + 3\sin t$

41. $-4, 4$ **43.** both are positive **45.** $4, 4$ **47.** $\sqrt{2}, -1$

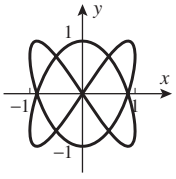
49. $\dfrac{\sqrt{2}(2+\sqrt{2})}{2},\ \dfrac{2}{3-2\sqrt{2}}$ **51.** $y = -e^{-2}x + 2e^{-1}$

53. **(a)** $0, \pi, 2\pi$ **(b)** $\pi/2, 3\pi/2$

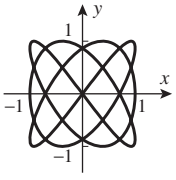
55. **(a)** **(b)** $y = -2x,\ y = 2x$



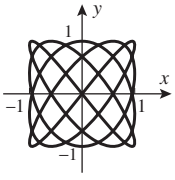
$a = 1,\ b = 2$



$a = 2,\ b = 3$

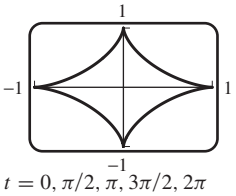


$a = 3,\ b = 4$



$a = 4,\ b = 5$

57. $y = 3x - 27,\ y = -3x + 27$ **59.**



$t = 0, \pi/2, \pi, 3\pi/2, 2\pi$

61. **(a)** $\dfrac{dy}{dx} = \dfrac{3\sin t}{1 - 3\cos t}$ **(b)** $\theta \approx -0.4345$

63. **(a)** ellipses with fixed center, varying axes of symmetry

(b) ellipses with varying center, fixed shape, size, and orientation

(c) circles of radius 1 with centers on line $y = x - 1$

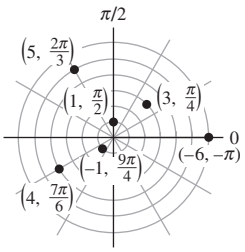
65. $\frac{1}{3}(5\sqrt{5} - 8)$ **67.** 3π **69.** $\frac{\sqrt{10}}{2}(e^2 - e^{-2})$

73. **(b)** $x = \cos t + \cos 2t,\ y = \sin t + \sin 2t$ **(c)** yes

75. $S = \dfrac{205\pi\sqrt{41} - 625}{6}$ **77.** $S = \sqrt{2}\pi$

Exercise Set 10.2 (Page 716)

1.



3. **(a)** $(2\sqrt{3}, 2)$ **(b)** $(-4, 4\sqrt{3})$ **(c)** $(3\sqrt{3}, 3)$ **(d)** $(0, 0)$ **(e)** $(-7\sqrt{3}/2, 7/2)$ **(f)** $(-5, 0)$

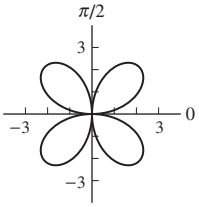
5. **(a)** $(7, \pi), (7, -\pi)$ **(b)** $(4, 11\pi/6), (4, -\pi/6)$ **(c)** $(2, 3\pi/2), (2, -\pi/2)$ **(d)** $(8\sqrt{2}, 5\pi/4), (8\sqrt{2}, -3\pi/4)$ **(e)** $(6, 2\pi/3), (6, -4\pi/3)$ **(f)** $(2\sqrt{2}, \pi/4), (2\sqrt{2}, -7\pi/4)$

7. **(a)** $(5, 0.92730)$ **(b)** $(10, 2.214297)$ **(c)** $(1.27155, 2.47582)$

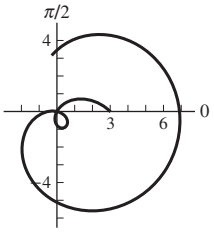
9. **(a)** circle **(b)** line **(c)** circle **(d)** line

11. **(a)** $r = 3\sec\theta$ **(b)** $r = \sqrt{14}$ **(c)** $r = -6\sin\theta$ **(d)** $r^2\cos\theta\sin\theta = 4/9$

13.



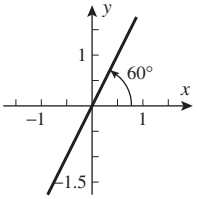
15.



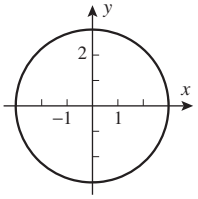
17. **(a)** $r = 5$ **(b)** $r = 6\cos\theta$ **(c)** $r = 1 - \cos\theta$

19. **(a)** $r = 3\sin 2\theta$ **(b)** $r = 3 + 2\sin\theta$ **(c)** $r^2 = 9\cos 2\theta$

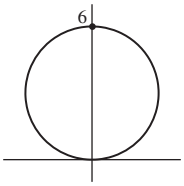
21.



23.

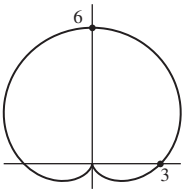


25.



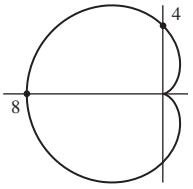
Circle

27.



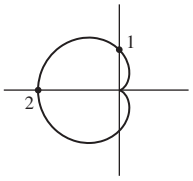
Cardioid

29.



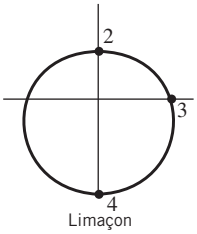
Cardioid

31.



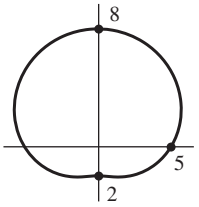
Cardioid

33.



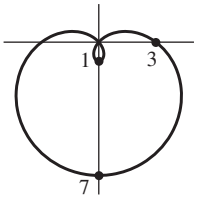
Limaçon

35.



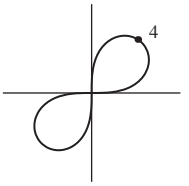
Limaçon

37.



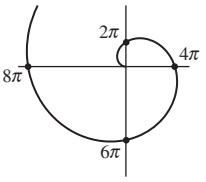
Limaçon

39.



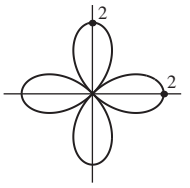
Lemniscate

41.



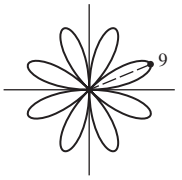
Spiral

43.



Four-petal rose

45.



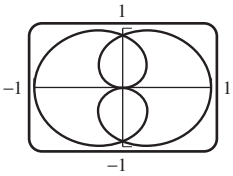
Eight-petal rose

Responses to True–False questions may be abridged to save space.

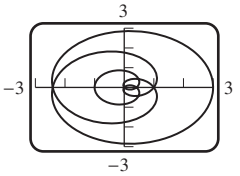
47. True; $\left(-1, \frac{\pi}{3}\right)$ describes the same point as $\left(1, \frac{\pi}{3} + \pi\right)$, which describes the same point as $\left(1, \frac{\pi}{3} + \pi - 2\pi\right) = \left(1, -\frac{2\pi}{3}\right)$.

49. False; $-1 < \sin 2\theta < 0$ for $\pi/2 < \theta < \pi$, so this portion of the graph is in the fourth quadrant.

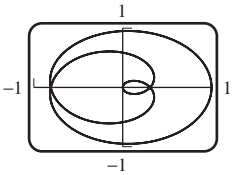
51. $0 \le \theta \le 4\pi$



53. $0 \le \theta \le 8\pi$

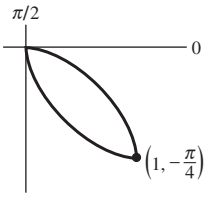


55. $0 \le \theta \le 5\pi$

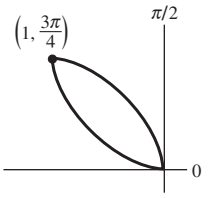


57. **(a)** $-4\pi < \theta < 4\pi$

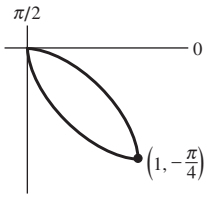
61. **(a)**



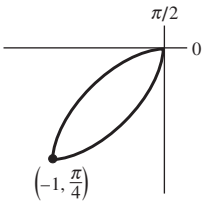
(b)



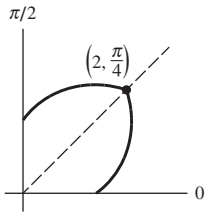
(c)



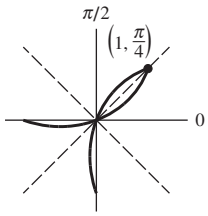
(d)



63. **(a)**



(b)



67. **(a)** $1 + \frac{\sqrt{2}}{2}(\sin\theta - \cos\theta)$ **(b)** $1 - \cos\theta$ **(c)** $1 - \sin\theta$ **(d)** $1 + \frac{\sqrt{2}}{2}(\cos\theta - \sin\theta)$

69. $(3/2, \pi/3)$ **73.** $\sqrt{2}$

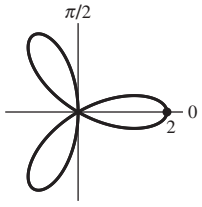
Exercise Set 10.3 (Page 726)

1. $-\sqrt{3}$ **3.** $\dfrac{\tan 3 - 3}{3\tan 3 + 1}$ **5.** $-\sqrt{3}$ **7.** $1, 0, -1$

9. horizontal: $(3a/2, \pi/3)$, $(0, \pi)$, $(3a/2, 5\pi/3)$;
vertical: $(2a, 0)$, $(a/2, 2\pi/3)$, $(a/2, 4\pi/3)$

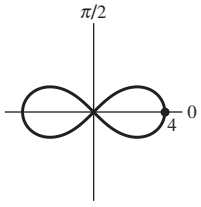
11. $(0, 0)$, $(\sqrt{2}/4, \pi/4)$, $(\sqrt{2}/4, 3\pi/4)$

13.



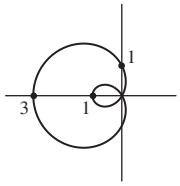
$\theta = \pi/2, \pm\pi/6$

15.



$\theta = \pm\pi/4$

17.



$\theta = \pm\pi/3$

19. $L = 2\pi a$ **21.** $L = 8a$

23. **(b)** ≈ 2.42

(c)

n	2	3	4	5	6	7
L	2.42211	2.22748	2.14461	2.10100	2.07501	2.05816

n	8	9	10	11	12	13	14
L	2.04656	2.03821	2.03199	2.02721	2.02346	2.02046	2.01802

n	15	16	17	18	19	20
L	2.01600	2.01431	2.01288	2.01167	2.01062	2.00971

25. **(a)** $\int_{\pi/2}^{\pi} \frac{1}{2}(1-\cos\theta)^2 d\theta$ **(b)** $\int_0^{\pi/2} 2\cos^2\theta\, d\theta$
(c) $\int_0^{\pi/2} \frac{1}{2}\sin^2 2\theta\, d\theta$ **(d)** $\int_0^{2\pi} \frac{1}{2}\theta^2\, d\theta$
(e) $\int_{-\pi/2}^{\pi/2} \frac{1}{2}(1-\sin\theta)^2\, d\theta$ **(f)** $\int_0^{\pi/4} \cos^2 2\theta\, d\theta$

27. **(a)** $\frac{9}{4}\pi a^2$ **(b)** $\frac{9}{4}\pi a^2$ **29.** 6π **31.** 9π **33.** $\pi - 3\sqrt{3}/2$
35. $\pi/2 - \frac{1}{4}$ **37.** $10\pi/3 - 4\sqrt{3}$ **39.** π **41.** $9\sqrt{3}/2 - 10\pi$
43. $(\pi + 3\sqrt{3})/4$ **45.** $\pi - 2$

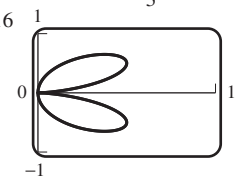
Responses to True–False questions may be abridged to save space.

47. True; apply Theorem 10.3.1: $\cos\frac{\theta}{2}\Big|_{\theta=3\pi} = 0$ and $\frac{dr}{d\theta}\Big|_{\theta=3\pi} = \frac{1}{2} \neq 0$, so the line $\theta = 3\pi$ (the x-axis) is tangent to the curve at the origin.

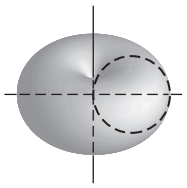
49. False; the area of the sector is $\frac{\theta}{2\pi}\cdot\pi r^2 = \frac{1}{2}\theta r^2$.

51. **(b)** a^2 **(c)** $2\sqrt{3} - \frac{2\pi}{3}$ **53.** $8\pi^3 a^2$

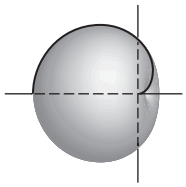
59. $\pi/16$



65. π^2



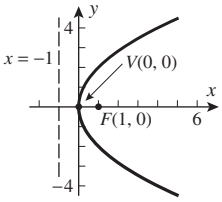
67. $32\pi/5$



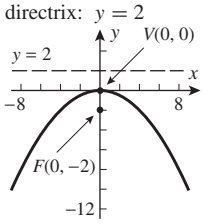
Exercise Set 10.4 (Page 744)

1. **(a)** $x = y^2$ **(b)** $-3y = x^2$ **(c)** $\frac{x^2}{9} + \frac{y^2}{4} = 1$ **(d)** $\frac{x^2}{4} + \frac{y^2}{9} = 1$
(e) $y^2 - x^2 = 1$ **(f)** $\frac{x^2}{4} - \frac{y^2}{4} = 1$

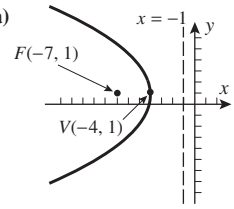
3. **(a)** focus: $(1, 0)$;
vertex: $(0, 0)$;
directrix: $x = -1$



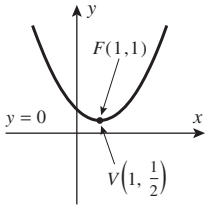
(b) focus: $(0, -2)$;
vertex: $(0, 0)$;
directrix: $y = 2$



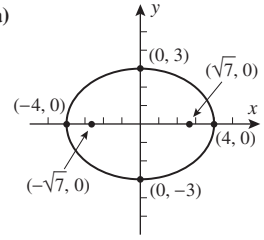
5. **(a)**



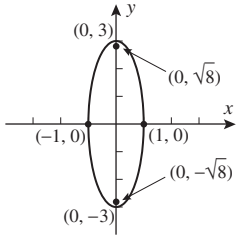
(b)



7. **(a)**

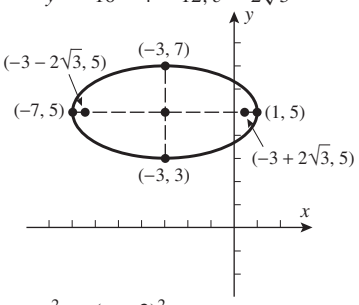


(b)



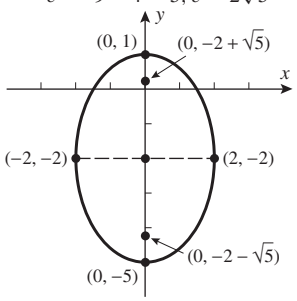
9. **(a)** $\dfrac{(x+3)^2}{16} + \dfrac{(y-5)^2}{4} = 1$

$y^2 = 16 - 4 = 12,\ c = 2\sqrt{3}$

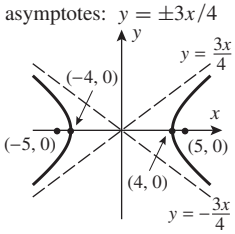


(b) $\dfrac{x^2}{4} + \dfrac{(y+2)^2}{9} = 1$

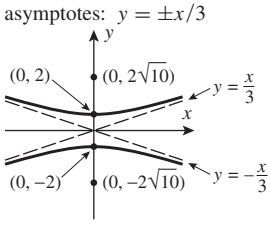
$c^2 = 9 - 4 = 5,\ c = 2\sqrt{5}$



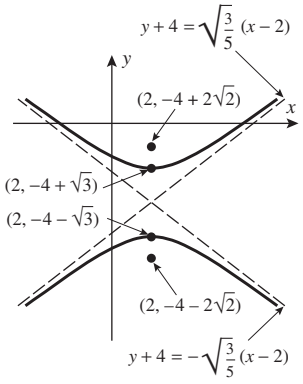
11. **(a)** vertices: $(\pm 4, 0)$;
foci: $(\pm 5, 0)$;
asymptotes: $y = \pm 3x/4$



(b) vertices: $(0, \pm 2)$;
foci: $(0, \pm 2\sqrt{10})$;
asymptotes: $y = \pm x/3$

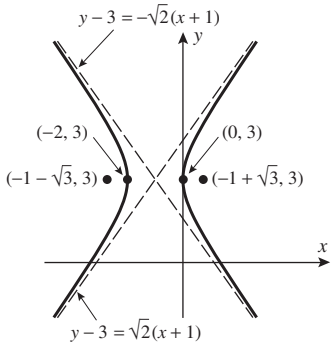


13. **(a)** $c^2 = 3 + 5 = 8,\ c = 2\sqrt{2}$



(b) $\dfrac{(x+1)^2}{1} - \dfrac{(y-3)^2}{2} = 1$

$c^2 = 1 + 2 = 3,\ c = \sqrt{3}$



15. **(a)** $y^2 = 20x$ **(b)** $x^2 = -2y$ **17.** $y^2 = x - 2$

19. **(a)** $\dfrac{x^2}{25} + \dfrac{y^2}{4} = 1$ **(b)** $\dfrac{x^2}{16} + \dfrac{y^2}{25} = 1$

21. **(a)** $\dfrac{x^2}{81/8} + \dfrac{y^2}{36} = 1$ **(b)** $\dfrac{(x+1)^2}{9} + \dfrac{(y-2)^2}{10} = 1$

23. **(a)** $\dfrac{x^2}{4} - \dfrac{y^2}{5} = 1$ **(b)** $\dfrac{y^2}{4} - \dfrac{x^2}{9} = 1$

25. **(a)** $\dfrac{y^2}{9} - \dfrac{x^2}{16} = 1,\ \dfrac{x^2}{16} - \dfrac{y^2}{9} = 1$ **(b)** $\dfrac{x^2}{9/5} - \dfrac{y^2}{36/5} = 1$

Responses to True–False questions may be abridged to save space.

27. False; the distance from the parabola's focus to its directrix is $2p$; see Figure 10.4.6.

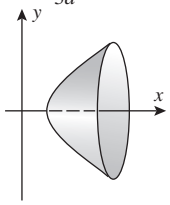
29. False; the description matches a parabola.

31. **(a)** 16 ft **(b)** $8\sqrt{3}$ ft **35.** $\frac{1}{16}$ ft

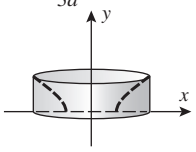
39. $\frac{1}{32}(x-5)^2 + \frac{1}{36}(y-3)^2 = 1$

43. $L = D\sqrt{1+p^2},\ T = \frac{1}{2}pD$ **45.** (64.612, 200)

47. **(a)** $V = \dfrac{\pi b^2}{3a^2}(b^2 - 2a^2)\sqrt{a^2+b^2} + \dfrac{2}{3}ab^2\pi$



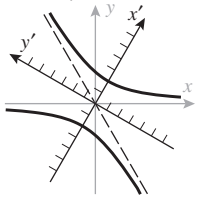
(b) $V = \dfrac{2b^4}{3a}\pi$



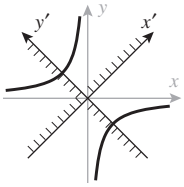
53. **(a)** $(x-2)^2 - 5(y+1)^2 = 8$, hyperbola
(b) $x^2 - 3(y+1)^2 = 0,\ x = \pm\sqrt{3}(y+1)$, two lines
(c) $4(x+2)^2 + 8(y+1)^2 = 8$, ellipse
(d) $3(x+2)^2 + (y+1)^2 = 0$, the point $(-2, -1)$ (degenerate case)
(e) $(x+4)^2 + 2y = 2$, parabola
(f) $5(x+4)^2 + 2y = -14$, parabola

Exercise Set 10.5 (Page 753)

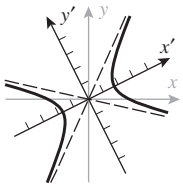
1. (a) $x' = -2 + \sqrt{3}$, $y' = 1 + 2\sqrt{3}$ (b) $3x'^2 - y'^2 = 12$ (c)



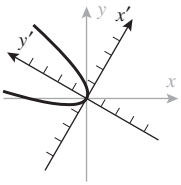
3. $y'^2 - x'^2 = 18$, hyperbola



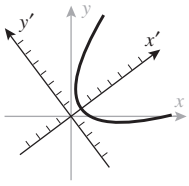
5. $\frac{1}{3}x'^2 - \frac{1}{2}y'^2 = 1$, hyperbola



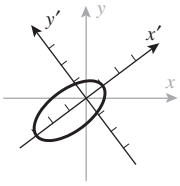
7. $y' = x'^2$, parabola



9. $y'^2 = 4(x' - 1)$, parabola



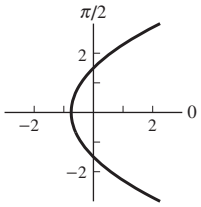
11. $\frac{1}{4}(x' + 1)^2 + y'^2 = 1$, ellipse

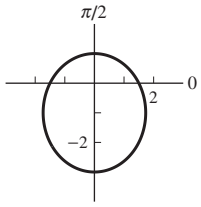


13. $x^2 + xy + y^2 = 5$
19. vertex: $(0, 0)$; focus: $(-1/\sqrt{2}, 1/\sqrt{2})$; directrix: $y = x - \sqrt{2}$
21. vertex: $(8/5, 6/5)$; focus: $(12/5, 9/5)$; directrix: $4x + 3y = 5$
23. foci: $\pm(4\sqrt{7}/5, 3\sqrt{7}/5)$; vertices: $\pm(16/5, 12/5)$; ends: $\pm(-9/5, 12/5)$
25. foci: $(1 - \sqrt{5}/2, -\sqrt{3} + \sqrt{15}/2)$, $(1 + \sqrt{5}/2, -\sqrt{3} - \sqrt{15}/2)$; vertices: $(-1/2, \sqrt{3}/2)$, $(5/2, -5\sqrt{3}/2)$; ends: $(1 + \sqrt{3}, 1 - \sqrt{3})$, $(1 - \sqrt{3}, -1 - \sqrt{3})$
27. foci: $\pm(\sqrt{15}, \sqrt{5})$; vertices: $\pm(2\sqrt{3}, 2)$; asymptotes: $y = \dfrac{5\sqrt{3} \pm 8}{11}x$
29. foci: $\left(-\dfrac{4}{\sqrt{5}} \pm 2\sqrt{\dfrac{13}{5}}, \dfrac{8}{\sqrt{5}} \pm \sqrt{\dfrac{13}{5}}\right)$; vertices: $(2/\sqrt{5}, 11/\sqrt{5})$, $(-2\sqrt{5}, \sqrt{5})$; asymptotes: $y = 7x/4 + 3\sqrt{5}$, $y = -x/8 + 3\sqrt{5}/2$

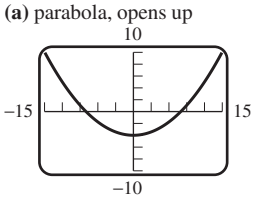
Exercise Set 10.6 (Page 761)

1. (a) $e = 1, d = \frac{3}{2}$ (b) $e = \frac{1}{2}, d = 3$

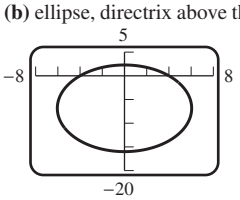




3. (a) parabola, opens up



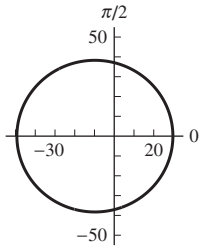
(b) ellipse, directrix above the pole



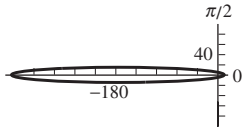
5. (a) $r = \dfrac{12}{4 + 3\cos\theta}$ (b) $r = \dfrac{1}{1 + \cos\theta}$ (c) $r = \dfrac{12}{1 + 2\sin\theta}$
7. (a) $r_0 = 4, r_1 = 12$; $\frac{x^2}{48} + \frac{(y+4)^2}{64} = 1$
(b) $r_0 = \frac{1}{3}, r_1 = 1$; $\frac{9}{4}\left(x - \frac{1}{3}\right)^2 + 3y^2 = 1$
9. (a) $r_0 = 1, r_1 = 3$; $(y - 2)^2 - \dfrac{x^2}{3} = 1$
(b) $r_0 = 1, r_1 = 5$; $\dfrac{(x + 3)^2}{4} - \dfrac{y^2}{5} = 1$
11. (a) $r = \dfrac{15}{2 + \cos\theta}$ (b) $r = \dfrac{64}{25 - 15\sin\theta}$
13. $r = \dfrac{5\sqrt{2} + 5}{1 + \sqrt{2}\cos\theta}$ or $r = \dfrac{5\sqrt{2} - 5}{1 + \sqrt{2}\cos\theta}$

Responses to True–False questions may be abridged to save space.

19. True; the eccentricity e of an ellipse satisfies $0 < e < 1$ (Theorem 10.6.1).
21. False; eccentricity correlates to the "flatness" of an ellipse, which is independent of the distance between its foci.
23. (a) $T \approx 248$ yr
(b) $r_0 \approx 4{,}449{,}675{,}000$ km, $r_1 \approx 7{,}400{,}325{,}000$ km
(c) $r \approx \dfrac{37.05}{1 + 0.249\cos\theta}$ AU (d)



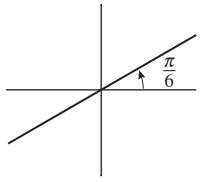
25. (a) $a \approx 178.26$ AU
(b) $r_0 \approx 0.8735$ AU, $r_1 \approx 355.64$ AU
(c) $r \approx \dfrac{1.74}{1 + 0.9951\cos\theta}$ AU (d)



27. 563 km, 4286 km

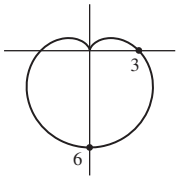
Chapter 10 Review Exercises (Page 763)

1. $x = \sqrt{5}\cos t$, $y = -\sqrt{5}\sin t$ $(0 \le t \le 3\pi/2)$
3. (a) $1/6, 1/6$
5. (a) $t = \pi/2 + n\pi$ for $n = 0, \pm1, \ldots$ (b) $t = n\pi$ for $n = 0, \pm1, \ldots$
7. (a) $(3\sqrt{2}, 3\sqrt{2})$ (b) $(7/\sqrt{2}, -7/\sqrt{2})$ (c) $(4\sqrt{2}, 4\sqrt{2})$ (d) $(7, 0)$ (e) $(0, -2)$ (f) $(0, 0)$
9. (a) $(5, 0.6435)$ (b) $(\sqrt{29}, 5.0929)$ (c) $(1.2716, 0.6658)$
11. (a) parabola (b) hyperbola (c) line (d) circle
13.



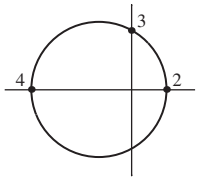
Line

15.



Cardioid

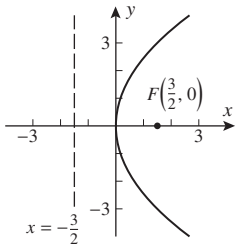
17.



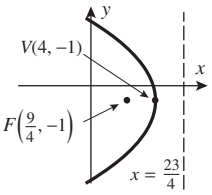
Limaçon

19. (a) $-2, 1/4$ (b) $-3\sqrt{3}/4, 3\sqrt{3}/4$
21. (a) The top is traced from right to left as t goes from 0 to π. The bottom is traced from right to left as t goes from π to 2π, except for the loop, which is traced counterclockwise as t goes from $\pi + \sin^{-1}(1/4)$ to $2\pi - \sin^{-1}(1/4)$. (b) $y = 1$
(c) horizontal: $t = \pi/2, 3\pi/2$; vertical: $t = \pi + \sin^{-1}(1/\sqrt[3]{4})$, $2\pi - \sin^{-1}(1/\sqrt[3]{4})$
(d) $r = 4 + \csc\theta$, $\theta = \pi + \sin^{-1}(1/4)$, $\theta = 2\pi - \sin^{-1}(1/4)$

23. $A = 6\pi$ **25.** $A = \dfrac{5\pi}{12} - \dfrac{\sqrt{3}}{2}$ **27.**

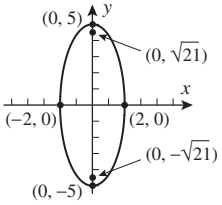


29.



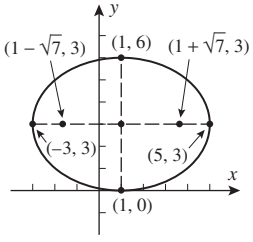
focus: $(9/4, -1)$;
vertex: $(4, -1)$;
directrix: $x = 23/4$

31.

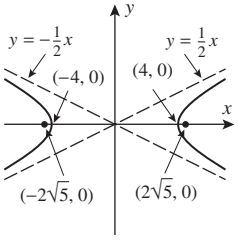


foci: $(0, \pm\sqrt{21})$;
vertices: $(0, \pm 5)$;
ends: $(\pm 2, 0)$

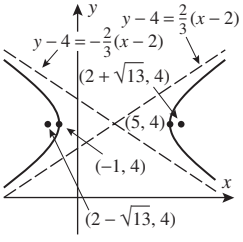
33.



35.



37.



39. $x^2 = -20y$ **41.** $y^2 - x^2 = 4$

43. **(b)** $x = \dfrac{v_0^2}{g}\sin\alpha\cos\alpha$; $y = y_0 + \dfrac{v_0^2\sin^2\alpha}{2g}$

45. $\theta = \pi/4$; $5(y')^2 - (x')^2 = 8$; hyperbola

47. $\theta = \tan^{-1}(1/2)$; $y' = (x')^2$; parabola

49. **(a)** (i) ellipse; (ii) right; (iii) 1 **(b)** (i) hyperbola (ii) left; (iii) 1/3 **(c)** (i) parabola; (ii) above; (iii) 1/3 **(d)** (i) parabola; (ii) above; (iii) 3

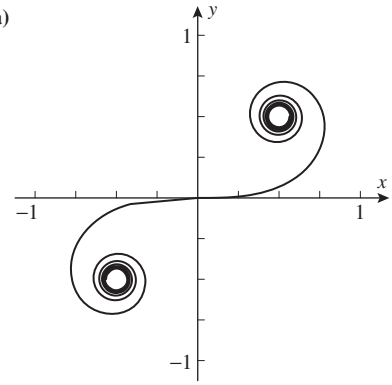
51. **(a)** $\dfrac{(x+3)^2}{25} + \dfrac{(y-2)^2}{9} = 1$ **(b)** $(x+2)^2 = -8y$
(c) $\dfrac{(y-5)^2}{4} - 16(x+1)^2 = 1$

53. 15.86543959

▶ Chapter 10 Making Connections (Page 766)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

1. **(a)**



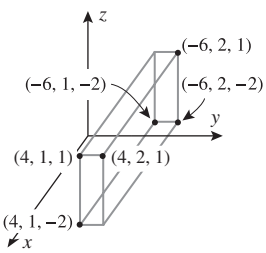
(c) $L = \displaystyle\int_{-1}^{1}\left[\cos^2\left(\frac{\pi t^2}{2}\right) + \sin^2\left(\frac{\pi t^2}{2}\right)\right]dt = 2$

2. **(a)** P: $(b\cos t, b\sin t)$;
Q: $(a\cos t, a\sin t)$;
R: $(a\cos t, b\sin t)$

▶ Exercise Set 11.1 (Page 771)

1. **(a)** (0, 0, 0), (3, 0, 0), (3, 5, 0), (0, 5, 0), (0, 0, 4), (3, 0, 4), (3, 5, 4), (0, 5, 4)
(b) (0, 1, 0), (4, 1, 0), (4, 6, 0), (0, 6, 0), (0, 1, −2), (4, 1, −2), (4, 6, −2), (0, 6, −2)

3. (4, 2, −2), (4, 2, 1), (4, 1, 1), (4, 1, −2), (−6, 1, 1), (−6, 2, 1), (−6, 2, −2), (−6, 1, −2)



5. **(a)** point **(b)** line parallel to the y-axis
(c) plane parallel to the yz-plane

9. radius $\sqrt{59}$, center (2, 1, −5) **11.** **(b)** (2, 1, 6) **(c)** area 147/2

13. **(a)** $(x-1)^2 + y^2 + (z+1)^2 = 9$
(b) $(x+1)^2 + (y-3)^2 + (z-2)^2 = 19$
(c) $\left(x + \frac{1}{2}\right)^2 + (y-2)^2 + (z-2)^2 = \frac{5}{4}$

15. $(x-2)^2 + (y-1)^2 + (z+3)^2 = r^2$;
(a) $r^2 = 9$ **(b)** $r^2 = 1$ **(c)** $r^2 = 4$

Responses to True–False questions may be abridged to save space.

19. False; see Figure 11.1.6.

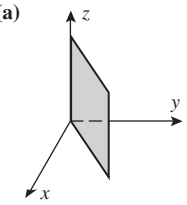
21. True; see Figure 11.1.3.

23. sphere, center (−5, −2, −1), radius 8

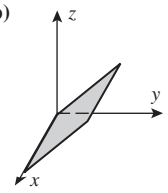
25. sphere; center $\left(\frac{1}{2}, \frac{3}{4}, -\frac{5}{4}\right)$, radius $\dfrac{3\sqrt{6}}{4}$

27. no graph

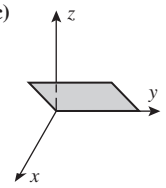
29. **(a)**



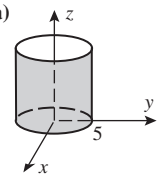
(b)



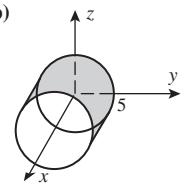
(c)



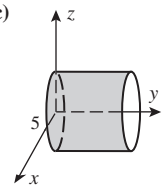
31. **(a)**



(b)

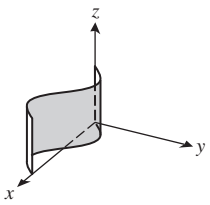


(c)

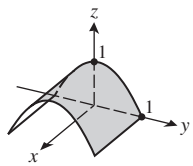


33. **(a)** $-3y + z = 0$ **(b)** $-2x + z = 0$ **(c)** $(x-1)^2 + (y-1)^2 = 4$ **(d)** $(x-1)^2 + (z-1)^2 = 1$

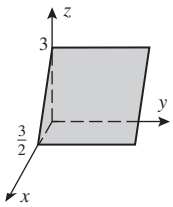
35.



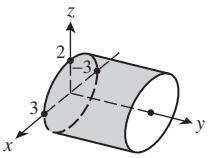
37.



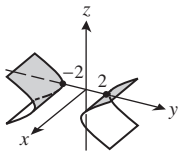
39.



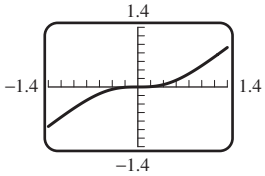
41.

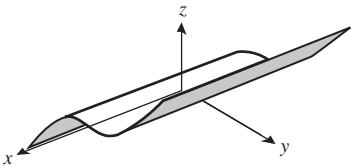


43.



45.





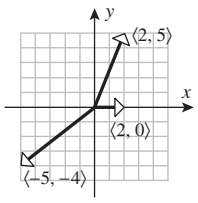
47. largest distance, $4 + \sqrt{6}$; smallest distance, $4 - \sqrt{6}$

49. all points outside the circular cylinder $(y+3)^2 + (z-4)^2 = 36$

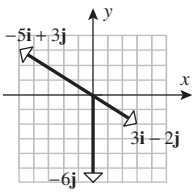
51. $r = (2 - \sqrt{3})R$ **53.** **(b)** $y^2 + z^2 = e^{2x}$

Exercise Set 11.2 (Page 782)

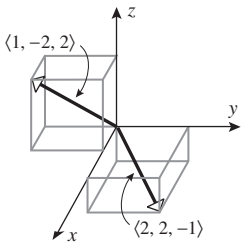
1. **(a-c)**



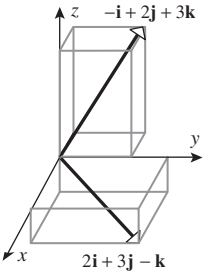
(d-f)



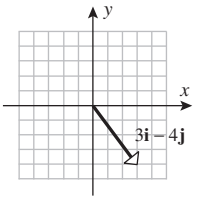
3. **(a,b)**

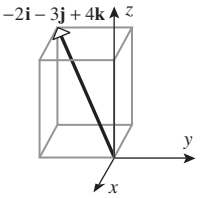


(c,d)



5. **(a)** $\langle 3, -4\rangle$ **(b)** $\langle -2, -3, 4\rangle$





7. **(a)** $\langle -1, 4\rangle$ **(b)** $\langle -7, 2\rangle$ **(c)** $\langle -2, 4, 2\rangle$

9. **(a)** $(5, -1)$ **(b)** $(7, -3, -5)$

11. **(a)** $-\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$ **(b)** $21\mathbf{i} + 9\mathbf{j} - 7\mathbf{k}$ **(c)** $-\mathbf{i} - 5\mathbf{j} - 2\mathbf{k}$ **(d)** $39\mathbf{i} - 3\mathbf{j} - 6\mathbf{k}$ **(e)** $-2\mathbf{i} - 16\mathbf{j} - 18\mathbf{k}$ **(f)** $-\mathbf{i} + 13\mathbf{j} - 2\mathbf{k}$

13. **(a)** $\sqrt{2}$ **(b)** $2\sqrt{17}$ **(c)** $\sqrt{21}$ **(d)** $\sqrt{14}$

15. **(a)** $2\sqrt{3}$ **(b)** $\sqrt{14} + \sqrt{2}$ **(c)** $2\sqrt{14} + 2\sqrt{2}$ **(d)** $2\sqrt{37}$ **(e)** $(1/\sqrt{6})\mathbf{i} + (1/\sqrt{6})\mathbf{j} - (2/\sqrt{6})\mathbf{k}$ **(f)** 1

Responses to True–False questions may be abridged to save space.

17. False; $\|\mathbf{i} + \mathbf{j}\| = \sqrt{2} \neq 1 + 1 = 2$

19. True; one in the same direction and one in the opposite direction.

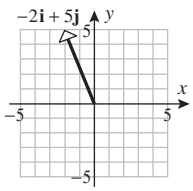
21. **(a)** $(-2/\sqrt{29})\mathbf{i} + (5/\sqrt{29})\mathbf{j}$ **(b)** $(-3\mathbf{i} + 2\mathbf{j} - \mathbf{k})/\sqrt{14}$ **(c)** $(4\mathbf{i} + \mathbf{j} - \mathbf{k})/(3\sqrt{2})$

23. **(a)** $\langle -\frac{3}{2}, 2\rangle$ **(b)** $\dfrac{1}{\sqrt{5}}\langle 7, 0, -6\rangle$

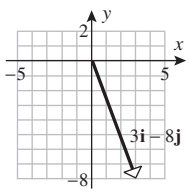
25. **(a)** $\langle 2\sqrt{2}, 2\sqrt{2}\rangle$ **(b)** $\langle 0, 2\rangle$ **(c)** $\langle -5/2, 5\sqrt{3}/2\rangle$ **(d)** $\langle \sqrt{3}/2, 1/2\rangle$

27. $\langle (\sqrt{3} - \sqrt{2})/2, (1 + \sqrt{2})/2\rangle$

29. **(a)** $\langle -2, 5\rangle$



(b) $\langle 3, -8\rangle$



31. $\langle -2, 3\rangle$ **33.** $\mathbf{u} = \frac{5}{7}\mathbf{i} + \frac{2}{7}\mathbf{j} + \frac{1}{7}\mathbf{k}$, $\mathbf{v} = \frac{8}{7}\mathbf{i} - \frac{1}{7}\mathbf{j} - \frac{4}{7}\mathbf{k}$

35. $\sqrt{5}, 3$ **37.** **(a)** $\pm\frac{5}{3}$ **(b)** 2

39. **(a)** $\langle 1/\sqrt{10}, 3/\sqrt{10}\rangle, \langle -1/\sqrt{10}, -3/\sqrt{10}\rangle$ **(b)** $\langle 1/\sqrt{2}, -1/\sqrt{2}\rangle, \langle -1/\sqrt{2}, 1/\sqrt{2}\rangle$ **(c)** $\pm\dfrac{1}{\sqrt{26}}\langle 5, 1\rangle$

41. **(a)** the circle of radius 1 about the origin
(b) the closed disk of radius 1 about the origin
(c) all points outside the closed disk of radius 1 about the origin

43. **(a)** the (hollow) sphere of radius 1 about the origin
(b) the closed ball of radius 1 about the origin
(c) all points outside the closed ball of radius 1 about the origin

45. magnitude $= 30\sqrt{5}$ lb, $\theta \approx 26.57°$

47. magnitude ≈ 207.06 N, $\theta = 45°$

49. magnitude ≈ 94.995 N, $\theta \approx 28.28°$

51. magnitude ≈ 9.165 lb, angle $\approx -70.890°$

53. ≈ 183.02 lb, 224.13 lb

55. **(a)** $c_1 = 1, c_2 = \frac{1}{2}$

Exercise Set 11.3 (Page 792)

1. **(a)** -5; $\cos\theta = -1/\sqrt{5}$ **(b)** -3; $\cos\theta = -3/\sqrt{58}$ **(c)** 0; $\cos\theta = 0$ **(d)** -20; $\cos\theta = -20/(3\sqrt{70})$

3. **(a)** obtuse **(b)** acute **(c)** obtuse **(d)** orthogonal

5. $\sqrt{2}/2, 0, -\sqrt{2}/2, -1, -\sqrt{2}/2, 0, \sqrt{2}/2$

7. **(a)** vertex B **(b)** $82°, 60°, 38°$ **13.** $r = 7/5$

15. **(a)** $\alpha = \beta \approx 55°, \gamma \approx 125°$ **(b)** $\alpha \approx 48°, \beta \approx 132°, \gamma \approx 71°$

19. **(a)** $\approx 35°$ **(b)** $90°$

21. $64°, 41°, 60°$ **23.** $68°, 56°, 42°$

25. **(a)** $\left\langle \frac{2}{3}, \frac{4}{3}, \frac{4}{3} \right\rangle, \left\langle \frac{4}{3}, -\frac{7}{3}, \frac{5}{3} \right\rangle$
(b) $\left\langle \frac{38}{17}, \frac{57}{17}, \frac{38}{17} \right\rangle, \left\langle \frac{30}{17}, -\frac{74}{17}, \frac{81}{17} \right\rangle$

27. **(a)** $\langle -2/5, -1/5 \rangle + \langle -13/5, 26/5 \rangle$ **(b)** $\left\langle 0, -\frac{8}{5}, \frac{4}{5} \right\rangle + \left\langle -2, \frac{13}{5}, \frac{26}{5} \right\rangle$
(c) $\mathbf{v} = \langle 1, 4, 1 \rangle$ is orthogonal to $\mathbf{b}$.

Responses to True–False questions may be abridged to save space.

29. True; $\mathbf{v} + \mathbf{w} = \mathbf{0}$ implies $0 = \mathbf{v} \cdot (\mathbf{v} + \mathbf{w}) = \|\mathbf{v}\|^2 \neq 0$, a contradiction.

31. True; see Equation (12). **33.** $\sqrt{564/29}$ **35.** 210 N

37. $-5\sqrt{3}$ J **45.** **(a)** $40°$ **(b)** $x \approx -0.682328$

Exercise Set 11.4 (Page 803)

1. **(a)** $-\mathbf{j} + \mathbf{k}$ **3.** $\langle 9, 9, 9 \rangle$ **5.** $\langle -3, -8, -4 \rangle$

7. **(a)** $\langle -20, -67, -9 \rangle$ **(b)** $\langle -78, 52, -26 \rangle$
(c) $\langle 0, -56, -392 \rangle$ **(d)** $\langle 0, 56, 392 \rangle$

9. $\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0$ **11.** $\pm\frac{1}{\sqrt{6}}\langle 2, 1, 1 \rangle$

Responses to True–False questions may be abridged to save space.

13. True; see Theorem 11.4.5(c).

15. False; let $\mathbf{v} = \mathbf{u} = \mathbf{i}$ and let $\mathbf{w} = 2\mathbf{i}$.

17. $\sqrt{30}$ **19.** $\sqrt{339}/2$

21. 80 **23.** -4 **25.** 25.20 **27.** **(a)** yes **(b)** yes **(c)** no

29. **(a)** 9 **(b)** $\sqrt{122}$ **(c)** $\sin^{-1}\left(\frac{9}{14}\right)$

31. **(a)** $2\sqrt{141/29}$ **(b)** $\sqrt{8}$ **33.** $\frac{2}{3}$ **37.** $\theta = \pi/4$

39. **(a)** $10\sqrt{2}$ lb·ft, direction of rotation about P is counterclockwise looking along $\overrightarrow{PQ} \times \mathbf{F} = -10\mathbf{i} + 10\mathbf{k}$ toward its initial point
(b) 10 lb·ft, direction of rotation about P is counterclockwise looking along $-10\mathbf{i}$ toward its initial point
(c) 0 lb·ft, no rotation about P

41. ≈ 36.19 N·m **45.** $-8\mathbf{i} - 20\mathbf{j} + 2\mathbf{k}, -8\mathbf{i} - 8\mathbf{k}$ **49.** 1.887850

Exercise Set 11.5 (Page 810)

1. **(a)** $L_1: x = 1, y = t$, $L_2: x = t, y = 1$, $L_3: x = t, y = t$
(b) $L_1: x = 1, y = 1, z = t$, $L_2: x = t, y = 1, z = 1$, $L_3: x = 1, y = t, z = 1$, $L_4: x = t, y = t, z = t$

3. **(a)** $x = 4 + t, y = -2 + 3t$; line segment: $0 \le t \le 1$
(b) $x = 5 - 3t, y = -2 + 6t, z = 1 + 2t$; line segment: $0 \le t \le 1$

5. **(a)** $x = 2 + t, y = -3 - 5t$ **(b)** $x = t, y = -t, z = 1 + 2t$

7. **(a)** $P(2, -1)$, $\mathbf{v} = 3\mathbf{i} + 2\mathbf{j}$ **(b)** $P(-1, 2, 4)$, $\mathbf{v} = 5\mathbf{i} + 7\mathbf{j} - 8\mathbf{k}$

9. **(a)** $\langle -3, 4 \rangle + t\langle 1, 6 \rangle$; $-3\mathbf{i} + 4\mathbf{j} + t(\mathbf{i} + 6\mathbf{j})$
(b) $\langle 2, -3, 0 \rangle + t\langle -1, 5, 1 \rangle$; $2\mathbf{i} - 3\mathbf{j} + t(-\mathbf{i} + 5\mathbf{j} + \mathbf{k})$

Responses to True–False questions may be abridged to save space.

11. False; the lines could be skew.

13. False; see part (b) of the solution to Example 3.

15. $x = -5 + 2t, y = 2 + 5t$ **17.** $x = 3 + 4t, y = -4 + 3t$

19. $x = -1 + 3t, y = 2 - 4t, z = 4 + 2t$

21. $x = -1 + 2t, y = -t, z = 5 + 3t$

23. **(a)** $x = 7$ **(b)** $y = \frac{7}{3}$ **(c)** $x = \dfrac{-1 \pm \sqrt{85}}{6}, y = \dfrac{43 \mp \sqrt{85}}{18}$

25. $(-2, 8, 0)$; $(-2, 0, -8)$; The line does not intersect the yz-plane.

27. $(0, 4, -2), (4, 0, 6)$ **29.** $(1, -1, 2)$ **33.** The lines are parallel.

35. The points do not lie on the same line.

39. $\langle x, y \rangle = \langle -1, 2 \rangle + t\langle 1, 1 \rangle$

41. the point $1/n$ of the way from $(-2, 0)$ to $(1, 3)$

43. the line segment joining the points $(1, 0)$ and $(-5, 9)$

45. $(5, 2)$ **47.** $2\sqrt{5}$ **49.** distance $= \sqrt{35/6}$

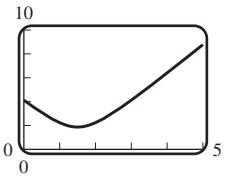
51. **(a)** $x = x_0 + (x_1 - x_0)t, y = y_0 + (y_1 - y_0)t, z = z_0 + (z_1 - z_0)t$
(b) $x = x_1 + at, y = y_1 + bt, z = z_1 + ct$

53. **(b)** $\langle x, y, z \rangle = \langle 1 + 2t, -3 + 4t, 5 + t \rangle$

55. **(b)** $84°$ **(c)** $x = 7 + t, y = -1, z = -2 + t$

57. $x = t, y = 2 + t, z = 1 - t$

59. **(a)** $\sqrt{17}$ cm **(b)**



(d) $\sqrt{14}/2$ cm

Exercise Set 11.6 (Page 819)

1. $x = 3, y = 4, z = 5$ **3.** $2x + 3y + 2z = 26$ **5.** $z = 0$

7. $x - y = 0$ **9.** $y + z = 1$ **11.** $2y - z = 1$

13. **(a)** parallel **(b)** perpendicular **(c)** neither

15. **(a)** parallel **(b)** neither **(c)** perpendicular

17. **(a)** point of intersection is $(3, 3, 3)$ **(b)** no intersection

19. $48°$

Responses to True–False questions may be abridged to save space.

21. True; each will be the negative of the other.

23. True; the direction vector of L must be orthogonal to both normal vectors.

25. $3x - 2y + 7z = 0$ **27.** $4x - 13y + 21z = -14$

29. $x + 2y - 5z = 10$ **31.** $x + 5y + 3z = -6$

33. $x + y + 2z = 9$ **35.** $x = 5 - 2t, y = 5t, z = -2 + 11t$

37. $8x + y + 10z = 28$ **39.** yes

41. $x = -\frac{11}{7} - 23t, y = -\frac{12}{7} + t, z = -7t$

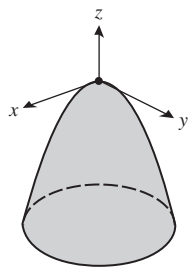
43. $\frac{4}{3}$ **45.** $5/\sqrt{54}$ **47.** $25/\sqrt{126}$

49. $(x - 2)^2 + (y - 1)^2 + (z + 3)^2 = \frac{121}{14}$ **51.** $5/\sqrt{12}$

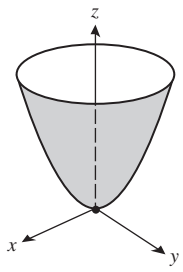
Exercise Set 11.7 (Page 830)

1. **(a)** elliptic paraboloid, $a = 2, b = 5$
(b) hyperbolic paraboloid, $a = 1, b = 5$
(c) hyperboloid of one sheet, $a = b = c = 6$
(d) circular cone, $a = b = 1$ **(e)** elliptic paraboloid, $a = 2, b = 1$
(f) hyperboloid of two sheets, $a = b = c = 1$

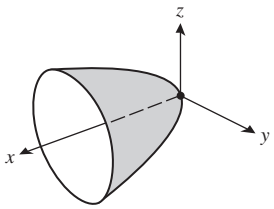
3. **(a)** $-z = x^2 + y^2$, circular paraboloid opening down the negative z-axis



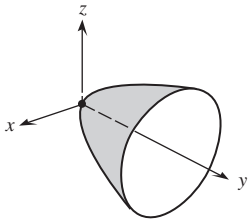
(b) $z = x^2 + y^2$, circular paraboloid, no change
(c) $z = x^2 + y^2$, circular paraboloid, no change
(d) $z = x^2 + y^2$, circular paraboloid, no change



(e) $x = y^2 + z^2$,
circular paraboloid opening along the positive x-axis

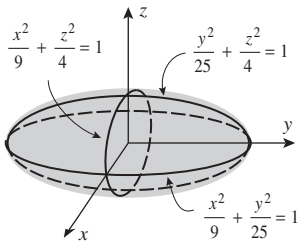


(f) $y = x^2 + z^2$,
circular paraboloid opening along the positive y-axis

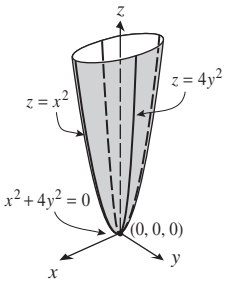


5. **(a)** hyperboloid of one sheet, axis is y-axis
(b) hyperboloid of two sheets separated by yz-plane
(c) elliptic paraboloid opening along the positive x-axis
(d) elliptic cone with x-axis as axis
(e) hyperbolic paraboloid straddling the x-axis
(f) paraboloid opening along the negative y-axis

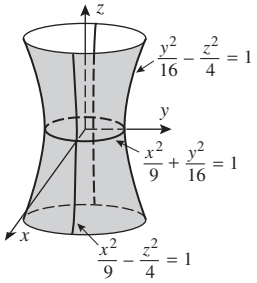
7. **(a)** $x = 0$: $\dfrac{y^2}{25} + \dfrac{z^2}{4} = 1$;
$y = 0$: $\dfrac{x^2}{9} + \dfrac{z^2}{4} = 1$;
$z = 0$: $\dfrac{x^2}{9} + \dfrac{y^2}{25} = 1$



(b) $x = 0$: $z = 4y^2$;
$y = 0$: $z = x^2$;
$z = 0$: $x = y = 0$



(c) $x = 0$: $\dfrac{y^2}{16} - \dfrac{z^2}{4} = 1$;
$y = 0$: $\dfrac{x^2}{9} - \dfrac{z^2}{4} = 1$;
$z = 0$: $\dfrac{x^2}{9} + \dfrac{y^2}{16} = 1$



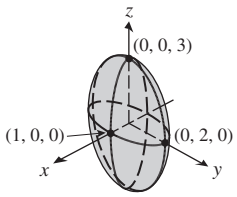
9. **(a)** $4x^2 + 2z^2 = 3$; ellipse **(b)** $y^2 + z^2 = 3$; circle
(c) $y^2 + z^2 = 128$; circle **(d)** $9x^2 - y^2 = 20$; hyperbola
(e) $z = 9x^2 + 16$; parabola **(f)** $9x^2 + 4y^2 = 4$; ellipse

Responses to True–False questions may be abridged to save space.

11. False; "quadric" refers to second powers.

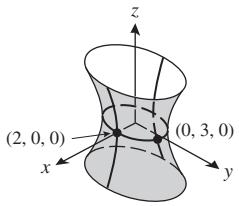
13. False; there need not exist a line such that all cross sections orthogonal to the line are circular.

15.



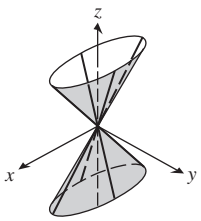
Ellipsoid

17.



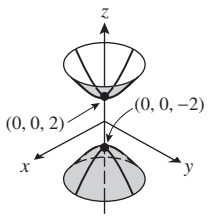
Hyperboloid of one sheet

19.



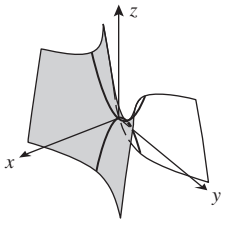
Elliptic cone

21.



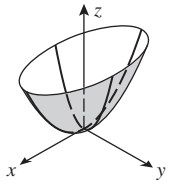
Hyperboloid of two sheets

23.



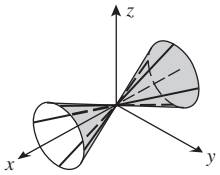
Hyperbolic paraboloid

25.



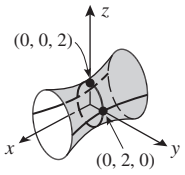
Elliptic paraboloid

27.



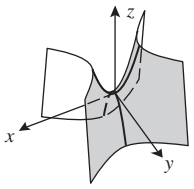
Circular cone

29.



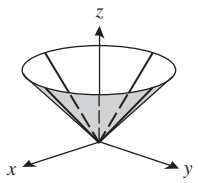
Hyperboloid of one sheet

31.

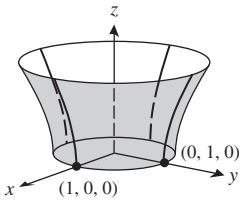


Hyperbolic paraboloid

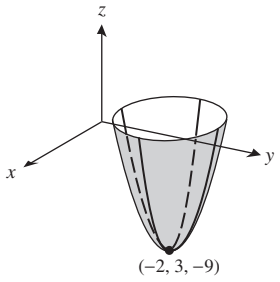
33.



35.

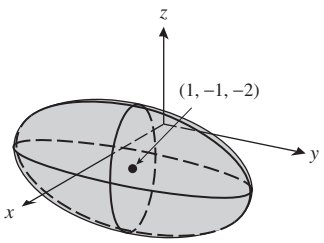


37.



Circular paraboloid

39.



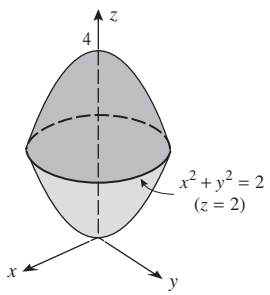
Ellipsoid

41. **(a)** $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ **(b)** 6, 4 **(c)** $(\pm\sqrt{5}, 0, \sqrt{2})$
(d) The focal axis is parallel to the x-axis.

43. **(a)** $\dfrac{y^2}{9} - \dfrac{x^2}{9} = 1$ **(b)** $(0, \pm 3, 9)$ **(c)** $(0, \pm 3\sqrt{2}, 9)$
(d) The focal axis is parallel to the y-axis.

45. **(a)** $z + 9 = y^2$ **(b)** $(3, 0, -9)$ **(c)** $\left(3, 0, -\frac{35}{4}\right)$
(d) The focal axis is parallel to the z-axis.

47. circle of radius $\sqrt{2}$ in the plane $z = 2$, centered at $(0, 0, 2)$



49. $y = 4(x^2 + z^2)$ **51.** $z = (x^2 + y^2)/8$ (circular paraboloid)

▶ Exercise Set 11.8 (Page 837)

1. **(a)** $(4, \pi/6, -4)$ **(b)** $(5\sqrt{2}, 3\pi/4, 6)$
(c) $(2, \pi/2, 0)$ **(d)** $(8, 5\pi/3, 6)$

3. **(a)** $(3, 3\sqrt{3}, 3)$ **(b)** $(-4\sqrt{2}, 4\sqrt{2}, -2)$
(c) $(5, 0, 4)$ **(d)** $(-7, 0, -9)$

5. **(a)** $(4\sqrt{2}, \pi/3, 3\pi/4)$ **(b)** $(2, 7\pi/4, \pi/4)$
(c) $(6, \pi/2, \pi/3)$ **(d)** $(8, 5\pi/6, \pi/2)$

7. **(a)** $(3, \sqrt{3}, 2)$ **(b)** $(7, 0, 0)$
(c) $(0, 0, 1)$ **(d)** $(0, -2, 0)$

9. **(a)** $(4\sqrt{3}, \pi/3, \pi/6)$ **(b)** $(\sqrt{2}, \pi/4, 3\pi/4)$
(c) $(2, 3\pi/4, \pi/2)$ **(d)** $(4\sqrt{3}, 1, 2\pi/3)$

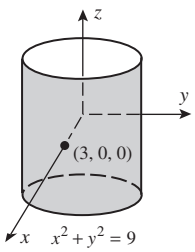
11. **(a)** $(5\sqrt{3}/2, \pi/4, 5/2)$ **(b)** $(0, 5\pi/6, -2)$
(c) $(0, 0, 3)$ **(d)** $(4, \pi/6, 0)$

Responses to True–False questions may be abridged to save space.

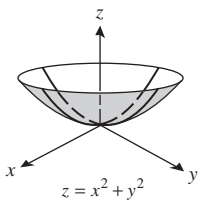
15. True; see Figure 11.8.1*b*.

17. True; see Figures 11.8.3 and 11.8.4.

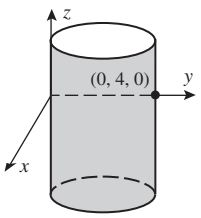
19.



21.

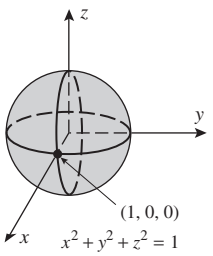


23.



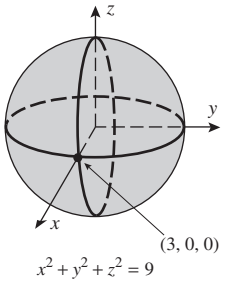
$x^2 + (y - 2)^2 = 4$

25.



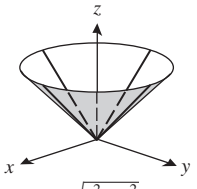
$x^2 + y^2 + z^2 = 1$

27.



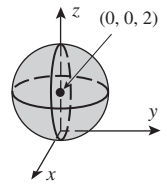
$x^2 + y^2 + z^2 = 9$

29.



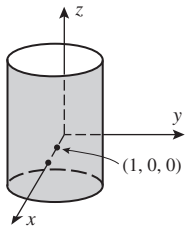
$z = \sqrt{x^2 + y^2}$

31.



$x^2 + y^2 + (z - 2)^2 = 4$

33.



$(x - 1)^2 + y^2 = 1$

35. **(a)** $z = 7$ **(b)** $\rho = 7 \sec\phi$ **37.** **(a)** $z = 3r^2$ **(b)** $\rho = \frac{1}{3}\csc\phi\cot\phi$

39. **(a)** $r = 3$ **(b)** $\rho = 3\csc\phi$ **41.** **(a)** $r^2 + z^2 = 4$ **(b)** $\rho = 4$

43. **(a)** $2r\cos\theta + 3r\sin\theta + 4z = 2$
(b) $2\rho\sin\phi\cos\theta + 3\rho\sin\phi\sin\theta + 4\rho\cos\phi = 2$

45. **(a)** $r^2\cos^2\theta = 16 - z^2$ **(b)** $\rho^2(1 - \sin^2\phi\sin^2\theta) = 16$

47. all points on or above the paraboloid $z = x^2 + y^2$ that are also on or below the plane $z = 4$

49. all points on or between concentric spheres of radii 2 and 5 centered at the origin

51. spherical: $(4000, \pi/6, \pi/6)$; rectangular: $(1000\sqrt{3}, 1000, 2000\sqrt{3})$

53. **(a)** $(10, \pi/2, 1)$ **(b)** $(0, 10, 1)$ **(c)** $(\sqrt{101}, \pi/2, \tan^{-1} 10)$

▶ Chapter 11 Review Exercises (Page 838)

3. **(b)** $-1/2, \pm\sqrt{3}/2$ **(d)** true

5. **(a)** $r^2 = 16$ **(b)** $r^2 = 9$ **(c)** $r^2 = 4$

7. $(8, 5)$

9. **(a)** $-\frac{3}{4}$ **(b)** $\frac{1}{7}$ **(c)** $(48 \pm 25\sqrt{3})/39$ **(d)** $c = \frac{4}{3}$

13. 13 ft·lb **15.** **(a)** $\frac{5}{2}\sqrt{2}$ **(b)** $\frac{5}{3}\sqrt{2}$

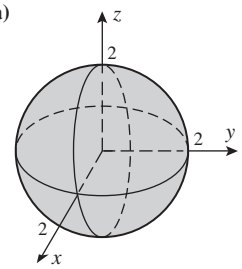
17. **(a)** 29 **(b)** $\dfrac{29}{\sqrt{65}}$ **19.** $x = 5 + t, y = 2 - t, z = 3$

21. $x + 5y - z - 6 = 0$ **23.** $a_1a_2 + b_1b_2 + c_1c_2 = 0$

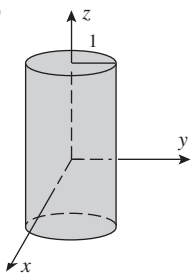
25. **(a)** hyperboloid of one sheet **(b)** sphere **(c)** circular cone

27. **(a)** $z = x^2 - y^2$ **(b)** $xz = 1$

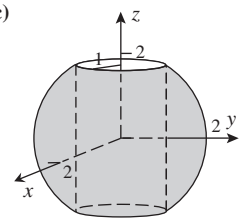
29. **(a)**



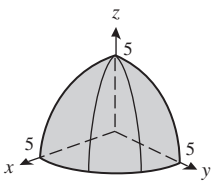
(b)



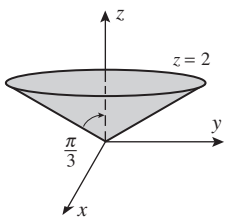
(c)



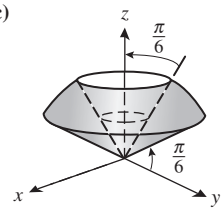
31. **(a)**



(b)



(c)



▶ Chapter 11 Making Connections (Page 840)

Answers are provided in the Student Solutions Manual.

▶ Exercise Set 12.1 (Page 845)

1. $(-\infty, +\infty)$; $\mathbf{r}(\pi) = -4\pi\mathbf{j}$ **3.** $[4, \infty)$; $\mathbf{r}(3) = -\mathbf{i} - \ln 7\mathbf{j} + \mathbf{k}$

5. $\mathbf{r} = 3\cos t\mathbf{i} + (2t + \cos 2t)\mathbf{j}$ **7.** $x = 3t^2$, $y = -2\ln t$

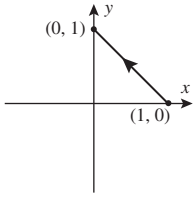
9. the line in 2-space through $(3, 0)$ with direction vector $\mathbf{a} = -\mathbf{i} + 5\mathbf{j}$

11. the line in 3-space through the point $(0, -3, 1)$ and parallel to the vector $2\mathbf{i} - 5\mathbf{k}$

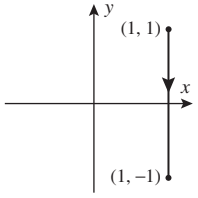
13. an ellipse centered at $(0, 0, 2)$ in the plane $z = 2$

15. **(a)** slope $-\frac{5}{2}$ **(b)** $\left(\frac{5}{2}, 0, \frac{3}{2}\right)$

17. **(a)**

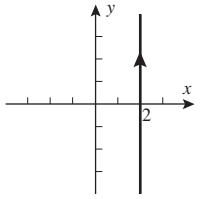


(b)

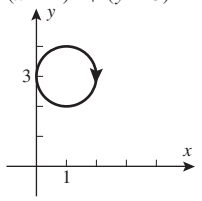


19. $\mathbf{r} = (1 - t)(3\mathbf{i} + 4\mathbf{j})$, $0 \le t \le 1$

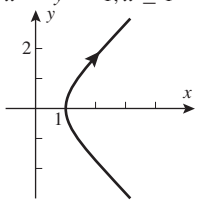
21. $x = 2$



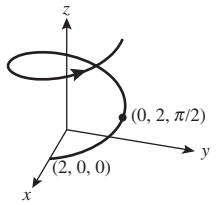
23. $(x - 1)^2 + (y - 3)^2 = 1$



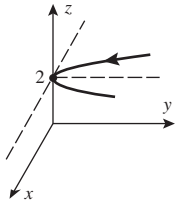
25. $x^2 - y^2 = 1$, $x \ge 1$



27.



29.

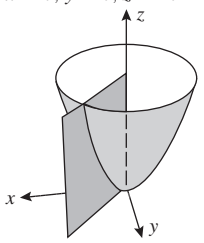


Responses to True–False questions may be abridged to save space.

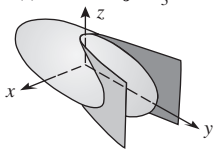
31. True; $\mathbf{r}(t) = (1 - t)\mathbf{r}_0 + t\mathbf{r}_1 (0 \le t \le 1)$ represents the line segment in 3-space that is traced from $\mathbf{r}_0$ to $\mathbf{r}_1$.

33. False; the natural domain of a vector-valued function is the *intersection* of the domains of its component functions.

35. $x = t$, $y = t$, $z = 2t^2$



37. $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} \pm \frac{1}{3}\sqrt{81 - 9t^2 - t^4}\mathbf{k}$ **43.** $c = 3/(2\pi)$



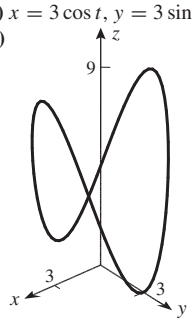
47. **(a)** III, since the curve is a subset of the plane $y = -x$

(b) IV, since only x is periodic in t and y, z increase without bound

(c) II, since all three components are periodic in t

(d) I, since the projection onto the yz-plane is a circle and the curve increases without bound in the x-direction

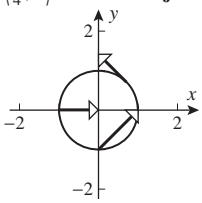
49. **(a)** $x = 3\cos t,\ y = 3\sin t,\ z = 9\cos^2 t$
(b)



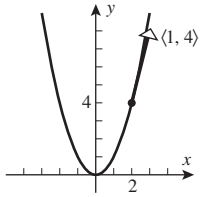
Exercise Set 12.2 (Page 856)

1. $\langle \frac{1}{4}, 0\rangle$ **3.** $10\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$ **5.** **(a)** continuous **(b)** not continuous

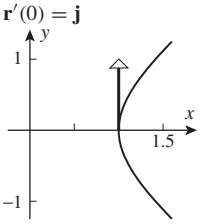
7.



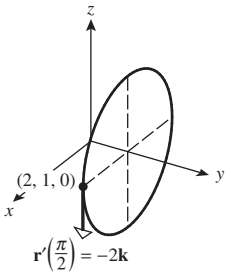
9. $(\cos t)\mathbf{j}$ **11.** $\mathbf{r}'(2) = \langle 1, 4\rangle$



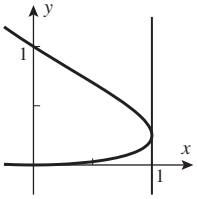
13. $\mathbf{r}'(0) = \mathbf{j}$



15. $\mathbf{r}'(\pi/2) = -2\mathbf{k}$



17.



19. $x = e^2 + 2et,\ y = 1 - \frac{t}{e}$
21. $x = 1 - \sqrt{3}\pi t,\ y = \sqrt{3} + \pi t,\ z = 2 + 6t$
23. $\mathbf{r} = (-\mathbf{i} + 2\mathbf{j}) + t\left(3\mathbf{i} + \frac{3}{4}\mathbf{j}\right)$
25. $\mathbf{r} = (4\mathbf{i} - \mathbf{j}) + t(-4\mathbf{i} + \mathbf{j} + 4\mathbf{k})$
27. **(a)** $\mathbf{i} - \mathbf{j} + \mathbf{k}$ **(b)** $-\mathbf{i} + \mathbf{k}$ **(c)** 0
29. $7t^6;\ 18t^5\mathbf{i} - 10t^4\mathbf{j}$
31. $5t\mathbf{i} + 2t^2\mathbf{j} + \mathbf{C}$

33. $(\sin t)\mathbf{i} - (\cos t)\mathbf{j} + \mathbf{C}$ **35.** $\mathbf{j}$ **37.** $(17\sqrt{17} - 1)/12$
39. $\frac{52}{3}\mathbf{i} + 4\mathbf{j}$

Responses to True–False questions may be abridged to save space.

41. False; for example, $\mathbf{r}(t) = \langle t, |t|\rangle$ is continuous at $t = 0$, but the specified limit doesn't exist at $t = 0$.

43. True; see the definition of $\int_a^b \mathbf{r}(t)\,dt$.

45. $(t^2 + 1)\mathbf{i} + (t^3 - 2)\mathbf{j}$
47. $y(t) = \left(\frac{1}{2}t^2 + 2\right)\mathbf{i} + (e^t - 1)\mathbf{j}$
49. **(a)** $(-2, 4, 6)$ and $(1, 1, -3)$ **(b)** $76°, 71°$ **51.** $68°$

Exercise Set 12.3 (Page 866)

1. smooth **3.** not smooth, $\mathbf{r}'(1) = \mathbf{0}$ **5.** $L = \frac{3}{4}$ **7.** $L = e - e^{-1}$
9. $L = 58$ **11.** $L = 2\pi\sqrt{13}$ **13.** $\mathbf{r}'(\tau) = 5\mathbf{i} + 10(5\tau + 1)\mathbf{j}$
15. $\mathbf{r}'(\tau) = 2\tau e^{\tau^2}\mathbf{i} - 8\tau e^{-\tau^2}\mathbf{j}$

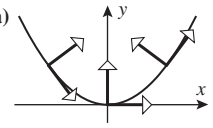
Responses to True–False questions may be abridged to save space.

17. False; $\int_a^b \|\mathbf{r}'(t)\|\,dt$ is a scalar that represents the arc length of the curve in 2-space traced by $\mathbf{r}(t)$ from $t = a$ to $t = b$ (Theorem 12.3.1).

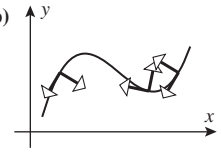
19. False; $\mathbf{r}'$ isn't defined at the point corresponding to the origin.
21. **(a)** $x = \dfrac{s}{\sqrt{2}},\ y = \dfrac{s}{\sqrt{2}}$ **(b)** $x = y = z = \dfrac{s}{\sqrt{3}}$
23. **(a)** $x = 1 + \dfrac{s}{3},\ y = 3 - \dfrac{2s}{3},\ z = 4 + \dfrac{2s}{3}$ **(b)** $\left\langle \frac{28}{3}, -\frac{41}{3}, \frac{62}{3}\right\rangle$
25. $x = 4 + \cos s,\ y = 5 + \sin s,\ 0 \le s \le 2\pi$
27. $x = \frac{1}{3}[(3s + 1)^{2/3} - 1]^{3/2},\ y = \frac{1}{2}[(3s + 1)^{2/3} - 1],\ s \ge 0$
29. $x = \left(\dfrac{s}{\sqrt{2}} + 1\right)\cos\left[\ln\left(\dfrac{s}{\sqrt{2}} + 1\right)\right],$
$y = \left(\dfrac{s}{\sqrt{2}} + 1\right)\sin\left[\ln\left(\dfrac{s}{\sqrt{2}} + 1\right)\right],$ $0 \le s \le \sqrt{2}(e^{\pi/2} - 1)$
33. $x = 2a\cos^{-1}[1 - s/(4a)]$
$-2a(1 - [1 - s/(4a)]^2)^{1/2}(2[1 - s/(4a)]^2 - 1),$
$y = \dfrac{s(8a - s)}{8a}$ for $0 \le s \le 8a$
35. **(a)** $9/2$ **(b)** $9 - 2\sqrt{6}$ **37.** **(a)** $\sqrt{3}(1 - e^{-2})$ **(b)** $4\sqrt{5}$
39. **(a)** $g(\tau) = \pi(\tau)$ **(b)** $g(\tau) = \pi(1 - \tau)$ **41.** 44 in
43. **(a)** $2t + \dfrac{1}{t}$ **(b)** $2t + \dfrac{1}{t}$ **(c)** $15 + 2\ln 2$

Exercise Set 12.4 (Page 872)

1. **(a)**



(b)



5. $\mathbf{T}(1) = \dfrac{2}{\sqrt{5}}\mathbf{i} + \dfrac{1}{\sqrt{5}}\mathbf{j},\ \mathbf{N}(1) = \dfrac{1}{\sqrt{5}}\mathbf{i} - \dfrac{2}{\sqrt{5}}\mathbf{j}$
7. $\mathbf{T}\left(\dfrac{\pi}{3}\right) = -\dfrac{\sqrt{3}}{2}\mathbf{i} + \dfrac{1}{2}\mathbf{j},\ \mathbf{N}\left(\dfrac{\pi}{3}\right) = -\dfrac{1}{2}\mathbf{i} - \dfrac{\sqrt{3}}{2}\mathbf{j}$
9. $\mathbf{T}\left(\dfrac{\pi}{2}\right) = -\dfrac{4}{\sqrt{17}}\mathbf{i} + \dfrac{1}{\sqrt{17}}\mathbf{k},\ \mathbf{N}\left(\dfrac{\pi}{2}\right) = -\mathbf{j}$
11. $\mathbf{T}(0) = \dfrac{1}{\sqrt{3}}\mathbf{i} + \dfrac{1}{\sqrt{3}}\mathbf{j} + \dfrac{1}{\sqrt{3}}\mathbf{k},\ \mathbf{N}(0) = -\dfrac{1}{\sqrt{2}}\mathbf{i} + \dfrac{1}{\sqrt{2}}\mathbf{j}$
13. $x = s,\ y = 1$ **15.** $\mathbf{B} = \frac{4}{5}\cos t\,\mathbf{i} - \frac{4}{5}\sin t\,\mathbf{j} - \frac{3}{5}\mathbf{k}$ **17.** $\mathbf{B} = -\mathbf{k}$
19. $\mathbf{T}\left(\dfrac{\pi}{4}\right) = \dfrac{\sqrt{2}}{2}(-\mathbf{i} + \mathbf{j}),\ \mathbf{N}\left(\dfrac{\pi}{4}\right) = -\dfrac{\sqrt{2}}{2}(\mathbf{i} + \mathbf{j}),$
$\mathbf{B}\left(\dfrac{\pi}{4}\right) = \mathbf{k}$; rectifying: $x + y = \sqrt{2}$; osculating: $z = 1$;
normal: $-x + y = 0$

Responses to True–False questions may be abridged to save space.

21. False; $\mathbf{T}(t)$ points in the direction of increasing parameter but may not be orthogonal to $\mathbf{r}(t)$. For example, if $\mathbf{r}(t) = \langle t, t\rangle$, then $\mathbf{T}(t) = \langle 1/\sqrt{2}, 1/\sqrt{2}\rangle$ is parallel to $\mathbf{r}(t)$.

23. True; $\mathbf{T}(s) = \mathbf{r}'(s)$, the unit tangent vector, and $\mathbf{N}(s) = \dfrac{\mathbf{r}''(s)}{\|\mathbf{r}''(s)\|}$, the unit normal vector, are orthogonal, so $\mathbf{r}'(s)$ and $\mathbf{r}''(s)$ are orthogonal.

Exercise Set 12.5 (Page 879)

1. $\kappa \approx 2$ **3.** **(a)** I is the curvature of II. **(b)** I is the curvature of II.
5. $\dfrac{6}{|t|(4 + 9t^2)^{3/2}}$ **7.** $\dfrac{6e^t}{(4e^{4t} + e^{-2t})^{3/2}}$ **9.** $\frac{5}{26}$ **11.** $\dfrac{1}{2\cosh^2 t}$
13. $\kappa = \frac{6}{13},\ \rho = \frac{13}{6}$ **15.** $\kappa = \dfrac{\sqrt{2}}{3},\ \rho = \dfrac{3\sqrt{2}}{2}$ **17.** $\kappa = \frac{1}{4}$

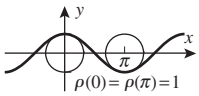
Responses to True–False questions may be abridged to save space.

19. False; see Definition 12.5.1: the curvature of the graph of $\mathbf{r}(s)$ is $\|\mathbf{r}''(s)\|$, the length of $\mathbf{r}''(s)$.

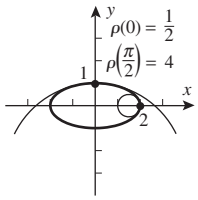
21. True; see Example 1: a circle of radius a has constant curvature $1/a$.

25. 1 **27.** $\dfrac{e^{-2}}{(1 + e^{-4})^{3/2}}$ **29.** $\dfrac{6}{13\sqrt{13}}$ **31.** $\dfrac{16}{17\sqrt{17}}$

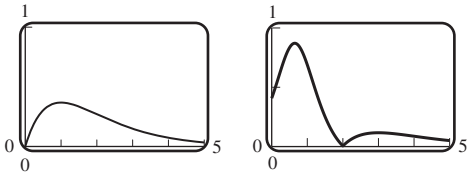
33. **(a)**



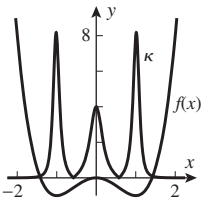
(b)



35.



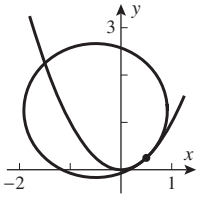
37. **(a)** $\kappa = \dfrac{|12x^2 - 4|}{[1 + (4x^3 - 4x)^2]^{3/2}}$ **(b)**



(c) $\rho = \frac{1}{4}$ for $x = 0$ and $\rho = \frac{1}{8}$ when $x = \pm 1$

41. $\dfrac{3}{4}$ **43.** $\dfrac{18}{27}$ **45.** $\rho = 2|p|$ **47.** $(3, 0), (-3, 0)$

51. **(b)** $\rho = \sqrt{2}$ **(c)**



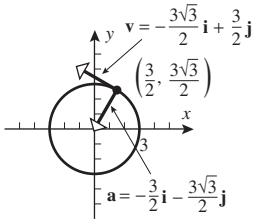
55. $a = \dfrac{1}{2r}$

63. $\tau = \dfrac{2}{(t^2 + 2)^2}$

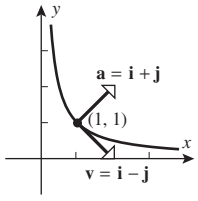
65. $\tau = -\dfrac{\sqrt{2}}{(e^t + e^{-t})^2}$

Exercise Set 12.6 (Page 891)

1. $\mathbf{v}(t) = -3\sin t\,\mathbf{i} + 3\cos t\,\mathbf{j}$
$\mathbf{a}(t) = -3\cos t\,\mathbf{i} - 3\sin t\,\mathbf{j}$
$\|\mathbf{v}(t)\| = 3$



3. $\mathbf{v}(t) = e^t\mathbf{i} - e^{-t}\mathbf{j}$
$\mathbf{a}(t) = e^t\mathbf{i} + e^{-t}\mathbf{j}$
$\|\mathbf{v}(t)\| = \sqrt{e^{2t} + e^{-2t}}$

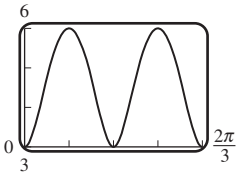


5. $\mathbf{v} = \mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$, $\|\mathbf{v}\| = \sqrt{2i}$, $\mathbf{a} = \mathbf{j} + 4\mathbf{k}$

7. $\mathbf{v} = -2\mathbf{i} + \mathbf{k}$, $\|\mathbf{v}\| = \sqrt{5}$, $\mathbf{a} = -2\mathbf{j}$

13. minimum speed $3\sqrt{2}$ when $\mathbf{r} = 24\mathbf{i} + 8\mathbf{j}$

15. **(a)**



(b) maximum speed = 6, minimum speed = 3
(d) The maximum speed first occurs when $t = \pi/6$.

17. $\mathbf{v}(t) = (1 - \sin t)\mathbf{i} + (\cos t - 1)\mathbf{j}$;
$\mathbf{r}(t) = (t + \cos t - 1)\mathbf{i} + (\sin t - t + 2)\mathbf{j}$

19. $\mathbf{v}(t) = (1 - \cos t)\mathbf{i} + \sin t\,\mathbf{j} + e^t\mathbf{k}$;
$\mathbf{r}(t) = (t - \sin t)\mathbf{i} + (2 - \cos t)\,\mathbf{j} + (e^t - 1)\mathbf{k}$

21. $9°$ **23.** **(a)** $1.5\mathbf{i} + 4.8\mathbf{j} - 3\mathbf{k}$ **(b)** $\mathbf{r}_0 = -0.7\mathbf{i} - 2.9\mathbf{j} + 4.8\mathbf{k}$

25. $\Delta\mathbf{r} = 12\mathbf{i} + \left(\frac{56}{3}\right)\mathbf{j}$, $s = (40\sqrt{5} - 16\sqrt{2})/3$

27. $\Delta\mathbf{r} = 3\mathbf{i} - \frac{3}{4}\mathbf{j} + \sqrt{2}\ln 4\mathbf{k}$; $s = \frac{15}{4}$

31. **(a)** $a_T = 0$, $a_N = \sqrt{2}$ **(b)** $a_T\mathbf{T} = \mathbf{0}$, $a_N\mathbf{N} = \mathbf{i} + \mathbf{j}$ **(c)** $1/\sqrt{2}$

33. **(a)** $a_T = 2\sqrt{5}$, $a_N = 2\sqrt{5}$ **(b)** $a_T\mathbf{T} = 2\mathbf{i} + 4\mathbf{j}$, $a_N\mathbf{N} = 4\mathbf{i} - 2\mathbf{j}$
(c) $2/\sqrt{5}$

35. **(a)** $a_T = -7/\sqrt{6}$, $a_N = \sqrt{53/6}$
(b) $a_T\mathbf{T} = -\frac{7}{6}(\mathbf{i} - 2\mathbf{j} + \mathbf{k})$, $a_N\mathbf{N} = \frac{13}{6}\mathbf{i} + \frac{19}{3}\mathbf{j} + \frac{7}{6}\mathbf{k}$ **(c)** $\dfrac{\sqrt{53}}{6\sqrt{6}}$

37. $a_T = -3$, $a_N = 2$, $\mathbf{T} = -\mathbf{j}$, $\mathbf{N} = \mathbf{i}$ **39.** -1

41. $a_N = 1.764 \times 10^{11}$ km/s^2

43. $a_N = 32/(1 + 4x^2)^{3/2}$ **45.** $a_N = 0$

Responses to True–False questions may be abridged to save space.

47. True; the velocity and unit tangent vectors have the same direction, so are parallel.

49. False; in this case the velocity and acceleration vectors will be parallel, but they may have opposite direction.

53. ≈ 257.20 N

55. $40\sqrt{3}$ ft **57.** 800 ft/s **59.** $\approx 7°$ or $83°$ **61.** **(c)** ≈ 14.942 ft

63. **(a)** $\rho \approx 176.78$ m **(b)** $\frac{125}{4}$ m

65. **(b)** R is maximum when $\alpha = 45°$, maximum value v_0^2/g

67. **(a)** 3.22 s **(b)** 278.9 ft

69. **(a)** $v_0 \approx 83$ ft/s, $\alpha \approx 8°$ **(b)** 268.76 ft

Exercise Set 12.7 (Page 901)

7. 7.69 km/s **9.** 10.80 km/s

11. **(a)** minimum distance = 220,680 mi, maximum distance = 246,960 mi **(b)** 27.5 days

13. **(a)** 17,224 mi/h **(b)** $e \approx 0.071$, apogee altitude = 819 mi

Chapter 12 Review Exercises (Page 902)

3. the circle of radius 3 in the plane $z = 2$, with center $(0, 0, 2)$

5. a parabola in the plane $x = 4$, vertex at $(4, 0, -1)$, opening upward

11. $x = 1 + t$, $y = -t$, $z = t$ **13.** $-(\cos t)\mathbf{i} + (\sin t)\mathbf{j} + \mathbf{C}$

15. $y(t) = (t^3 + 1)\,\mathbf{i} + (t^2 + 1)\mathbf{j}$ **17.** 15/4

19. $\mathbf{r}(s) = \dfrac{s - 3}{3}\mathbf{i} + \dfrac{12 - 2s}{3}\mathbf{j} + \dfrac{9 + 2s}{3}\mathbf{k}$ **25.** 1/5 **27.** 0

29. **(a)** speed **(b)** distance traveled
(c) distance of the particle from the origin

33. **(a)** $\mathbf{r}(t) = \left(\frac{1}{6}t^4 + t\right)\mathbf{i} + \left(\frac{1}{2}t^2 + 2t\right)\mathbf{j} - \left(\frac{1}{4}\cos 2t + t - \frac{1}{4}\right)\mathbf{k}$
(b) 3.475 **35.** 10.65 km/s **37.** 24.78 ft

Chapter 12 Making Connections (Page 904)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

1. **(c)** (i) $\mathbf{N} = \frac{1}{\sqrt{5}}\mathbf{i} - \frac{2}{\sqrt{5}}\mathbf{j}$ (ii) $\mathbf{N} = -\mathbf{j}$

2. **(b)** (i) $\mathbf{N} = -\sin t\,\mathbf{i} - \cos t\,\mathbf{j}$
(ii) $\mathbf{N} = \dfrac{-(4t + 18t^3)\mathbf{i} + (2 - 18t^4)\mathbf{j} + (6t + 12t^3)\mathbf{k}}{2\sqrt{81t^8 + 117t^6 + 54t^4 + 13t^2 + 1}}$

3. **(c)** $\kappa(s) \to +\infty$, so the spiral winds ever tighter.

4. semicircle: 53.479 ft; quarter-circle: 60.976 ft; point: 64.001 ft

Exercise Set 13.1 (Page 914)

1. **(a)** 5 **(b)** 3 **(c)** 1 **(d)** -1 **(e)** $16a^3 + 1$ **(f)** $a^3b^2 - a^2b^3 + 1$

3. **(a)** $x^2 - y^2 + 3$ **(b)** $3x^3y^5 + 3$ **5.** $x^3e^{x^3(2y-1)}$

7. **(a)** $t^2 + 3t^{10}$ **(b)** 0 **(c)** 4

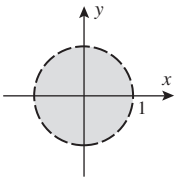
9. **(a)** WCI = 17.2°F **(b)** WCI = 22.6°F

11. **(a)** 59% **(b)** 73.5% **(c)** 60.6%

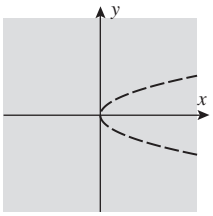
13. **(a)** 57 **(b)** -9 **(c)** 3 **(d)** $a^6 + 3$ **(e)** $-t^8 + 3$
(f) $(a + b)(a - b)^2b^3 + 3$

15. $(y + 1)e^{2x^2(y+1)z^2}$ **17.** **(a)** $120\sqrt{\pi}$ **(b)** $n(n + 1)/2$

19.



21.



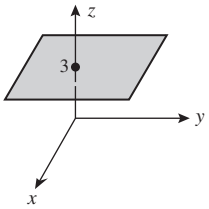
23. **(a)** all points above or on the line $y = -2$ **(b)** all points on or within the sphere $x^2 + y^2 + z^2 = 25$ **(c)** all points in 3-space

Responses to True–False questions may be abridged to save space.

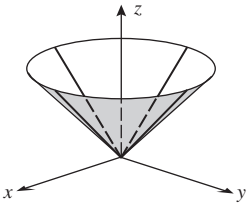
25. True; the interval [0, 1] is the intersection of the domains of $\sin^{-1} t$ and $\sqrt{t}$.

27. False; the natural domain is an infinite solid cylinder.

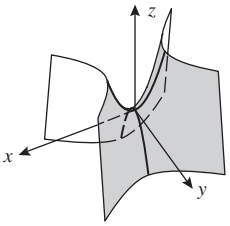
29.



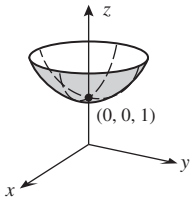
31.



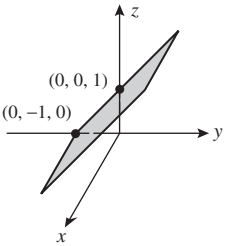
33.



35.



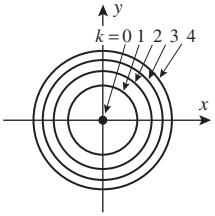
37.



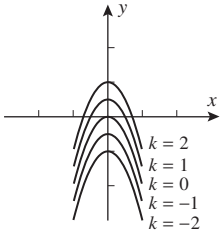
39. **(a)** $1 - x^2 - y^2$
(b) $\sqrt{x^2 + y^2}$
(c) $x^2 + y^2$

41. **(a)** A
(b) B
(c) increase
(d) decrease
(e) increase
(f) decrease

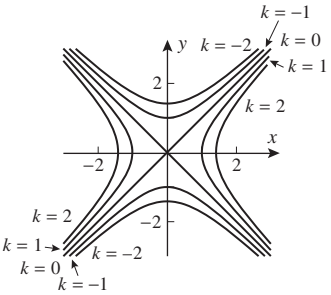
43.



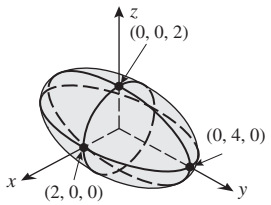
45.



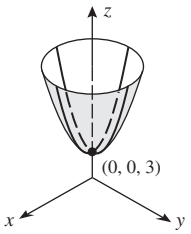
47.



49.



51.



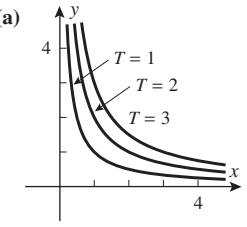
53. concentric spheres, common center at (2, 0, 0)

55. concentric cylinders, common axis the y-axis

57. **(a)** $x^2 - 2x^3 + 3xy = 0$ **(b)** $x^2 - 2x^3 + 3xy = 0$
(c) $x^2 - 2x^3 + 3xy = 14$

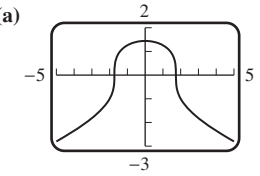
59. **(a)** $x^2 + y^2 - z = 5$ **(b)** $x^2 + y^2 - z = -3$ **(c)** $x^2 + y^2 - z = 0$

61. **(a)**

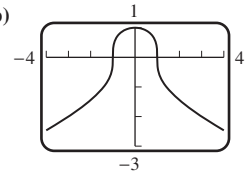


(b) the path $xy = 4$

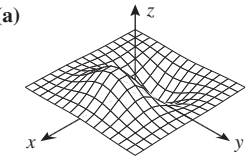
63. **(a)**



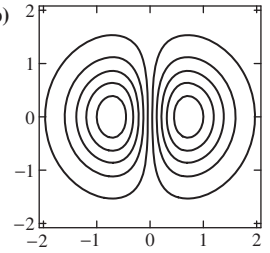
(b)



65. **(a)**



(b)



67. **(a)** The graph of g is the graph of f shifted one unit in the negative x-direction.
(b) The graph of g is the graph of f shifted one unit up the z-axis.
(c) The graph of g is the graph of f shifted one unit down the y-axis and then inverted with respect to the plane $z = 0$.

▶ Exercise Set 13.2 (Page 925)

1. 37 **3.** $-\frac{8}{3}$ **5.** 0

7. **(a)** along $x = 0$ limit does not exist
(b) along $x = 0$ limit does not exist

9. $\frac{1}{2}$ **11.** 0 **13.** 0 **15.** limit does not exist **17.** $\frac{48}{5}$ **19.** 0

21. limit does not exist **23.** 0 **25.** 0 **27.** 0

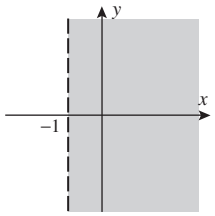
Responses to True–False questions may be abridged to save space.

29. True; by the definition of open set.

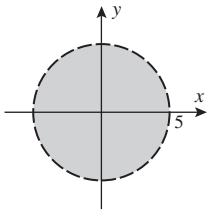
31. False; let $f(x, y) = \begin{cases} 1, & x \le 0 \\ -1, & x > 0 \end{cases}$ and let $g(x, y) = -f(x, y)$.

33. **(a)** no **(d)** no; yes **37.** $-\pi/2$ **39.** no

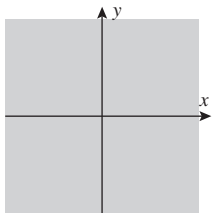
41.



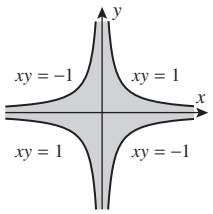
43.



45.



47.



49. all of 3-space

51. all points not on the cylinder $x^2 + z^2 = 2$

Exercise Set 13.3 (Page 936)

1. **(a)** $12x^2y^2$ **(b)** $8x^3y$ **(c)** $12y^2$ **(d)** $12x^2$ **(e)** $8y$ **(f)** $8x^3$ **(g)** 48 **(h)** 16

3. **(a)** $\frac{3}{8}$ **(b)** $\frac{1}{4}$ **5.** **(a)** $-4\cos 11$ **(b)** $2\cos 11$

7. $\partial z/\partial x = -4$; $\partial z/\partial y = \frac{1}{2}$ **9.** **(a)** 4.9 **(b)** 1.2

11. $z = f(x, y)$ has II as its graph, f_x has I as its graph, and f_y has III as its graph.

Responses to True–False questions may be abridged to save space.

13. True; on $y = 2$, $f(x, 2) = c$ is a constant function of x.

15. True; z must be a linear function of x and y.

17. $12xy^3e^{x^2y^3}$, $18x^2y^2e^{x^2y^3}$

19. $x^4/(y^{3/5} + x) + 4x^3\ln(1 + xy^{-3/5})$, $-\frac{3}{5}x^5/(y^{8/5} + xy)$

21. $-\dfrac{y(x^2 - y^2)}{(x^2 + y^2)^2}$, $\dfrac{x(x^2 - y^2)}{(x^2 + y^2)^2}$

23. $(3/2)x^2y(4x - 7)(3x^4y - 7x^3y)^{-1/2}$
$(1/2)x^3(3x - 7)(3x^4y - 7x^3y)^{-1/2}$

25. $\dfrac{y^{-1/2}}{y^2 + x^2}$, $-\dfrac{xy^{-3/2}}{y^2 + x^2} - \dfrac{3}{2}y^{-5/2}\tan^{-1}\left(\dfrac{x}{y}\right)$

27. $-\frac{5}{3}y^2\sec^2 x(y^2\tan x)^{-8/3}$, $-\frac{10}{3}y\tan x(y^2\tan x)^{-8/3}$

29. $-8, -21$ **31.** $1/\sqrt{17}, 8/\sqrt{17}$

33. **(a)** $2xy^4z^3 + y$ **(b)** $4x^2y^3z^3 + x$ **(c)** $3x^2y^4z^2 + 2z$ **(d)** $2y^4z^3 + y$ **(e)** $32z^3 + 1$ **(f)** 438

35. $f_x = 3z/x$, $f_y = z/y$, $f_z = \ln(x^3y\cos z) - z\tan z$

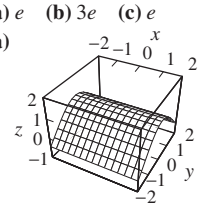
37. $f_x = -y^2z^4/(1 + x^2y^4z^8)$, $f_y = -2xyz^4/(1 + x^2y^4z^8)$, $f_z = -4xy^2z^3/(1 + x^2y^4z^8)$

39. $2yze^z\cos(xz)$, $2e^z\sin(xz)$, $2ye^z(\sin(xz) + 2x\cos(xz))$

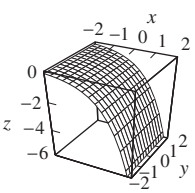
41. $x/\sqrt{x^2 + y^2 + z^2}$, $y/\sqrt{x^2 + y^2 + z^2}$, $z/\sqrt{x^2 + y^2 + z^2}$

43. **(a)** e **(b)** $3e$ **(c)** e

45. **(a)**



(b)



47. 6

49. -2

51. **(a)** $\partial V/\partial r = 2\pi rh$ **(b)** $\partial V/\partial h = \pi r^2$ **(c)** 64π **(d)** 64π

53. **(a)** $\dfrac{1}{4}\dfrac{\text{lb}}{\text{in}^2\cdot\text{K}}$ **(b)** $-2\dfrac{\text{in}^5}{\text{lb}}$

55. **(a)** $\dfrac{\partial V}{\partial l} = 6$ **(b)** $\dfrac{\partial V}{\partial w} = 15$ **(c)** $\dfrac{\partial V}{\partial h} = 10$

59. **(a)** $\pm\sqrt{6}/4$ **61.** $-x/z, -y/z$

63. $-\dfrac{2x + yz^2\cos(xyz)}{xyz\cos(xyz) + \sin(xyz)}$; $-\dfrac{xz^2\cos(xyz)}{xyz\cos(xyz) + \sin(xyz)}$

65. $-x/w, -y/w, -z/w$

67. $-\dfrac{yzw\cos(xyz)}{2w + \sin(xyz)}, -\dfrac{xzw\cos(xyz)}{2w + \sin(xyz)}, -\dfrac{xyw\cos(xyz)}{2w + \cos(xyz)}$

69. $e^{x^2}, -e^{y^2}$

71. $f_x(x, y) = 2xy^3\sin(x^6y^9)$, $f_y(x, y) = 3x^2y^2\sin(x^6y^9)$

73. **(a)** $-\dfrac{\cos y}{4\sqrt{x^3}}$ **(b)** $-\sqrt{x}\cos y$ **(c)** $-\dfrac{1}{2\sqrt{x}}\sin y$ **(d)** $-\dfrac{1}{2\sqrt{x}}\sin y$

75. $-32y^3$ **77.** $-2e^{2x}\sin y$ **79.** $\dfrac{28}{(4x - 7y)^2}$ **81.** $\dfrac{2(x - y)}{(x + y)^3}$

83. **(a)** $\dfrac{\partial^3 f}{\partial x^3}$ **(b)** $\dfrac{\partial^3 f}{\partial y^2\partial x}$ **(c)** $\dfrac{\partial^4 f}{\partial x^2\partial y^2}$ **(d)** $\dfrac{\partial^4 f}{\partial y\partial x^3}$

85. **(a)** $30xy^4 - 4$ **(b)** $60x^2y^3$ **(c)** $60x^3y^2$

87. **(a)** -30 **(b)** -125 **(c)** 150

89. **(a)** $15x^2y^4z^7 + 2y$ **(b)** $35x^3y^4z^6 + 3y^2$ **(c)** $21x^2y^5z^6$ **(d)** $6xy^5z^7$ **(e)** $140x^3y^3z^6 + 6y$ **(f)** $30xy^4z^7$ **(g)** $105x^2y^4z^6$ **(h)** $105x^2y^4z^6$

97. $\dfrac{\partial f}{\partial v} = 8vw^3x^4y^6$, $\dfrac{\partial f}{\partial w} = 12v^2w^2x^4y^6$, $\dfrac{\partial f}{\partial x} = 16v^2w^3x^3y^6$, $\dfrac{\partial f}{\partial y} = 24v^2w^3x^4y^5$

99. $\dfrac{\partial f}{\partial v_1} = \dfrac{2v_1}{v_3^2 + v_4^2}$, $\dfrac{\partial f}{\partial v_2} = \dfrac{-2v_2}{v_3^2 + v_4^2}$, $\dfrac{\partial f}{\partial v_3} = \dfrac{-2v_3(v_1^2 - v_2^2)}{(v_3^2 + v_4^2)^2}$, $\dfrac{\partial f}{\partial v_4} = \dfrac{-2v_4(v_1^2 - v_2^2)}{(v_3^2 + v_4^2)^2}$

101. **(a)** 0 **(b)** 0 **(c)** 0 **(d)** 0 **(e)** $2(1 + yw)e^{yw}\sin z\cos z$ **(f)** $2xw(2 + yw)e^{yw}\sin z\cos z$

103. $-i\sin(x_1 + 2x_2 + \cdots + nx_n)$

105. **(a)** xy-plane, $12x^2 + 6x$ **(b)** $y \neq 0$, $-3x^2/y^2$

107. $f_x(2, -1) = 11$, $f_y(2, -1) = -8$

109. **(b)** does not exist if $y \neq 0$ and $x = -y$

Exercise Set 13.4 (Page 947)

1. 5.05 **3.** 4.14 **9.** $dz = 9\,dx - 2\,dy$ **11.** $dz = 5x^4y^2\,dx + 2x^5y\,dy$

13. $dz = \dfrac{y}{1 + x^2y^2}\,dx + \dfrac{x}{1 + x^2y^2}\,dy$ **15.** $dw = 8\,dx - 5\,dy + 4\,dz$

17. $dw = 3x^2y^2z^2\,dx + 2x^3yz^2\,dy + 2x^3y^2z\,dz$

19. $dw = \dfrac{yz}{1 + x^2y^2z^2}\,dx + \dfrac{xz}{1 + x^2y^2z^2}\,dy + \dfrac{xy}{1 + x^2y^2z^2}\,dz$

21. $df = 0.08$, $\Delta f = 0.0807$ **23.** $df = 0.03$, $\Delta f \approx 0.029412$

25. $df = 0.96$, $\Delta f \approx 0.97929$

Responses to True–False questions may be abridged to save space.

27. False; see the discussion at the beginning of this section.

29. True; see Theorems 13.4.3 and 13.4.4.

31. The increase in the area of the rectangle is given by the sum of the areas of the three small rectangles, and the total differential is given by the sum of the areas of the upper left and lower right rectangles.

33. **(a)** $L = \frac{1}{5} - \frac{4}{125}(x - 4) - \frac{3}{125}(y - 3)$ **(b)** 0.000176603

35. **(a)** $L = 0$ **(b)** 0.0024

37. **(a)** $L = 6 + 6(x - 1) + 3(y - 2) + 2(z - 3)$ **(b)** -0.000481

39. **(a)** $L = e + e(x - 1) - e(y + 1) - e(z + 1)$ **(b)** 0.01554

45. 0.5 **47.** $1, 1, -1, 2$ **49.** $(1, 2)$ **51.** $(1, 1, 1)$ **53.** 7%

55. $r\%$ **57.** 0.3%

59. **(a)** $(r + s)\%$ **(b)** $(r + s)\%$ **(c)** $(2r + 3s)\%$ **(d)** $\left(3r + \dfrac{s}{2}\right)\%$

61. $\approx 39\ \text{ft}^2$

Exercise Set 13.5 (Page 956)

1. $51t^{16}$ **3.** $3t^{-2}\sin(1/t)$ **5.** $-\frac{17}{4}t^{13/4}e^{1-t^{17/4}}$ **7.** $\dfrac{dw}{dt} = 145t^{28}$

9. $-\cos t$ **11.** 11.2688 **13.** 0

17. $24u^2v^2 + 16uv^3 - 2v + 3$, $16u^3v + 24u^2v^2 - 2u + 3$

19. $-\dfrac{2\sin u}{5\sin v}, -\dfrac{2\cos u\cos v}{5\sin^2 v}$ **21.** $e^u, 0$

23. $3r^2\sin\theta\cos^2\theta - 4r^3\sin^3\theta\cos\theta,$
$-2r^3\sin^2\theta\cos\theta + r^4\sin^4\theta + r^3\cos^3\theta - 3r^4\sin^2\theta\cos^2\theta$

25. $\dfrac{x^2+y^2}{4x^2y}, \dfrac{-(x^2+y^2)}{4xy^2}$ **27.** $\dfrac{\partial z}{\partial r} = \dfrac{2r\sin^2\theta}{r^2\sin^2\theta+1}, \dfrac{\partial z}{\partial\theta} = \dfrac{2r^2\sin\theta\cos\theta}{r^2\sin^2\theta+1}$

29. $\dfrac{dw}{d\rho} = 2\rho(4\sin^2\phi+\cos^2\phi), \dfrac{\partial w}{\partial\phi} = 6\rho^2\sin\phi\cos\phi, \dfrac{dw}{d\theta} = 0$

31. $-\pi$ **33.** $\sqrt{3}e^{\sqrt{3}}, (2-4\sqrt{3})e^{\sqrt{3}}$

Responses to True–False questions may be abridged to save space.

35. False; the symbols ∂z and ∂x have no individual meaning.

37. False; consider $z = xy, x = t, y = t$.

39. $-\dfrac{2xy^3}{3x^2y^2+\cos y}$

41. $-\dfrac{ye^{xy}}{xe^{xy}+ye^y+e^y}$ **43.** $\dfrac{2x+yz}{6yz-xy}, \dfrac{xz-3z^2}{6yz-xy}$

45. $\dfrac{ye^x}{15\cos 3z+4}, \dfrac{e^x}{15\cos 3z+4}$

59. $\dfrac{\partial w}{\partial\rho} = (\sin\phi\cos\theta)\dfrac{\partial w}{\partial x} + (\sin\phi\sin\theta)\dfrac{\partial w}{\partial y} + (\cos\phi)\dfrac{\partial w}{\partial z},$
$\dfrac{\partial w}{\partial\phi} = (\rho\cos\phi\cos\theta)\dfrac{\partial w}{\partial x} + (\rho\cos\phi\sin\theta)\dfrac{\partial w}{\partial y} - (\rho\sin\phi)\dfrac{\partial w}{\partial z},$
$\dfrac{\partial w}{\partial\theta} = -(\rho\sin\phi\sin\theta)\dfrac{\partial w}{\partial x} + (\rho\sin\phi\cos\theta)\dfrac{\partial w}{\partial y}$

63. **(a)** $\dfrac{dw}{dt} = \displaystyle\sum_{i=1}^{4}\dfrac{\partial w}{\partial x_i}\dfrac{dx_i}{dt}$ **(b)** $\dfrac{\partial w}{\partial v_j} = \displaystyle\sum_{i=1}^{4}\dfrac{\partial w}{\partial x_i}\dfrac{\partial x_i}{\partial v_j}, j = 1, 2, 3$

Exercise Set 13.6 (Page 968)

1. $\frac{81}{4}\sqrt{2}$ **3.** 0 **5.** $-\frac{704}{3}$ **7.** $-\frac{410}{949}$ **9.** 0 **11.** $-\frac{25}{\sqrt{2}}$

13. $\sqrt{2}/2$ **15.** $72/\sqrt{14}$ **17.** $-8/63$ **19.** $1/2+\sqrt{3}/8$ **21.** $2\sqrt{2}$

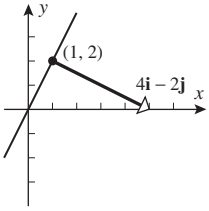
23. $1/\sqrt{5}$ **25.** $-\frac{3}{2}e$ **27.** $3/\sqrt{11}$ **29.** **(a)** 5 **(b)** 10 **(c)** $-5\sqrt{5}$

31. III **33.** $4\mathbf{i}-10\mathbf{j}$

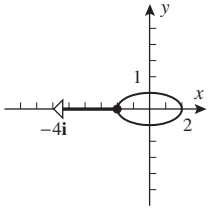
35. $\nabla w = \dfrac{x}{x^2+y^2+z^2}\mathbf{i} + \dfrac{y}{x^2+y^2+z^2}\mathbf{j} + \dfrac{z}{x^2+y^2+z^2}\mathbf{k}$

37. $-540\mathbf{i}-216\mathbf{j}$ **39.** $5(\mathbf{i}+\mathbf{j}+\mathbf{k})$

41.



43.



45. $\pm(-4\mathbf{i}+\mathbf{j})/\sqrt{17}$ **47.** $\mathbf{u} = (3\mathbf{i}-\mathbf{j})/\sqrt{10}, \|\nabla f(-1,1)\| = 4\sqrt{10}$

49. $\mathbf{u} = (4\mathbf{i}+3\mathbf{j})/5, \|\nabla f(4,-3)\| = 1$ **51.** $\dfrac{1}{\sqrt{2}}(\mathbf{i}-\mathbf{j}), 3\sqrt{2}$

53. $\dfrac{1}{\sqrt{2}}(-\mathbf{i}+\mathbf{j}), \dfrac{1}{\sqrt{2}}$

55. $\mathbf{u} = (-\mathbf{i}+3\mathbf{j})/\sqrt{10}, -\|\nabla f(-1,3)\| = -2\sqrt{10}$

57. $\mathbf{u} = (3\mathbf{i}-\mathbf{j})/\sqrt{10}, -\|\nabla f(\pi/6,\pi/4)\| = -\sqrt{5}$

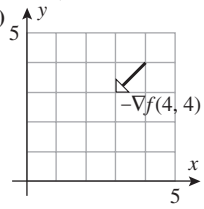
59. $(\mathbf{i}-11\mathbf{j}+12\mathbf{k})/\sqrt{266}, -\sqrt{266}$

Responses to True–False questions may be abridged to save space.

61. False; they are equal. **63.** False; let $\mathbf{u} = \mathbf{i}$ and let $f(x, y) = y$.

65. $8/\sqrt{29}$

67. **(a)** $\approx 1/\sqrt{2}$
(b)



69. $16x^2+y^2 = 16$

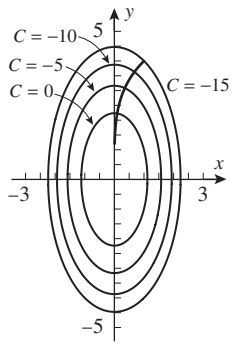
71. $36/\sqrt{17}$

73. **(a)** $2e^{-\pi/2}\mathbf{i}$

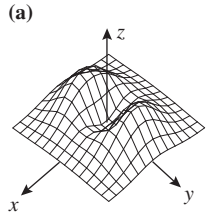
75. $-\frac{5}{3}(2\mathbf{i}-\mathbf{j}-2\mathbf{k})$

81. $x(t) = e^{-8t}, y(t) = 4e^{-2t}$

83.



85. **(a)**



(c) $\nabla f = [2x - 2x(x^2+3y^2)]e^{-(x^2+y^2)}\mathbf{i} + [6y - 2y(x^2+3y^2)]e^{-(x^2+y^2)}\mathbf{j}$
(d) $x = y = 0$ or $x = 0, y = \pm 1$ or $x = \pm 1, y = 0$

Exercise Set 13.7 (Page 975)

1. **(a)** $x+y+2z = 6$ **(b)** $x = 2+t, y = 2+t, z = 1+2t$
(c) 35.26°

3. tangent plane: $3x+4z = 25$;
normal line: $x = 3-(3t/4), y = 0, z = 4-t$

5. tangent plane: $48x-14y-z = 66$;
normal line: $x = 1+48t, y = -2-14t, z = 10-t$

7. tangent plane: $x-y-ez = -e$;
normal line: $x = 1+\frac{1}{e}t, y = 1-\frac{1}{e}t, z = 1-t$

9. tangent plane: $3y-z = -3$;
normal line: $x = \pi/6, y = 3t, z = 3-t$

11. **(a)** all points on the x-axis or y-axis **(b)** $(0, -2, -4)$

13. $(-1, -2, -3)$ **15.** **(a)** $(-2, 1, 5), (0, 3, 9)$ **(b)** $\dfrac{4}{3\sqrt{14}}, \dfrac{4}{\sqrt{222}}$

Responses to True–False questions may be abridged to save space.

17. False; they need only be parallel.

19. True; see Formula (15) of Section 13.4.

21. $\pm\dfrac{1}{\sqrt{1218}}(\mathbf{i}-16\mathbf{j}-31\mathbf{k})$ **25.** $(1, 2/3, 2/3), (-1, -2/3, -2/3)$

27. $x = 1+8t, y = -1+5t, z = 2+6t$

29. $x = 3+4t, y = -3-4t, z = 4-3t$

Exercise Set 13.8 (Page 985)

1. **(a)** minimum at $(2, -2)$, no maxima
(b) maximum at $(0, 0)$, no minima **(c)** no maxima or minima

3. minimum at $(3, -1)$, no maxima **5.** relative minimum at $(0, 0)$

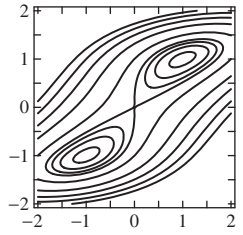
7. relative minimum at $(0, 0)$; saddle points at $(\pm 2, 1)$

9. saddle point at $(0, -2)$ **11.** relative minimum at $(4, -2)$

13. relative minima at $(-2, -2)$ and $(2, 2)$ **15.** saddle point at $(-1, 0)$

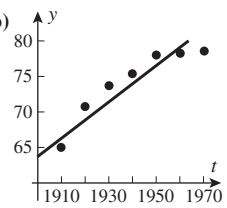
17. no critical points **19.** relative maximum at $(-2, 0)$

21. saddle point at $(0, 0)$; relative minima at $(1, 1)$ and $(-1, -1)$

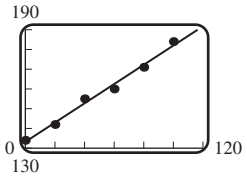


Responses to True–False questions may be abridged to save space.

23. False; let $f(x, y) = y$.
25. True; this follows from Theorem 13.8.6.
27. **(b)** relative minimum at $(0, 0)$
31. absolute maximum 0,
absolute minimum -12
33. absolute maximum 3,
absolute minimum -1
35. absolute maximum $\frac{33}{4}$,
absolute minimum $-\frac{1}{4}$
37. 12, 12, 12
39. maximum at $(1, 2, 2)$
41. $2a/\sqrt{3}, 2a/\sqrt{3}, 2a/\sqrt{3}$
43. length and width 2 ft, height 4 ft
45. **(a)** $x = 0$: minimum -3, maximum 0;
$x = 1$: minimum 3, maximum $13/3$;
$y = 0$: minimum 0, maximum 4;
$y = 1$: minimum -3, maximum 3
(b) $y = x$: minimum 0, maximum 3;
$y = 1 - x$: maximum 4, minimum -3
(c) minimum -3, maximum $13/3$
47. length and width $\sqrt[3]{2V}$, height $\sqrt[3]{2V}/2$ **51.** $y = \frac{3}{4}x + \frac{19}{12}$
53. $y = 0.5x + 0.8$
55. **(a)** $y = 63.73 + 0.2565t$ **(b)**
(c) about 84 years

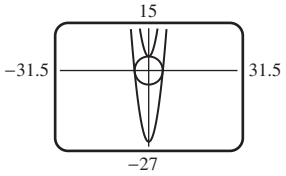


57. **(a)** $P = \dfrac{2798}{21} + \dfrac{171}{350}T$ **(b)**
(c) $T \approx -272.7096°\text{C}$



Exercise Set 13.9 (Page 996)

1. **(a)** 4 **3.** **(a)**

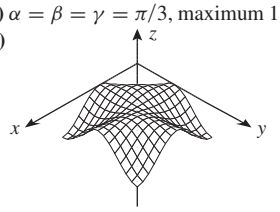


(c) maximum $\frac{101}{4}$,
minimum -5

5. maximum $4\sqrt{2}$ at $(-2\sqrt{2}, -2)$ and $(2\sqrt{2}, 2)$,
minimum $-4\sqrt{2}$ at $(-2\sqrt{2}, 2)$ and $(2\sqrt{2}, -2)$
7. maximum $\sqrt{2}$ at $(1/\sqrt{2}, 0)$, minimum $-\sqrt{2}$ at $(-1/\sqrt{2}, 0)$
9. maximum 6 at $\left(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3}\right)$, minimum -6 at $\left(-\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right)$
11. maximum is $1/(3\sqrt{3})$ at $(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$,
$(1/\sqrt{3}, -1/\sqrt{3}, -1/\sqrt{3})$, $(-1/\sqrt{3}, 1/\sqrt{3}, -1/\sqrt{3})$, and
$(-1/\sqrt{3}, -1/\sqrt{3}, 1/\sqrt{3})$; minimum is $-1/(3\sqrt{3})$ at
$(1/\sqrt{3}, 1/\sqrt{3}, -1/\sqrt{3})$, $(1/\sqrt{3}, -1/\sqrt{3}, 1/\sqrt{3})$,
$(-1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$, and $(-1/\sqrt{3}, -1/\sqrt{3}, -1/\sqrt{3})$

Responses to True–False questions may be abridged to save space.

13. False; a Lagrange multiplier is a scalar.
15. False; we must solve three equations in three unknowns.
17. $\left(\frac{2}{5}, -\frac{4}{5}\right)$ **19.** $\left(\frac{1}{3}, \frac{2}{3}, \frac{1}{3}\right)$
21. $(3, 6)$ is closest and $(-3, -6)$ is farthest **23.** $5(\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$
25. 9, 9, 9 **27.** $(\pm\sqrt{5}, 0, 0)$ **29.** length and width 2 ft, height 4 ft
33. **(a)** $\alpha = \beta = \gamma = \pi/3$, maximum $1/8$
(b)



Chapter 13 Review Exercises (Page 997)

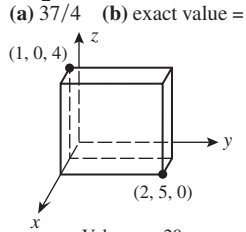
1. **(a)** xy **(b)** $e^{rs}\ln(r + s)$
5. **(a)** not defined on line $y = x$ **(b)** not continuous
9. **(a)** 10 Pa/min **(b)** $500/3$ Pa/min
15. df (the differential of f) is an approximation for Δf (the change in f)
17. $dV = -0.2\ \text{m}^3$; $\Delta V = -0.1973333\ \text{m}^3$ **19.** 2
21. $\dfrac{-f_y^2 f_{xx} + 2f_x f_y f_{xy} - f_x^2 f_{yz}}{f_y^3}$ **25.** $\dfrac{7}{2} + \dfrac{4}{5}\ln 2$ **27.** $-7/\sqrt{5}$
29. $(0, 0, 2)$, $(1, 1, 1)$, $(-1, -1, 1)$ **31.** $\left(-\frac{1}{2}, -1, \frac{25}{4}\right)$
33. relative minimum at $(15, -8)$
35. saddle point at $(0, 0)$, relative minimum at $(1, 1)$
37. absolute maximum of 4 at $(\pm 1, \pm 2)$, absolute minimum of 0 at
$(\pm\sqrt{2}, 0)$ and $(0, \pm 2\sqrt{2})$
39. $I_1 : I_2 : I_3 = \dfrac{1}{R_1} : \dfrac{1}{R_2} : \dfrac{1}{R_3}$
41. **(a)** $\partial P/\partial L = c\alpha L^{\alpha-1}K^{\beta}$, $\partial P/\partial K = c\beta L^{\alpha}K^{\beta-1}$

Chapter 13 Making Connections (Page 999)

Answers are provided in the Student Solutions Manual.

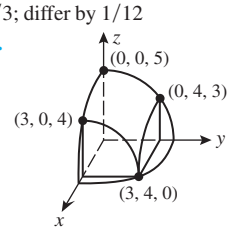
Exercise Set 14.1 (Page 1007)

1. 11 **3.** 16 **5.** 3 **7.** 4 **9.** $2 - \ln 3$
11. $\dfrac{1 - \ln 2}{2}$ **13.** 0 **15.** $\frac{1}{3}$
17. **(a)** $37/4$ **(b)** exact value $= 28/3$; differ by $1/12$
19.



Volume = 20

21.



Responses to True–False questions may be abridged to save space.

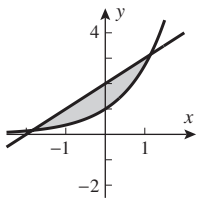
23. False; ΔA_k is the area of such a rectangular region.
25. False; $\displaystyle\iint_R f(x, y)\,dA = \int_1^5 \int_2^4 f(x, y)\,dy\,dx$.
29. $\frac{63}{2}$ **31.** 16 **33.** $\dfrac{1}{3\pi}$ **35.** $1 - \dfrac{2}{\pi}$ **37.** $\frac{14}{3}$ °C **39.** 1.381737122
41. first integral equals $\frac{1}{2}$, second equals $-\frac{1}{2}$; no

Exercise Set 14.2 (Page 1015)

1. $\frac{2}{15}$ **3.** 18 **5.** $\dfrac{\pi}{2}$ **7.** $\frac{4}{3}$
9. **(a)** $\displaystyle\int_0^2 \int_0^{x^2} f(x, y)\,dy\,dx$ **(b)** $\displaystyle\int_0^4 \int_{\sqrt{y}}^2 f(x, y)\,dx\,dy$
11. **(a)** $\displaystyle\int_1^2 \int_{-2x+5}^3 f(x, y)\,dy\,dx + \int_2^4 \int_1^3 f(x, y)\,dy\,dx +$
$\displaystyle\int_4^5 \int_{2x-7}^3 f(x, y)\,dy\,dx$ **(b)** $\displaystyle\int_1^3 \int_{(5-y)/2}^{(y+7)/2} f(x, y)\,dx\,dy$
13. **(a)** $\frac{16}{3}$ **(b)** 38 **15.** 576 **17.** 0 **19.** $\dfrac{\sqrt{26} - 1}{2}$ **21.** $\frac{50}{3}$

23. $-\frac{2}{15}$ **25.** $\dfrac{1-\cos 8}{3}$

27. **(a)**



(b) $(-1.8414, 0.1586)$, $(1.1462, 3.1462)$
(c) -0.4044
(d) -0.4044

29. $\sqrt{2}-1$ **31.** 32

Responses to True–False questions may be abridged to save space.

33. False; $\int_0^1\int_{x^2}^{2x} f(x,y)\,dy\,dx$ integrates $f(x,y)$ over the region between the graphs of $y=x^2$ and $y=2x$ for $0\le x\le 1$ and results in a number, but $\int_{x^2}^{2x}\int_0^1 f(x,y)\,dx\,dy$ produces an expression involving x.

35. False; although R is symmetric across the x-axis, the integrand may not be.

37. 12 **39.** 64π **41.** 170 **43.** 18π **45.** 8π

47. $\int_0^{\sqrt{2}}\int_{y^2}^{2} f(x,y)\,dx\,dy$ **49.** $\int_1^{e^3}\int_{\ln x}^{3} f(x,y)\,dy\,dx$

51. $\int_0^{\pi/2}\int_0^{\sin x} f(x,y)\,dy\,dx$ **53.** $\dfrac{1-e^{-16}}{8}$ **55.** $\dfrac{e^8-1}{3}$

57. **(a)** 0 **(b)** $\tan 1$ **59.** 0 **61.** $\dfrac{\pi}{4}-\dfrac{\ln 2}{2}$ **63.** $\frac{2}{3}$°C **65.** 0.676089

▶ Exercise Set 14.3 (Page 1024)

1. $\frac{\sqrt{2}}{24}$ **3.** $\frac{4}{9}a^3$ **5.** $\frac{1}{12}$ **7.** $\dfrac{3\pi}{2}$ **9.** $\dfrac{7\pi}{16}$ **11.** $\int_{\pi/6}^{5\pi/6}\int_2^{4\sin\theta} f(r,\theta)r\,dr\,d\theta$

13. $8\int_0^{\pi/2}\int_1^3 r\sqrt{9-r^2}\,dr\,d\theta$ **15.** $2\int_0^{\pi/2}\int_0^{\cos\theta}(1-r^2)r\,dr\,d\theta$

17. $\dfrac{64\sqrt{2}}{3}\pi$ **19.** $\dfrac{5\pi}{32}$ **21.** $\dfrac{27\pi}{16}$ **23.** $(1-e^{-1})\pi$ **25.** $\dfrac{\pi}{8}\ln 10$ **27.** 2π

29. $\frac{16}{9}$ **31.** $\dfrac{\pi}{2}\left(1-\dfrac{1}{\sqrt{1+a^2}}\right)$ **33.** $\dfrac{\pi}{4}(\sqrt{5}-1)$

Responses to True–False questions may be abridged to save space.

35. True; the disk is given in polar coordinates by $0\le r\le 2, 0\le\theta\le 2\pi$.

37. False; the integrand is missing a factor of r:

$$\iint_R f(r,\theta)\,dA=\int_0^{\pi/2}\int_1^2 f(r,\theta)r\,dr\,d\theta.$$

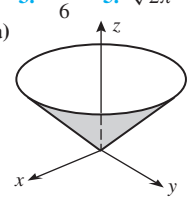
39. $\pi a^2 h$ **41.** $\dfrac{1}{5}+\dfrac{\pi}{2}$

43. **(a)** $\dfrac{4}{3}\pi a^2 c$ **(b)** $\approx 1.0831682\times 10^{21}\ \text{m}^3$ **45.** $2a^2$

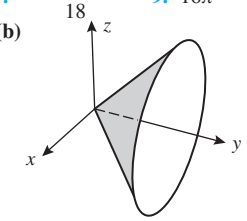
▶ Exercise Set 14.4 (Page 1036)

1. 3π **3.** $\dfrac{\sqrt{7}}{6}$ **5.** $\sqrt{2}\pi$ **7.** $\dfrac{(5\sqrt{5}-1)\pi}{18}$ **9.** 16π

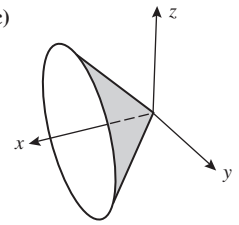
11. **(a)**



(b)



(c)



13. **(a)** $x=u, y=v, z=3+\frac{3}{2}u-2v$ **(b)** $x=u, y=v, z=u^2$

15. **(a)** $x=\sqrt{7}\cos u, y=\sqrt{7}\sin u, z=v; 0\le u\le 2\pi, 0\le v\le 1$
(b) $x=2\cos u, y=v, z=2\sin u; 0\le u\le 2\pi, 1\le v\le 5$

17. $x=u, y=\sin u\cos v, z=\sin u\sin v$

19. $x=r\cos\theta, y=r\sin\theta, z=\dfrac{1}{3+r^2}$

21. $x=r\cos\theta, y=r\sin\theta, z=2r^3\cos\theta\sin^2\theta$

23. $x=r\cos\theta, y=r\sin\theta, z=\sqrt{9-r^2}; r\le 2\sqrt{2}$

25. $x=\dfrac{1}{2}\rho\cos\theta, y=\dfrac{1}{2}\rho\sin\theta, z=\dfrac{\sqrt{3}}{2}\rho$ **27.** $z=x-y$; a plane

29. $(x/3)^2+(y/2)^2=1; 2\le z\le 4$; part of an elliptic cylinder

31. $(x/2)^2+(y/5)^2=z^2; 0\le z\le 1$; part of an elliptic cone

33. **(a)** $x=r\cos\theta, y=r\sin\theta, z=r, 0\le r\le 2$;
$x=u, y=v, z=\sqrt{u^2+v^2}, 0\le u^2+v^2\le 4$

35. **(a)** $0\le u\le 3, 0\le v\le\pi$ **(b)** $0\le u\le 4, -\pi/2\le v\le\pi/2$

37. **(a)** $0\le\phi\le\pi/2, 0\le\theta\le 2\pi$ **(b)** $0\le\phi\le\pi, 0\le\theta\le\pi$

39. $2x+4y-z=5$ **41.** $z=0$ **43.** $x-y+\dfrac{\sqrt{2}}{2}z=\dfrac{\pi\sqrt{2}}{8}$

45. $\dfrac{(17\sqrt{17}-5\sqrt{5})\pi}{6}$

Responses to True–False questions may be abridged to save space.

47. False; the surface area is $S=\iint_R\sqrt{[f_x(x,y)]^2+[f_y(x,y)]^2+1}\,dA$.

49. True; see the discussion preceding Definition 14.4.1.

51. $4\pi a^2$ **55.** $4\pi^2 ab$ **57.** 9.099

59. $(x/a)^2+(y/b)^2+(z/c)^2=1$; ellipsoid

61. $(x/a)^2+(y/b)^2-(z/c)^2=-1$; hyperboloid of two sheets

▶ Exercise Set 14.5 (Page 1045)

1. 16 **3.** $\frac{1}{24}$ **5.** $\frac{81}{5}$ **7.** $-\frac{32}{15}$ **9.** $\pi(\pi-3)/8$ **11.** $\frac{27}{32}$ **13.** 4.712

15. 4 **17.** $\frac{256}{15}$

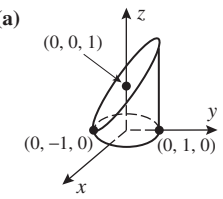
19. **(a)** $\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_{4x^2+y^2}^{4-3y^2} f(x,y,z)\,dz\,dy\,dx$

(b) $\int_{-1}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}}\int_{4x^2+y^2}^{4-3y^2} f(x,y,z)\,dz\,dx\,dy$

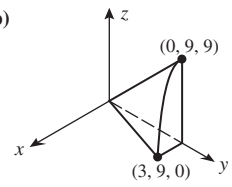
21. $4\int_0^1\int_0^{\sqrt{1-x^2}}\int_{4x^2+y^2}^{4-3y^2} dz\,dy\,dx$

23. $2\int_{-3}^{3}\int_0^{\frac{1}{3}\sqrt{9-x^2}}\int_0^{x+3} dz\,dy\,dx$

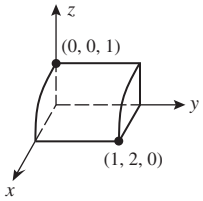
25. **(a)**



(b)



(c)



Responses to True–False questions may be abridged to save space.

27. True; apply Fubini's Theorem (Theorem 14.5.1).

29. False;

$$\iiint_G f(x, y, z)\, dV = \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} f(x, y, z)\, dz\, dy\, dx.$$

33. $\frac{3}{4}$ **35.** 3.291

37. **(a)** $\int_0^a \int_0^{b(1-x/a)} \int_0^{c(1-x/a-y/z)} dz\, dy\, dx$ is one example.

39. **(a)** $\int_0^2 \int_0^{\sqrt{4-x^2}} \int_0^6 f(x, y, z)\, dz\, dy\, dx$

(b) $\int_0^9 \int_0^{3-\sqrt{x}} \int_y^{3-\sqrt{x}} f(x, y, z)\, dz\, dy\, dx$

(c) $\int_0^2 \int_0^{4-x^2} \int_y^{8-y} f(x, y, z)\, dz\, dy\, dx$

Exercise Set 14.6 (Page 1056)

1. -2π **3.** $\dfrac{\pi}{8}$

5. The region is bounded by the xy-plane and the upper half of a sphere of radius 1 centered at the origin; $f(r, \theta, z) = z$.

7. The region is the portion of the first octant inside a sphere of radius 1 centered at the origin; $f(\rho, \theta, \phi) = \rho \cos \phi$.

9. 128π **11.** $\frac{152}{3}\pi + \frac{80}{3}\pi\sqrt{5}$ **13.** $\dfrac{64\pi}{3}(2 - \sqrt{3})$ **15.** $\dfrac{11\pi a^3}{3}$

17. $\dfrac{\pi a^6}{48}$ **19.** $\dfrac{32(2\sqrt{2} - 1)\pi}{15}$

Responses to True–False questions may be abridged to save space.

21. False; the factor r^2 should be r [Formula (6)]:

$$\iiint_G f(x, y, z)\, dV = \iiint_{\substack{\text{appropriate}\\ \text{limits}}} f(r\cos\theta, r\sin\theta, z) r\, dz\, dr\, d\theta.$$

23. True; G is the spherical wedge bounded by the spheres $\rho = 1$ and $\rho = 3$, the half-planes $\theta = 0$ and $\theta = 2\pi$, and above the cone $\phi = \pi/4$, so

$$(\text{volume of } G) = \iiint_G dV = \int_0^{\pi/4} \int_0^{2\pi} \int_1^3 \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi.$$

25. **(a)** $\frac{5}{2}(-8 + 3\ln 3)\ln(\sqrt{5} - 2)$ **(b)** $f(x, y, z) = \dfrac{y^3}{x^3\sqrt{1+z^2}}$; G is the cylindrical wedge $1 \le r \le 4$, $\dfrac{\pi}{6} \le \theta \le \dfrac{\pi}{3}$, $-2 \le z \le 2$

27. $\dfrac{4\pi a^3}{3}$ **29.** $\dfrac{2\pi}{3}(\sqrt{2} - 1)$

Exercise Set 14.7 (Page 1068)

1. -20 **3.** $\cos(u - v)$ **5.** $x = \frac{6}{17}u + \frac{5}{17}v$, $y = -\frac{1}{17}u + \frac{2}{17}v$; $\frac{1}{17}$

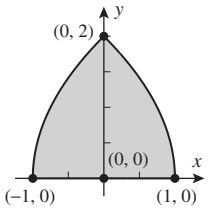
7. $x = \dfrac{\sqrt{u+v}}{\sqrt{2}}$, $y = \dfrac{\sqrt{v-u}}{\sqrt{2}}$; $\dfrac{1}{4\sqrt{v^2 - u^2}}$ **9.** 7 **11.** $-\dfrac{1}{v}$

Responses to True–False questions may be abridged to save space.

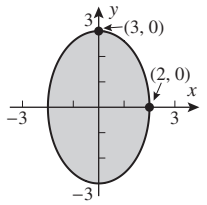
13. False; $|\partial(x, y)/\partial(u, v)| = \|\partial\mathbf{r}/\partial u \times \partial\mathbf{r}/\partial v\|$; evaluating this at (u_0, v_0) gives the area of the indicated parallelogram.

15. False; $\partial(x, y)/\partial(r, \theta) = r$.

17.



19.



21. $\frac{3}{2}\ln 3$ **23.** $1 - \frac{1}{2}\sin 2$ **25.** $\frac{800}{3}\pi$ **27.** $\dfrac{\pi}{24}(1 - \cos 1)$ **29.** $\frac{192}{5}\pi$

31. $u = \begin{cases} \cot^{-1}(x/y), & y \ne 0 \\ 0, & y = 0 \text{ and } x > 0 \\ \pi, & y = 0 \text{ and } x < 0 \end{cases}$

$v = \sqrt{x^2 + y^2}$; other answers possible

33. $u = (3/7)x - (2/7)y$, $v = (-1/7)x + (3/7)y$; other answers possible

35. $\frac{1}{4}\ln\frac{5}{2}$

37. $\dfrac{1}{2}\left[\ln(\sqrt{2} + 1) - \dfrac{\pi}{4}\right]$ **39.** $\frac{7}{50}$ **41.** $2\ln 2$ **45.** $21/8$

Exercise Set 14.8 (Page 1077)

1. $M = \frac{9}{10}$, center of gravity $\left(\frac{130}{189}, \frac{13}{27}\right)$

3. $M = a^4/8$, center of gravity $(8a/15, 8a/15)$

5. $\left(\frac{1}{2}, \frac{1}{2}\right)$ **7.** $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$

Responses to True–False questions may be abridged to save space.

9. True; recall this from Section 6.7.

11. False; the center of gravity of the lamina is $(\bar{x}, \bar{y}) = (M_y/M, M_x/M)$, where M_y and M_x are the lamina's first moments about the y- and x-axes, respectively, and M is the mass of the lamina.

15. $\left(\dfrac{128}{105\pi}, \dfrac{128}{105\pi}\right)$ **17.** $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$ **19.** $\left(\frac{1}{2}, 0, \frac{12}{5}\right)$

21. $(3a/8, 3a/8, 3a/8)$

23. $M = b^4/2$, center of gravity $(b/3, b/2, b/2)$

25. $M = \frac{1}{3}$, center of gravity $\left(0, \frac{16}{35}, \frac{1}{2}\right)$ **27.** **(a)** $\left(\frac{5}{8}, \frac{5}{8}\right)$ **(b)** $\left(\frac{2}{3}, \frac{1}{2}\right)$

29. (1.177406, 0.353554, 0.231557)

31. $\dfrac{27\pi}{4}$ **33.** $\pi k a^4$ **35.** $\left(0, 0, \frac{7}{16\sqrt{2} - 14}\right)$ **37.** $(3a/8, 3a/8, 3a/8)$

39. $4\pi(2 - \sqrt{2})$ **41.** $(0, 0, 8/15)$ **43.** $(0, 45/28, 0)$

47. $\frac{1}{2}\delta\pi a^4 h$ **49.** $\frac{1}{2}\delta\pi h(a_1^4 - a_1^4)$ **53.** $2\pi^2 abk$ **55.** $(a/3, b/3)$

Chapter 14 Review Exercises (Page 1080)

3. **(a)** $\iint_R dA$ **(b)** $\iiint_G dV$ **(c)** $\iint_R \sqrt{1 + \left(\dfrac{\partial z}{\partial x}\right)^2 + \left(\dfrac{\partial z}{\partial y}\right)^2}\, dA$

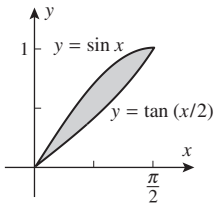
5. $\int_0^1 \int_{1-\sqrt{1-y^2}}^{1+\sqrt{1-y^2}} f(x, y)\, dx\, dy$

7. **(a)** $a = 2, b = 1, c = 1, d = 2$ or $a = 1, b = 2, c = 2, d = 1$ **(b)** 3

9. $-\dfrac{3}{2\sqrt{2}\pi}$

11. $\int_0^1 \int_{2y}^2 e^x e^y\, dx\, dy$

13.



15. $\frac{1}{3}(1 - \cos 64)$

17. a^2

19. $\frac{3}{2}$

21. 32π

23. **(a)** $\int_0^{2\pi} \int_0^{\pi/3} \int_0^a \rho^4 \sin^3\phi\, d\rho\, d\phi\, d\theta$

(b) $\int_0^{2\pi} \int_0^{\sqrt{3}a/2} \int_{r/\sqrt{3}}^{\sqrt{a^2-r^2}} r^3\, dz\, dr\, d\theta$

(c) $\int_{-\sqrt{3}a/2}^{\sqrt{3}a/2} \int_{-\sqrt{(3a^2/4)-x^2}}^{\sqrt{(3a^2/4)-x^2}} \int_{\sqrt{x^2+y^2}/\sqrt{3}}^{\sqrt{a^2-x^2-y^2}} (x^2 + y^2)\, dz\, dy\, dx$

25. $\dfrac{\pi a^3}{9}$ **27.** $\dfrac{13}{6}\sqrt{2} \approx 3.0641$ **29.** $2x + 6y - z = 10$

33. **(a)** $\dfrac{1}{2(u+w)}$ **(b)** $\frac{1}{2}(7\ln 7 - \ln 84,375)$ **35.** $\left(\frac{8}{5}, 0\right)$
37. $(0, 0, h/4)$

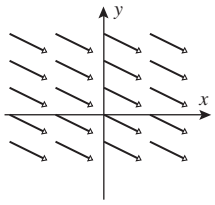
▶ Chapter 14 Making Connections (Page 1082)

Where correct answers to a Making Connections exercise may vary, no answer is listed. Sample answers for these questions are available on the Book Companion Site.

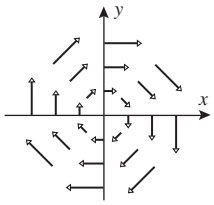
1. **(b)** $\dfrac{\pi}{4}$ **3.** **(a)** 1.173108605 **(b)** 1.173108605
4. **(a)** the sphere $0 \le x^2 + y^2 + z^2 \le 1$ **(b)** 4.934802202 **(c)** $\pi^2/2$
5. **(b)** 4.4506 **6.** $\frac{4}{35}\pi a^3$

▶ Exercise Set 15.1 (Page 1092)

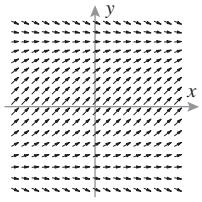
1. **(a)** III **(b)** IV **3.** **(a)** true **(b)** true **(c)** true
5.



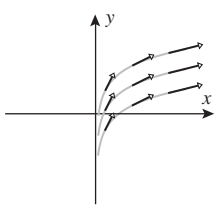
7.



9.



Responses to True–False questions may be abridged to save space.
11. False; the vector field has a nonzero **k**-component.
13. True; this is the curl of **F**.
15. **(a)** all x, y **(b)** all x, y **17.** div $\mathbf{F} = 2x + 2yz$, curl $\mathbf{F} = z^2\mathbf{i}$
19. div $\mathbf{F} = 0$, curl $\mathbf{F} = (40x^2z^4 - 16xy^3)\mathbf{i} + (14y^3z + 4y^4)\mathbf{j} - (16xz^5 + 21y^2z^2)\mathbf{k}$
21. div $\mathbf{F} = \dfrac{2}{\sqrt{x^2+y^2+z^2}}$, curl $\mathbf{F} = 0$ **23.** $5x$ **25.** 0
27. $\mathbf{i} + z\mathbf{j} + y\mathbf{k}$
39. $\nabla\cdot(k\mathbf{F}) = k\nabla\cdot\mathbf{F}$, $\nabla\cdot(\mathbf{F}+\mathbf{G}) = \nabla\cdot\mathbf{F} + \nabla\cdot\mathbf{G}$, $\nabla\cdot(\phi\mathbf{F}) = \phi\nabla\cdot\mathbf{F} + \nabla\phi\cdot\mathbf{F}$, $\nabla\cdot(\nabla\times\mathbf{F}) = 0$ **47.** **(b)** $x^2 + y^2 = K$
49. $\dfrac{dy}{dx} = \dfrac{1}{x}$, $y = \ln x + K$



▶ Exercise Set 15.2 (Page 1108)

1. **(a)** 1 **(b)** 0 **3.** 16
7. **(a)** $\frac{331}{108}\sqrt{10} - \frac{1}{36}\ln(3+\sqrt{10}) - \frac{4}{27}$ **(b)** 4 **(c)** $\frac{17}{2}$
9. **(a)** 4 **(b)** 4 **(c)** 4 **(d)** 4
Responses to True–False questions may be abridged to save space.
11. False; line integrals of functions are independent of the orientation of the curve. **13.** True; this is Equation (26).
15. $2\sqrt{5} - 2$ **17.** $\frac{46}{63}$ **19.** $3 - 2\pi$ **21.** 3
23. $-1 - (\pi/4)$ **25.** $1 - e^2$
27. **(a)** $\dfrac{63\sqrt{17}}{64} + \dfrac{1}{4}\ln(4+\sqrt{17}) - \dfrac{1}{8}\ln\dfrac{\sqrt{17}+1}{\sqrt{17}-1} - \dfrac{1}{4}\ln(\sqrt{2}+1) + \dfrac{1}{8}\ln\dfrac{\sqrt{2}+1}{\sqrt{2}-1}$ **(b)** $1/2 - \pi/4$
29. **(a)** -1 **(b)** -2 **31.** $\frac{5}{2}$ **33.** 0 **35.** $1 - e^{-1}$ **37.** $6\sqrt{3}$
39. $5k\tan^{-1}3$ **41.** $\frac{19}{35}$ **43.** $\frac{27}{28}$ **45.** $\frac{3}{4}$ **47.** $\dfrac{17\sqrt{17}-1}{4}$
49. **(b)** $S = \displaystyle\int_C z(t)\,dt$ **(c)** 4π **51.** $\lambda = -12$

▶ Exercise Set 15.3 (Page 1120)

1. conservative, $\phi = x^2 + y^2 + K$ **3.** not conservative
5. conservative, $\phi = x\cos y + y\sin x + K$
9. -8 **11.** $27e^2$ **13.** 32 **15.** $W = -8$ **17.** $W = e^2 - e^{-1}$
Responses to True–False questions may be abridged to save space.
19. False; the integral must be 0 for *all* closed curves C.
21. True; if $\nabla\phi$ is constant, then ϕ must be a linear function.
23. $\ln 2 - 1$ **25.** ≈ -0.307 **27.** no **33.** $h(x) = Ce^x$
35. **(a)** $W = -\dfrac{1}{\sqrt{14}} + \dfrac{1}{\sqrt{16}}$ **(b)** $W = -\dfrac{1}{\sqrt{14}} + \dfrac{1}{\sqrt{6}}$ **(c)** $W = 0$

▶ Exercise Set 15.4 (Page 1127)

1. 0 **3.** -6 **5.** π^2 **7.** 24π **9.** -4 **11.** -1 **13.** 0
Responses to True–False questions may be abridged to save space.
15. False; Green's Theorem applies to closed curves.
17. True; the integral is the area of the region bounded by C.
19. **(a)** ≈ -3.550999378 **(b)** ≈ -0.269616482 **21.** $\frac{3}{8}a^2\pi$ **23.** $\frac{1}{2}abt_0$
27. Formula (1) of Section 6.1 **29.** $\frac{128}{3}$ **31.** $-3\pi a^2$ **33.** $\left(\frac{8}{15}, \frac{8}{21}\right)$
35. $\left(0, \dfrac{4a}{3\pi}\right)$ **37.** the circle $x^4 + y^4 = 1$ **39.** 69

▶ Exercise Set 15.5 (Page 1136)

1. $40\pi\sqrt{2}$ **3.** $\dfrac{\pi}{4}$ **5.** $-2\sqrt{2}$ **7.** 9
Responses to True–False questions may be abridged to save space.
9. True; this follows from the definition.
11. False; the integral is the total mass of the lamina.
13. **(b)** $2\pi\left[1 - \sqrt{1-r^2} + \dfrac{r^2}{2}\right] \to 3\pi$ as $r \to 1^-$
(c) $\mathbf{r}(\phi,\theta) = \sin\phi\cos\theta\mathbf{i} + \sin\phi\sin\theta\mathbf{j} + \cos\phi\mathbf{k}$, $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi/2$;
$$\iint (1+z)\,dS = \int_0^{2\pi}\int_0^{\pi/2}(1+\cos\phi)\sin\phi\,d\phi\,d\theta = 3\pi$$
17. **(c)** $4\pi/3$
19. **(a)** $\dfrac{\sqrt{29}}{16}\displaystyle\int_0^6\int_0^{(12-2x)/3} xy(12-2x-3y)\,dy\,dx$
(b) $\dfrac{\sqrt{29}}{4}\displaystyle\int_0^3\int_0^{(12-4z)/3} yz(12-3y-4z)\,dy\,dz$
(c) $\dfrac{\sqrt{29}}{9}\displaystyle\int_0^3\int_0^{6-2z} xz(12-2x-4z)\,dx\,dz$
21. $\dfrac{18\sqrt{29}}{5}$
23. $\displaystyle\int_0^4\int_1^2 y^3z\sqrt{4y^2+1}\,dy\,dz$; $\dfrac{1}{2}\displaystyle\int_0^4\int_1^4 xz\sqrt{1+4x}\,dx\,dz$
25. $\dfrac{391\sqrt{17}}{15} - \dfrac{5\sqrt{5}}{3}$ **27.** $\frac{5}{3}\pi\delta_0$ **29.** $\frac{1}{3}(37\sqrt{37}-1)$ **31.** $M = \delta_0 S$
33. $(0, 0, 149/65)$ **35.** $\dfrac{363\sqrt{10}}{5}$ **37.** $\dfrac{\pi}{2}$ **39.** 57.895751

▶ Exercise Set 15.6 (Page 1146)

1. **(a)** zero **(b)** zero **(c)** positive **(d)** negative **(e)** zero **(f)** zero
3. **(a)** positive **(b)** zero **(c)** positive **(d)** zero **(e)** positive **(f)** zero
5. **(a)** $n = -\cos v\mathbf{i} - \sin v\mathbf{j}$ **(b)** inward **7.** 32π **9.** $\dfrac{52\pi}{3}$ **11.** 0
13. 21π **15.** $\frac{4}{9}$ **17.** **(a)** 8 **(b)** 24 **(c)** 0

Responses to True–False questions may be abridged to save space.

19. False; the Möbius strip has no orientation.

21. False; the net volume can be zero because as much fluid passes through the surface in the negative direction as in the positive direction.

23. $\frac{-52\pi}{3}$ **25.** **(a)** $0\ \text{m}^3/\text{s}$ **(b)** $0\ \text{kg/s}$ **27.** **(b)** $32/3$

29. **(a)** $4\pi a^{k+3}$ **(b)** $k = -3$ **31.** $a = 2, 3$

Exercise Set 15.7 (Page 1157)

1. 3 **3.** $\frac{4\pi}{3}$

Responses to True–False questions may be abridged to save space.

5. False; it equates a surface integral and a triple integral.

7. True; see subsection entitled Sources and Sinks.

9. 18 **11.** $4\pi a^2$ **13.** 352π **15.** $\frac{192\pi}{5}$ **17.** 8π

19. $\frac{4608}{35}$ **21.** 189π **33.** no sources or sinks

35. sources at all points except the origin, no sinks **37.** $\frac{7\pi}{4}$

Exercise Set 15.8 (Page 1164)

1. $\frac{3}{2}$ **3.** 0 **5.** 6π **7.** 16π **9.** 0 **11.** $2\pi a^2$

Responses to True–False questions may be abridged to save space.

13. True; see Theorem 15.8.1. **15.** False; the circulation is $\int_C \mathbf{F}\cdot\mathbf{T}\,ds$.

17. **(a)** $\frac{3}{2}$ **(b)** -1 **(c)** $-\frac{1}{\sqrt{2}}\mathbf{j} - \frac{1}{\sqrt{2}}\mathbf{k}$ **23.** $-\frac{5\pi}{4}$

Chapter 15 Review Exercises (Page 1166)

3. $\dfrac{1-x}{\sqrt{(1-x)^2+(3-y)^2}}\mathbf{i} + \dfrac{3-y}{\sqrt{(1-x)^2+(3-y)^2}}\mathbf{j}$ **5.** $\mathbf{i} + 2\mathbf{j} + \mathbf{k}$

7. **(a)** $\displaystyle\int_a^b \left[f(x(t), y(t))\frac{dx}{dt} + g(x(t), y(t))\frac{dy}{dt} \right] dt$

(b) $\displaystyle\int_a^b f(x(t), y(t))\sqrt{x'(t)^2 + y'(t)^2}\,dt$

11. 0 **13.** -7 **17.** **(a)** $h(x) = Cx^{-3/2}$ **(b)** $g(y) = C/y^3$

21. $A = \dfrac{1}{2}\displaystyle\int_\alpha^\beta r^2\,d\theta$

23. $\displaystyle\iint_R f(x(u,v), y(u,v), z(u,v))\|r_u \times r_v\|\,du\,dv$ **25.** yes **27.** 8π

31. -8π **35.** **(a)** conservative **(b)** not conservative

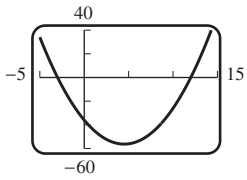
Chapter 15 Making Connections (Page 1168)

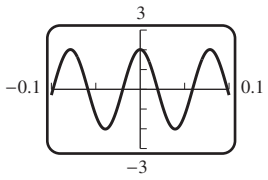
Answers are provided in the Student Solutions Manual.

Appendix A (Page A11)

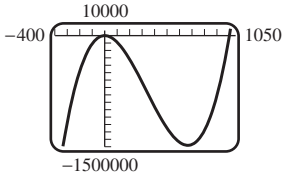
1. (e) **3.** (b), (c) **5.** $[-3, 3] \times [0, 5]$

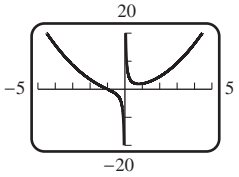
7. $[-5, 14] \times [-60, 40]$ **9.** $[-0.1, 0.1] \times [-3, 3]$





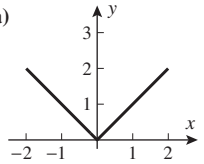
11. $[-400, 1050] \times [-1500000, 10000]$ **13.** $[-2, 2] \times [-20, 20]$



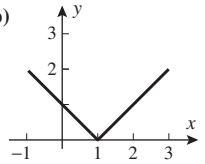


17. **(a)** $f(x) = \sqrt{16 - x^2}$ **(b)** $f(x) = -\sqrt{16 - x^2}$ **(e)** no

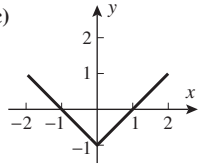
19. **(a)**



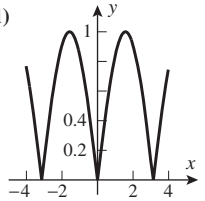
(b)



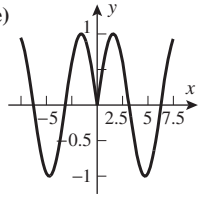
(c)



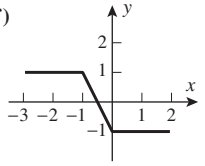
(d)



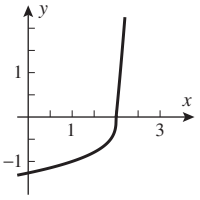
(e)



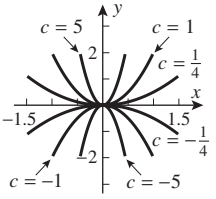
(f)



21.

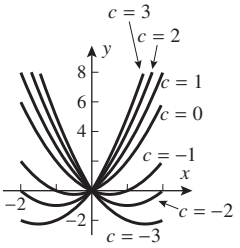


25. **(a)**



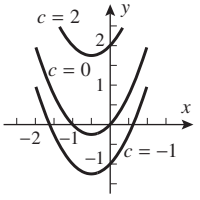
The graph is stretched in the vertical direction, and reflected across the x-axis if $c < 0$.

(b)



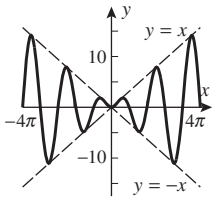
The graph is translated so its vertex is on the parabola $y = -x^2$.

(c)

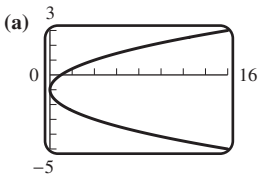


The graph is translated vertically.

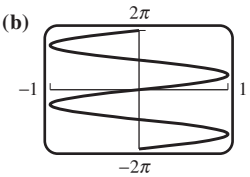
27.



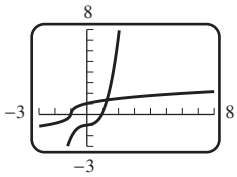
29. **(a)**



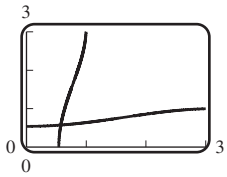
(b)



31.



33.



35. **(a)** $x = 4\cos t,\ y = 3\sin t$ **(b)** $x = -1 + 4\cos t,\ y = 2 + 3\sin t$

▶ Appendix B (Page A23)

1. **(a)** $\frac{5}{12}\pi$ **(b)** $\frac{13}{6}\pi$ **(c)** $\frac{1}{9}\pi$ **(d)** $\frac{23}{30}\pi$

3. **(a)** $12°$ **(b)** $(270/\pi)°$ **(c)** $288°$ **(d)** $540°$

5.

	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
(a)	$\sqrt{21}/5$	$2/5$	$\sqrt{21}/2$	$5/\sqrt{21}$	$5/2$	$2/\sqrt{21}$
(b)	$3/4$	$\sqrt{7}/4$	$3/\sqrt{7}$	$4/3$	$4/\sqrt{7}$	$\sqrt{7}/3$
(c)	$3/\sqrt{10}$	$1/\sqrt{10}$	3	$\sqrt{10}/3$	$\sqrt{10}$	$1/3$

7. $\sin\theta = 3/\sqrt{10}, \cos\theta = 1/\sqrt{10}$ **9.** $\tan\theta = \sqrt{21}/2, \csc\theta = 5/\sqrt{21}$

11. 1.8

13.

	θ	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
(a)	$225°$	$-1/\sqrt{2}$	$-1/\sqrt{2}$	1	$-\sqrt{2}$	$-\sqrt{2}$	1
(b)	$-210°$	$1/2$	$-\sqrt{3}/2$	$-1/\sqrt{3}$	2	$-2/\sqrt{3}$	$-\sqrt{3}$
(c)	$5\pi/3$	$-\sqrt{3}/2$	$1/2$	$-\sqrt{3}$	$-2/\sqrt{3}$	2	$-1/\sqrt{3}$
(d)	$-3\pi/2$	1	0	—	1	—	0

15.

	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
(a)	$4/5$	$3/5$	$4/3$	$5/4$	$5/3$	$3/4$
(b)	$-4/5$	$3/5$	$-4/3$	$-5/4$	$5/3$	$-3/4$
(c)	$1/2$	$-\sqrt{3}/2$	$-1/\sqrt{3}$	2	$-2/\sqrt{3}$	$-\sqrt{3}$
(d)	$-1/2$	$\sqrt{3}/2$	$-1/\sqrt{3}$	-2	$2/\sqrt{3}$	$-\sqrt{3}$
(e)	$1/\sqrt{2}$	$1/\sqrt{2}$	1	$\sqrt{2}$	$\sqrt{2}$	1
(f)	$1/\sqrt{2}$	$-1/\sqrt{2}$	-1	$\sqrt{2}$	$-\sqrt{2}$	-1

17. **(a)** 1.2679 **(b)** 3.5753

19.

	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
(a)	$a/3$	$\sqrt{9-a^2}/3$	$a/\sqrt{9-a^2}$	$3/a$	$3/\sqrt{9-a^2}$	$\sqrt{9-a^2}/a$
(b)	$a/\sqrt{a^2+25}$	$5/\sqrt{a^2+25}$	$a/5$	$\sqrt{a^2+25}/a$	$\sqrt{a^2+25}/5$	$5/a$
(c)	$\sqrt{a^2-1}/a$	$1/a$	$\sqrt{a^2-1}$	$a/\sqrt{a^2-1}$	a	$1/\sqrt{a^2-1}$

21. **(a)** $3\pi/4 \pm n\pi, n = 0, 1, 2, \ldots$
(b) $\pi/3 \pm 2n\pi$ and $5\pi/3 \pm 2n\pi, n = 0, 1, 2, \ldots$

23. **(a)** $\pi/6 \pm n\pi, n = 0, 1, 2, \ldots$
(b) $4\pi/3 \pm 2n\pi$ and $5\pi/3 \pm 2n\pi, n = 0, 1, 2, \ldots$

25. **(a)** $3\pi/4 \pm n\pi, n = 0, 1, 2, \ldots$
(b) $\pi/6 \pm n\pi, n = 0, 1, 2, \ldots$

27. **(a)** $\pi/3 \pm 2n\pi$ and $2\pi/3 \pm 2n\pi, n = 0, 1, 2, \ldots$
(b) $\pi/6 \pm 2n\pi$ and $11\pi/6 \pm 2n\pi, n = 0, 1, 2, \ldots$

29. $\sin\theta = 2/5, \cos\theta = -\sqrt{21}/5, \tan\theta = -2/\sqrt{21}$,
$\csc\theta = 5/2, \sec\theta = -5/\sqrt{21}, \cot\theta = -\sqrt{21}/2$

31. **(a)** $\theta = \pm n\pi, n = 0, 1, 2, \ldots$ **(b)** $\theta = \pi/2 \pm n\pi, n = 0, 1, 2, \ldots$
(c) $\theta = \pm n\pi, n = 0, 1, 2, \ldots$ **(d)** $\theta = \pm n\pi, n = 0, 1, 2, \ldots$
(e) $\theta = \pi/2 \pm n\pi, n = 0, 1, 2, \ldots$ **(f)** $\theta = \pm n\pi, n = 0, 1, 2, \ldots$

33. **(a)** $2\pi/3$ cm **(b)** $10\pi/3$ cm **35.** $\frac{2}{5}$

37. **(a)** $\dfrac{2\pi - \theta}{2\pi}R$ **(b)** $\dfrac{\sqrt{4\pi\theta - \theta^2}}{2\pi}R$ **39.** $\frac{21}{4}\sqrt{3}$ **41.** 9.2 ft

43. $h = d(\tan\beta - \tan\alpha)$ **45.** **(a)** $4\sqrt{5}/9$ **(b)** $-\frac{1}{9}$

47. $\sin 3\theta = 3\sin\theta\cos^2\theta - \sin^3\theta, \cos 3\theta = \cos^3\theta - 3\sin^2\theta\cos\theta$

61. **(a)** $\cos\theta$ **(b)** $-\sin\theta$ **(c)** $-\cos\theta$ **(d)** $\sin\theta$

69. **(a)** $153°$ **(b)** $45°$ **(c)** $117°$ **(d)** $89°$ **71.** **(a)** $60°$ **(b)** $117°$

▶ Appendix C (Page A32)

1. **(a)** $q(x) = x^2 + 4x + 2, r(x) = -11x + 6$
(b) $q(x) = 2x^2 + 4, r(x) = 9$
(c) $q(x) = x^3 - x^2 + 2x - 2, r(x) = 2x + 1$

3. **(a)** $q(x) = 3x^2 + 6x + 8, r(x) = 15$
(b) $q(x) = x^3 - 5x^2 + 20x - 100, r(x) = 504$
(c) $q(x) = x^4 + x^3 + x^2 + x + 1, r(x) = 0$

5.

x	0	1	−3	7
$p(x)$	−4	−3	101	5001

7. **(a)** $q(x) = x^2 + 6x + 13, r = 20$ **(b)** $q(x) = x^2 + 3x - 2, r = -4$

9. **(a)** $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 8, \pm 12, \pm 24$
(b) $\pm 1, \pm 2, \pm 5, \pm 10, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{5}{3}, \pm\frac{10}{3}$ **(c)** $\pm 1, \pm 17$

11. $(x+1)(x-1)(x-2)$ **13.** $(x+3)^3(x+1)$

15. $(x+3)(x+2)(x+1)^2(x-3)$ **17.** -3 **19.** $-2, -\frac{2}{3}, -1 \pm \sqrt{3}$

21. $-2, 2, 3$ **23.** $2, 5$ **25.** 7 cm

INDEX

Abel, Niels Henrik, 297
abscissa, Web-G1
absolute convergence, 641, 642
  ratio test for, 643, 644
absolute error, 116
  Euler's Method, 583
absolute extrema
  Extreme-Value Theorem, 267
  finding on closed and bounded sets, 982–985
  finding on finite closed intervals, 268
  finding on infinite intervals, 269, 270
  functions with one relative extrema, 270, 271
  on open intervals, 270
absolute extremum, 266, 977
absolute maximum, 266, 977
absolute minimum, 266, 977
absolute minimum values, 977
absolute value function, 5
  continuity, 114
  derivative of, 147
  properties, 5
absolute values, Web-F1–Web-F5
  and square roots, Web-F1, Web-F2
absolute zero, 988
acceleration, 290
  centripetal, 903
  constant, 378–380, Web-G11
  due to gravity, 127, 380
  instantaneous, 290
  motion along curves, 882
  normal and tangential components, 885–888
  rectilinear motion, 290
  sign of, 291
  tangential scalar/vector component, 886
  vector, 883
acceleration function, 290
addition
  formulas for sine and cosine, A20
  of functions, 15, 16
Agnesi, Maria, 841, 847
air resistance, 590
Alexeev, Vasili, 450
algebraic functions, 32
  integration formulas, 489
algebraic manipulation, of integrand, 488
alternating current, 395
alternating harmonic series, 639
alternating series, 638–641
alternating series test, 638, 645, 671
amplitude
  alternating current, 395
  simple harmonic motion, 180, Web-L5
  $\sin x$ and $\cos x$, 33
analytic geometry, 275, Web-E3
Anderson, Paul, 450
angle(s), A13–A15
  finding from trigonometric functions, A22
  of inclination, A23
  between planes, 816
  polar, 706
  rectangular coordinate system, A16–A18
  standard position, A16
  between vectors, 786–787
angular coordinate, 706
angular frequency, simple harmonic motion, 180
annuity, 301
antiderivative(s), 322, 323
  of vector-valued functions, 854
antidifferentiation, 320, 323
aphelion, 760, 900
apogee, 759, 900
  artificial Earth satellite, 210, 900
approximation(s)
  area under a curve, 319
  left, right, and midpoint, 533
  local linear, 943
  local linearity, 212–213
  local quadratic, 648
  Maclaurin and Taylor polynomial, 649, 651, 652, 655, 656
  midpoint, 534, 536, 537
  Riemann sum, 354
  of roots using Intermediate-Value Theorem, 116–117

of roots using Newton's Method, 297–300
of roots by zooming, 117
Simpson's rule, 537, 538, 540
trapezoidal, 534
arc functions, 44
arc length, 438, 440, 441, A14, A15
finding by numerical methods, 441
and improper integrals, 553
as a line integral, 1095
parametric curves, 443, 697
polar curve, 721, 722
from vector viewpoint, 859
arc length parametrization, 860
finding, 862–864
properties, 864, 865
Archimedean spiral, 714, 718
Archimedes, 316, 318
palimpsest, 318
area
antiderivative method, 320, 321
calculated as double integral, 1014
computing exact value of, 346
definition, 343, 344
finding with Green's Theorem, 1124–1125
function of two variables, 906
left endpoint approximation, 347
as a line integral, 1095, 1099, 1124–1125
midpoint approximation, 347
net signed, 347, 350, 355, 377
polar coordinates, 724, 725
with polar double integrals, 1022
rectangle method, 317, 321
right endpoint approximation, 347
surface of revolution, 444, 446
total, 366, 378
between two curves, 414, 416–418
under a curve, 344
area problem, 317
argument, 3
arithmetic mean (or average), 385, 757
aspect ratio distortion, A6
Astronomia Nova, 759
astronomy, planetary motions, 759, 896, 897, 899, 900
asymptote(s), 99, 254
a curve as an, 99
curvilinear, 258
horizontal, 32, 89
oblique, 258
slant, 258
vertical, 32, 76
asymptotes of a hyperbola, 732
finding, 738
asymptotic curves, 99
autonomous
differential equation, 585
auxiliary equation, Web-L2
average rate of change, 138
average value, 385, 386, 1008, 1017
and average velocity, 387
average velocity, 135, 385, 387
geometric interpretation, 136
axis (axes)
coordinate, Web-G1
of an ellipse, 732
of a hyperbola, 732
of a parabola, 732
polar, 705
of revolution, 425
rotation of, 749, 750

Badel, Hannskarl, 484
ball
closed, 920
open, 920
base *b* exponential function, 401
Bell, Alexander Graham, 59
Bernoulli equation, 595
Bernoulli, Daniel, 700
Bernoulli, Jakob, 767
Bernoulli, Jakob I, 698–700
Bernoulli, Johann (John), 3, 221, 767
Bernoulli, Johann (John) I, 698–700
Bernoulli, Nikolaus, 700
Bessel, Friedrich Wilhelm, 666
Bessel equation of order one, 689
Bessel equation of order zero, 689
Bessel functions, 666
derivative of, 679
Beverton–Holt model, 613–614
bifolium, 728
binomial coefficient, 677
binomial series, 674, 675
binormal vectors, 871
formula summary, 878
bipartite cubics, A12
Bolzano, Bernhard, 149
Bopp, Thomas, 762
boundary, 920
boundary point, 920
bounded functions, 359
bounded sets, 978
finding absolute extrema, 982
Bowditch, Nathaniel, 705
Bowditch curves, 705
Boyle's law, 30, 36
brachistochrone problem, 699
Brahe, Tycho, 759

branches, of hyperbola, 731
breakpoints, 6
butterfly curve, 717

carbon dating, 574
cardioid, 711
  area, 724
  families of, 713
  intersection with circle, 725
carrying capacity, 100, 563
Cartesian coordinate system, 767, Web-G1
Cassini, Giovanni Domenico, 719
Cassini ovals, 719
catenary, 475
Cauchy, Augustin, 649, 651
Cauchy–Riemann equations, 939
  polar form, 959
Cavalieri, Bonaventura, 431
Cavalieri's principle, 431
center
  circles, Web-H3
  ellipse, 731
  hyperbola, 732
center of curvature, 877
center of force, 896
center of gravity, 458, 1072
  lamina, 458–462, 1072, 1074
  solids, 1075, 1076
centimeter-gram-second (CGS), 450
central angle, cylindrical wedges, 1048
central force field, 896
central forces, 896
centripetal acceleration, 903
centripetal force, 903
centroid(s), 464
  geometric property, 462
  lamina, 1074
  of a region, 462–464
  of the region R, 1074
  solids, 1075, 1076
chain rule, 174
  alternative version, 175
  derivatives, 949–952, 954
  functions of n variables, 953
  generalized derivative formulas, 176
  implicit differentiation, 955
  implicit partial differentiation, 955–956
  Leibniz notation, 213
  partial derivatives, 952–953
  proof, A36
  two-variable, A43
change of parameter, 861, 862
change of variables
  double integrals, 1063–1065
  single integral, 1058
  triple integrals, 1065–1067
chaos
  and Newton's Method, 300
circle(s), 730, Web-H2–Web-H4
  of curvature, 877
  degenerate cases, Web-H4, Web-H5
  and discriminant, Web-K1
  families, 712, 713
  involute, 704
  osculating, 130
circular cone, 824–825
circular helix, 842
circular paraboloid, 824–825
circulation, 1163
circulation density, 1164
closed ball, 920
closed disk, 919
closed form, sigma notation, 343
closed interval(s), Web-E4
  absolute extrema on finite, 267
  continuity on, 112
closed parametric curves, 1113
closed sets, 919
  finding absolute extrema, 982–985
closed surfaces, 1148
clothoid, 766
Cobb–Douglas production model, 999
coefficient of friction, 154, 180
coefficient of linear expansion, 219
coefficient of volume expansion, 219
coefficients, A27
  correlation, Web-J2
  leading, A27
  polynomials, 31
cofunction, 181
Commentaries on the Motions of Mars, 759
common logarithms, 55
comparison test, 631, 632, 645
complete elliptic integral of the first kind, 560, 685
complete linear factorization, A28
Completeness Axiom, 611
completing the square, 20, Web-H4
complex numbers, Web-E1, Web-E2
compliance, Web-G12
component functions, 843
components of $r(t)$, 843
components, of vectors, 775
composition of functions, 17
  continuity, 114, 115
  derivatives, 174
  u-substitutions for, 332, 334–336

compounding interest
  annually, 301, 577
  continuously, 129, 577
  n times each year, 129
  quarterly, 577
  semiannually, 577
compressible fluids, 1140
computer algebra system(s) (CAS), 530, A1
  differentiation using, 178
  integration using, 338, 488, 528, 529
  linear systems, 519
  Maple, A1, A2
  Mathematica, A1, A2
  polynomial roots, A31
concave down, 235
concave up, 235
concavity, 235, 254
conchoid of Nicomedes, 764
concurrent forces, 781
conditionally convergent, 643
cone, 824, 830, 836
conic sections, 730–732
  applications, 743
  degenerate, 730, Web-K2
  focus–direction characterization, 754
  polar equation, 755, 756
  quadratics, 748–752, Web-K2
  reflection properties, 742, 743
  sketching in polar coordinates, 756–758
  translated, 740–742
conjugate axis, 732
connected domain, 1114
conservation of energy principle, 1119, 1168
conservative fields, 1087–1088
conservative vector fields, 1087–1088, 1112–1113
  path independence and, 1113–1114
  test for, 1115–1118
Constant Difference Theorem, 306, 307
constant function, 27, 80, 155, 233
constant of integration, 323
constant of proportionality, 29, 36, Web-G11
  inverse proportionality, 29, 36
constant polynomial, A27
constant surfaces, 832–833
constant term, polynomials, A27
constant u-curves, 1028, 1032, 1060
constant v-curves, 1028, 1032, 1060
constant(s)
  derivative, 155
  factor, in a derivative, 157
  factor, in a limit, 82
  and graph reflections, 22
  and graph translations, 20
  integration formulas, 489
  second-order linear homogeneous equations, Web-L2
constrained absolute maximum (minimum), 991–993
constrained relative maximum (minimum), 991–993
constrained-extremum principle
  three variables and one constraint, 993–994
  two variables and one constraint, 991–993
continuity, 110–117
  applications, 111
  compositions, 114–115
  and differentiability, 148, 943
  exponential function, 122, 400
  functions of three variables, 925
  functions of two variables, 922–923, 924
  Intermediate-Value Theorem, 115–117
  on an interval, 112
  inverse functions, 121
  from the left/right, 112
  logarithm function, 398, 403
  logarithmic function, 122
  polynomials and rational functions, 113
  proof of basic property, A35–A36
  trigonometric functions, 121
  vector-valued functions, 848
  of vector-valued functions, 849
continuity equation, for incompressible fluids, 1154
continuous compounding, 129, 577
continuous functions
  average value, 385–387
  properties, 113
contour maps, 909–911
contour plot, 909
  using technology, 911
convergence, 600, 610, 611, 614–616
  absolute, 641, 642
  and algebraic properties of series, 624, 625
  conditional, 643
  and eventual behavior of sequences, 610
  improper integral, 549–551
  infinite series, 616
  monotone sequences, 610, 611
  radius/interval of, 662, 664
  sequences, 600
convergence set, 662
convergence tests
  alternating series, 638–641
  comparison test, 631, 632
  integral test, 626, 628
  limit comparison test, 633, 634
  p-series, 627
  ratio test, 634, 635
  root test, 635
  summarized, 645

coordinate axes, Web-G1
coordinate line, Web-E3
coordinate planes, 768
coordinates, Web-E3, Web-G1, Web-G2
  Cartesian, 767, Web-G1
  converting between, 833–834
  cylindrical, 832
  polar, 705, 706
  spherical, 832, 836
Copernicus, 759
corner points, 147
Cornu, Marie Alfred, 766
Cornu spiral, 766, 904
correlation, Web-J1
correlation coefficient, Web-J2
cosecant, A15
  continuity, 121
  derivative, 170
  hyperbolic, 474
cosine, A15
  continuity, 121
  derivative of, 169, 173
  family, 32, 34
  formulas, A20, A22
  hyperbolic, 474
  integrating powers of, 500, 501, 505
  integrating products with sine, 501, 503
  rational functions of, 527
  trigonometric identities, A18–A20
cost function, 281
cotangent, A15
  continuity, 121
  derivative, 170
  hyperbolic, 474
Coulomb's law, 36, 1087
CRC Standard Mathematical Tables and Formulae, 488, 523
Crelle, Leopold, 297
critical point, 245, 979
cross product, 796
  algebraic properties, 797–798
  coordinate independent, 801
  derivatives, 852
  geometric properties, 798–799
cross-product term, 750, 751
cross section, 417, 421
cubic polynomial, 31, A27
curl, 1088–1090
  viewed as circulation, 1163
curl field, 1165
curvature, 873, 875, 877
  2-space interpretation, 877
curve(s)
  as asymptote, 99
  integral, 327, 562
  length of, 438–440, 697, 859, 860
  level, 909–911
  limits along, 917
  orthogonal, 191
  parametric, 694, Web-I2
  piecewise smooth, 1107
  polar, 707
  position versus time, 288
  smooth, 438
  velocity versus time, 377
curve fitting, 987
curve sketching, 254
curved lamina, 1130
curvilinear asymptotes, 475
cusp, 259
cycle, 395
cycloid, 698
  apple of discord, 699
cycloids
  prolate, 703
  role in mathematics history, 700
cylinder, 421, 422
cylindrical coordinates, 832
  converting, 833
  equations of surfaces in, 835–836
cylindrical elements of volume, 1048
cylindrical shells, 432–435
cylindrical surfaces, 770–771
cylindrical wedges, 1048, 1049

da Pisa, Leonardo (Fibonacci), 607
damping constant, Web-L8
decay constant, 571
  interpreting, 572
decibels, 59
decimal representations, Web-E2
decreases without bound, 75
decreasing functions, 233
  invertibility, 42
decreasing sequence, 607
definite integrals
  defined, 354
  and Fundamental Theorem of Calculus, 363, 365, 366, 369, 370
  integration by parts, 496
  properties of, 356–358
  substitution, 390–392
degenerate conic sections, 730, Web-K2
degree, polynomials, 31, A27
degree measure, A13
del ∇ operator, 1090

density
circulation, 1164
flux, 1154
lamina, 458, 1130
of lamina, 1071
mass, 469
weight, 469
density function, 1071
dependent variable, 3
derivative function, 143–145
defined, 143
domain, 143
"prime" notation, 143
slope producing, 144
derivative(s)
compositions, 174
constant function, 155
constant multiple rule, 157
defined, 143, 150, 151
difference quotient, 143
of dot and cross products, 852
equation of the tangent line, 144
exponential functions, 198
inverse trigonometric functions, 200
left-hand, 150
of a linear function, 145
logarithmic functions, 193
notation, 143, 150, 151
nth order, 160
one-sided, 150
of position function, 146
power rule, 156, 195
of a product, 163, 164
of a quotient, 165
of a reciprocal, 167
right-hand, 150
trigonometric functions, 169–172
of vector-valued functions, 849, 850
velocity as, 146
Descartes, René, 187, 275, 468, 698, Web-E3
determinants, 795
deterministic phenomena, Web-J1
difference quotient, 143
differentiability, 146, 148, 150, 151
and continuity, 943
and continuity, 148
continuity of first-order partial derivatives, 943
functions defined implicitly, 190
functions of two or three variables, 940–942
geometric interpretation, 147
lack of, 147
vector-valued functions, 849
of vector-valued functions, 850
differentiable
on (a, b), 146
everywhere, 146, 943
on an interval, 150
on R, 943
at x_0, 146
differential equation(s), 561–562
autonomous, 585
Bernoulli, 595
homogeneous, 595
initial-value problems, 328
and integration, 328
order of, 561
and rates of change, 561
separable, 568–571
slope/direction field for, 329, 580
solutions, 561
differentials, 213–214, 944, 945
error propagation, 215–216
formulas, 216–217
differentiation, 150
computer algebra systems, 178
implicit, 187–190, 952, 955
as inverse property of integration, 371
logarithmic, 194–195, 199
of power series, 678, 679
and related rates, 204–208
rules, 167
techniques of, 155, 156, 158–160, 163–167, 174
Diophantine analysis, 303
Diophantus, 275
direct proportionality, Web-G11, Web-G12
direction angles, 787–788
direction cosines, 787–788
direction field, 329, 580
direction of increasing parameter, 694, Web-I2
in 3-space, 841
directional derivatives, 960–963
directrix, 731, 754
discontinuities, 29, 110–111, Web-G4
infinite, 110
and integrability, 358
jump, 110
limits at, 119, 923
physical significance, 111
removable, 111, 119
at $x = c$, 110
discriminant, 752, Web-K1, Web-K2
disease, first-order differential equations modeling spread of, 564
disk
closed, 919
open, 919

displacement, 135
finding by integration, 372, 376
motion along curves, 884
from velocity versus time curve, 378
displacement vector, 773, 791
distance, 3-space, 768
distance formula, 768, Web-H1
distance traveled
finding by integration, 376
motion along curves, 884
rectilinear motion, 377
from velocity versus time curve, 378
divergence, 1088–1090, 1149
absolute, 641
as flux density, 1154
improper integral, 549–551
of infinite series, 616
of sequences, 600
divergence test, 623, 624, 645
Divergence Theorem, 1149–1151
finding flux, 1151–1153
inverse-square fields, 1155
division
of functions, 15, 16
by zero, Web-E2
domain, 7
effect of algebraic operations on, 8
functions of two or more variables, 907
inverse functions, 40, 44
natural, 7
physical considerations in applications, 9, 10
restricted, 44
restricting to make functions invertible, 44
vector-valued functions, 843
dot products, 785
algebraic properties, 785
coordinate independent, 801
derivatives, 852
sign of, 787
double integrals, 1002
area calculated as, 1014
change of variables, 1063, 1065
converting from rectangular to polar coordinates, 1023
evaluating, 1002–1005
limits of integration, 1011–1013, 1021
over nonrectangular regions, 1009–1011
in polar coordinates, 1019, 1020
properties of, 1006
in rectangular coordinates, 1020
reversing order of integration, 1013
double-angle formulas, A21
doubling time, 572–573
drag force, 590
dummy variables, 367, 368
dx, 213
dy, 213
dy/dx, 150, 213
dyne-centimeters (erg), 450
dynes, 450

e, 193, 400, 402
Earth
Mercator projections, 506
spherical coordinates, 836
Earth-observing satellite, 173
earthquake intensity, 59, 63
eccentricity, 754
as measure of flatness, 755
ecliptic, 896
economics, optimization problems, 281, 282
electric intensity vector, 970
electrostatic field, 1086–1087, 1138
elementary functions, 403
elements, of sets, Web-E4
ellipses, 730, 731
area, 746, 1125
and discriminant, Web-K1
eccentricity as measure of flatness, 755
focus–directrix property, 755
polar equation, 756
reflection properties, 743
sketching, 736, 737
sketching in polar coordinates, 756–758
standard equations, 734, 735
translated, 740
ellipsoids, 823, 824
identifying, 829
elliptic cones, 823–825
identifying, 829
elliptic paraboloids, 823, 824, 826
identifying, 829
elliptical orbits, 760, 895, 899
empty set, Web-E4
end behavior, 89
and limits, 91, 94
Endpaper Integral Table, 523–528
energy, 449
conservation of, 1119
kinetic, 449, 455
potential, 1119
relationship to work, 454–455
equal vectors, 774
equiangular spirals, 729
equilateral hyperbola, 29, 740
equilibrium, 460
equipotential curves, 917
erg, 450

error
  absolute, 116, 583
  in local linear approximations, 213
  measurement, 215
  percentage, 216, 583
  propagated, 215
  relative, 216
error function [erf(x)], 404, 530
error propagation, 215
escape speed, 899
Euler, Leonhard, 3, 622, 990
Euler's equidimensional equation, Web-L7
Euler's Method, 581–583
even function, 23
eventually holding properties, 610
evolute, 880
existence theorem, 267
exp, 54
exponential decay, 571–573
exponential functions, 53
  analyzing growth with L'Hôpital's rule, 223
  approximating with power series, 672
  with base b, 53
  continuity, 122
  derivatives, 198–199, 230
  end behavior, 95
  equation solving, 57, 58
  graph, 53
  integration formulas, 489
  natural, 54
  radioactive decay, 230
exponential growth, 60, 571–573
extrema, 977
Extreme-Value Theorem, 267, 274, 978
extremum (extrema), 244, 267, 268
  absolute, 266
  relative, 244, 254
extremum problem
  applied, 274–278, 280–282
  one-variable, 267
  three-variable with one constraint, 989, 993–995
  two variable with one constraint, 989, 991–993
extrusion, 770

Factor Theorem, A30
factorial (n!), 603, 650
factors, polynomials, A30
false gaps, graphing utilities, A8
false line segments, graphing utilities, A8
families
  of cardioids, 713, 714
  of circles, 712, 713
  of curves, 27
  of functions, 30–34
  of limaçons, 713, 714
  of rose curves, 713
  of spirals, 714
families of functions
  cosine, 32, 34
  exponential, 53
  sine, 32, 34
Fermat's last theorem, 275
Fermat's principle, 287
Fermat, Pierre de, 274, 698
Fibonacci sequence, 607
finite intervals, Web-E4
finite solid, 1039
first-degree equation
  x and y, Web-G10
first derivative test, 246, 247
first moment, of lamina, 1074
first octant, 768
first-order differential equations, 561, 562, 570, 586–591
  functions of two variables, 579–580
  linear, 586
  mixing problems, 589–590
  modeling with, 571–574, 589–591
  separable, 569–570
  slope fields, 580
first-order initial-value problem, 562
first-order model, pendulum period, 686
first-order partial derivatives, 933
flow fields, 1138
flow line, 1093
fluid, 467
  dynamics, 467
  force, vertical surface, 470–471
  homogeneous, 469
  pressure, 469
  statics, 467
fluid flow, 1084
  steady state, 1140
flux, 1140, 1142
  density, 1154
  of **F** across σ, 1142
  finding with Divergence Theorem, 1151–1153
  integrals, 1142
  outward, 1151–1153
fluxion, 1084
focal axis, 732
focus, 731, 754
  ellipse, 731
  hyperbola, 731
  parabola, 731
focus–directrix property, of conics, 754
Folium of Descartes, 186, 189, 728

Folsom points, 576
foot-pounds, 450
force(s)
constant, 449–451
fluid, 471
line of action, 805
pressure, 468–469
resultant of two concurrent forces, 780–781
variable, 451–453
force field
work performed by, 1105
force vector, 774
four-cusped hypocycloid, 26, 190
four-petal rose, 709, 713
fractals, and Newton's Method, 300
free-fall model, 380
integration, 380, 382
with resistance, 590–591
free motion, Web-L8
Frenet, Jean Frédéric, 871
Frenet frame, 871
Frenet–Serret formulas, 881
frequency
alternating current, 395
simple harmonic motion, 180, Web-L6
$\sin x$ and $\cos x$, 33
Fresnel, Augustin, 404
Fresnel cosine function, 404
Fresnel sine function, 404
Fubini, Guido, 1005
function(s), 1, 6
adding, 15
approximating with power series, 670–675
arithmetic operations on, 15
bounded, 359
compositions, 17, 18
constant, 27
decreasing, 42
defined by integrals, 403, 404
defined by power series, 665, 666
defined explicitly and implicitly, 185–186
described by tables, 908
differentiable, 146
domain, 7
elementary, 403
even, 23, 28, 29
expressing as composition, 18, 19
graphical information analysis, 1
graphing utilities, A1
graphs of, 4
homogeneous of degree n, 999
increase without bound, 60
increasing, 42
increasing and decreasing, 232–234
interpolating, Web-J1
inverse, 38–40
inversing, 42
invertible, 41
logarithmic, 55
natural domain, 7
odd, 23, 28, 29
one-to-one, 41
partial derivatives, 928
piecewise-defined, 6
properties, 4–10
range, 7
relative maxima and minima, 244
restriction, 44
smooth, 438, 858
of three variables, 907
of two variables, 579–580, 907
vertical line test, 4
ways of describing, 2
functions of n real variables, 907
functions of three variables
continuity, 924
gradient, 963
level surfaces, 911–912
multiple integrals, 1000, 1039–1042
partial derivatives, 932
functions of two variables
continuity, 922–923, 924
contour maps, 909–911
gradient, 963
graphing utilities, 912, 914
graphs, 908–909
maxima and minima, 977, 979–982, 984–985
multiple integrals, 1000–1003, 1005, 1006, 1009–1011
partial derivatives, 927, 930–932
Fundamental Theorem of Calculus, 362–364, 369, 370
dummy variables, 367, 368
and integrating rates of change, 371
relationship between definite and indefinite integrals, 365, 366
vector-form, 854
Fundamental Theorem of Work Integrals, 1112–1113

g, acceleration due to gravity, 127
Gabriel's horn, 555, 1030
Galileo, 698
Galileo, 763
Galois, Evariste, A31
Gamma function, 559
gases, 467
Gateway Arch, 484
Gauss, Carl Friedrich, 354, 1084, 1150
Gaussian curve, 1150

Gauss's law
for electric fields, 1156
for inverse-square fields, 1155
Gauss's Theorem, 1149–1151
general equation
circle, Web-H5
lines, Web-G10
general form, equation of plane, 814
general logarithms, 402, 403
general solution, 562
general term, 597, 623
geometric curves, Web-G2
geometric mean, 757, 758
geometric series, 617, 619
geosynchronous orbit, 900
gradient fields, 1087
gradients, 963–965
to find tangent lines to intersections of surfaces, 974
graphing using calculus and technology together, 262
graphing utilities
aspect ratio distortion, A6
contour plots, 911
errors of omission, A8
false gaps and line segments, A8
functions of two variables, 912, 914
inverse function graphing, A9, Web-I4
nonelementary functions, 407
parametric curve generation, 842, A9, Web-I3
polar curves, 715
rational functions, 261
resolution, A7
root approximation by zooming, 117
sampling error, A7
and true shape of graph, A9
graphs, Web-G2
conic sections in polar coordinates, 756–758
exponential functions, 54
functions, 4
functions of two variables, 908–909
hyperbolic functions, 475
inverse functions, 43
logarithmic functions, 56
parametric curves in 3-space, 842
polar coordinates, 707, 709
power functions, 29
properties of interest, 254
quadratic equation, Web-H5–Web-H7
quadric surfaces, 824–826
rational functions, 255, 261
reflections, 21
roots, 4
sequence, 599
stretches and compressions, 22
symmetry, 23
in 3-space, 769
translations, 20
vector-valued function of two variables, 1031
vector-valued functions, 844, 845
vertical tangents and cusps, 259
x-intercepts, 4
zeros, 4
gravitational field, 1084
great circle, 840
great circle distance, 840
Great Pyramid at Giza, 230
greatest integer function, 27
greatest lower bound, 612
Green, George, 1123
Green's Theorem, 1122–1123
finding area using, 1124–1125
finding work using, 1124
multiply connected regions, 1125–1126
and Stokes' Theorem, 1162
Gregory, James, 673
Gregory of St. Vincent, 397
grid lines, A2
growth constant, 571
interpreting, 572
growth rate, population, 182

Hale, Alan, 762
Hale–Bopp comet, 762
half-angle formulas, A22
half-closed intervals, Web-E4
half-life, 572–573
half-open intervals, Web-E4
Halley's comet, 760
hanging cables, 475, 765
harmonic motion, simple, 180, 567
harmonic series, 620
heat equation, 939
height
cylindrical wedges, 1048
right cylinder, 422
helicoid, 1038
Hippasus of Metapontum, Web-E1
homogeneous differential equation, 595
homogeneous functions, 999
homogeneous lamina, 458, 1071
Hooke, Robert, 452, Web-G12
Hooke's law, 452, Web-G12
horizontal asymptotes, 32, 89
polar curves, 719
horizontal line test, 42
horizontal range, 894
Hubble, Edwin P., Web-J8
Hubble's constant, Web-J8

Hubble's law, Web-J8
Humason, Milton L., Web-J8
hyperbolas, 730, 731
  and discriminant, Web-K1
  equilateral, 29, 740
  polar equation, 756
  reflection properties, 743
  sketching, 739, 740
  sketching in polar coordinates, 756, 757, 759
  standard equation, 737, 738
  translated, 740
hyperbolic cosecant, 474
hyperbolic cosine, 474
hyperbolic cotangent, 474
hyperbolic functions, 474
  applications arising in, 475
  derivative formulas, 477
  graphs, 475
  integration formulas, 477–478, 489
  inverse, 478–480
  and unit hyperbola, 476–477
hyperbolic identities, 476
hyperbolic navigation systems, 743
hyperbolic paraboloids, 823, 824, 826
  identifying, 829
hyperbolic secant, 474
hyperbolic sine, 474
hyperbolic spiral, 714, 719
hyperbolic substitution, 514
hyperbolic tangent, 474
hyperboloid, 836
hyperboloid of one sheet, 823, 824
  identifying, 829
hyperboloid of two sheets, 823, 825
  identifying, 829
hyperharmonic series, 627

ideal gas law, 927, 988
ill posed problems, 280
image of S under T, 1059
image of x under f, 3
implicit differentiation, 187–189, 955
  chain rule, 955
implicit functions, 185–186
implicit partial differentiation, 931, 932
  chain rule, 955–956
improper integrals, 547, 549–551, 553
  arc length and surface area, 553
  integrands with infinite discontinuities, 551
  over infinite intervals, 548–550
improper rational functions, integrating, 520, 521
incompressible fluids, 1140
  continuity equation, 1154
increases without bound, 75
increasing functions, 233
  invertibility, 42
increasing sequence, 607
increment, 150, 941
  Euler's method, 582
indefinite integrals, 323, 324
  and Fundamental Theorem of Calculus, 365, 366
  properties, 325–327
independence of path, 1113
independent variable, 3
indeterminate form, 85, 219, 222–225
  of type ∞/∞, 222
  of type $\infty - \infty$, 225
  of type $0/0$, 219
  of type 0^0, ∞^0, 1^∞, 225
  of type $0 \cdot \infty$, 224
index
  for sequence, 598
  of summation, 341
inequalities
  with absolute values, Web-F3, Web-F4
  algebraic properties, Web-E5, Web-E6
  solving, Web-E6–Web-E8
  triangle, Web-F5
infinite discontinuities, 110
  integrands with, 551
infinite intervals, Web-E4
  absolute extrema on, 269, 270
  integrals over, 548–550
infinite limits, 74, 105
  informal view, 75
infinite sequence, 596
infinite series, 596, 614–617, 619, 620
  absolute convergence, 641, 642
  algebraic properties of, 624–626
  alternating harmonic series, 639
  alternating series, 638–641
  binomial series, 674, 675
  comparison test for, 631–633, 645
  conditional convergence of, 643
  convergence/divergence of, 616
  convergence tests for, 623–628, 631–635, 638, 640–643, 645
  divergence test for, 624, 645
  general term of, 623
  geometric series, 617–619
  harmonic series, 620
  hyperharmonic series, 627
  integral test for, 626, 628
  limit comparison test for, 633, 634, 645, A40
  Maclaurin series, 660–663, 669–675, 681, 683, 684, 686
  nth partial sum, 616

power series, 661–666
p-series, 627
ratio test for, 634, 635
root test for, 635
roundoff error with, 672
sums of, 614–617
Taylor series, 659–663, 669, 670, 681, 683–685
terms of, 614
truncation error with, 672
infinite slope, 696, Web-G5
infinity, limits at, 89–96
inflection point at x_0, 236
inflection points, 236, 238, 254
applications, 238, 239
inhibited population growth, 563–564
initial conditions, 328, 562, Web-L4
initial point, vectors, 774
initial side, of angles, A13
initial-value problems, 328
Euler's Method for, 581–583
first-order differential equations, 562
second-order linear homogeneous differential equations, Web-L4
instantaneous acceleration, 290
motion along curves, 882
instantaneous rate of change, 67, 138
instantaneous speed
motion along curves, 882
instantaneous velocity, 136
definition, 136
geometric interpretation, 136
as a limit of average velocities, 136
motion along curves, 882
rectilinear motion, 136, 146
instantaneous velocity function, 146
integer powers, 52
integers, Web-E1
integrability and discontinuities, 358
integrable functions, 354
integrable on $[a, b]$, 354
integral(s)
complete elliptic, of the first kind, 560, 685
curves, 327, 562
double, 1002, 1006
functions defined by, 403, 404
iterated, 1003, 1009, 1010
Mean-Value Theorem for, 368, 369, 386
notation, 323
partial definite, with respect to x and y, 1003
relationship between definite and indefinite, 365, 366
Riemann, 354
sign ($\int$), 323, 324
single, 1006
surface, 1130
tables, 489, 490, 523–528
triple, 1039
of vector-valued functions, 853, 855
integral test, 626, 628, 645
integrand, 323
integrating factor, 587
integration, 323
constant of, 323
and differential equations, 328
and direction fields, 329
hyperbolic substitution, 514
as inverse property of differentiation, 371
iterated (or repeated), 1003, 1009, 1010
limits of, 354
notation for, 323, 324
partial, 1003
of power series, 679, 680
of rates of change, 371
reversing the order of, 1013
and slope fields, 329
u-substitution, 332, 333, 335, 336
integration by parts, 491–493
definite integrals, 496
LIATE, 493, 494
reduction formulas, 497
repeated, 493
tabular, 495
integration formulas, 324, 489, 490
inverse trigonometric functions, 337, 490
vector-value functions, 854
integration methods, 488, 490
intensity, of sound, 59
intercepts, Web-G4
interior, 920
interior point, 920
Intermediate-Value Theorem, 115
International Space Station, 452
International System of Units (SI), 450
interpolating function, Web-J1
intersection, intervals, Web-E5
interval of convergence, 662
finding, 662, 663
power series in $x - x_0$, 663, 664
intervals, Web-E4, Web-E5
intrinsic error, 540
inverse cosine function, 45
inverse function(s), 38, 39, 55
cancellation equations, 39
continuity, 121
domain, 40
domain and range, 40
existence, 41
finding, 40